FEDERAL STATE AUTONOMOUS EDUCATIONAL INSTITUTION
FOR HIGHER EDUCATION
NATIONAL RESEARCH UNIVERSITY HIGHER SCHOOL OF ECONOMICS
Faculty of Mathematics

Beldiev Ivan Sergeevich

Vertex Algebras Associated to Integral Lattices and Vector Bundles on the Projective Line

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Reviewer:
Candidate of Sciences
Makedonskyi Ievgen

Supervisor:
Doctor of Sciences, professor
Kirichenko Valentina Alekseevna

Advisor:
Doctor of Sciences, professor
Feigin Evgeny Borisovich

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Abstract

There is a general construction producing a vector bundle on any smooth algebraic curve from a finite-dimensional representation of the «positive half» Vir_+ of the Virasoro algebra. We study this construction in detail when the curve is the projective line. We prove, in particular, that in this case the bundle is completely determined by the action of the operators L_0 and L_1 and show how to compute it explicitly (i.e., to find the decomposition into the direct sum of line bundles) knowing the Jordan decomposition of L_1. After that, we compute explicitly these bundles for two families of representations of Vir_+: the Fock modules and the Demazure modules of level 1 corresponding to a multiple of the first fundamental weight for the affine Kac-Moody algebra $\widehat{\mathfrak{sl}}(n, \mathbb{C})$.

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1 Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let V be a representation of the «non-negative» half Vir_+ of the Virasoro algebra. In other words, V is equipped with an action of linear operators L_n, $n \in \mathbb{Z}_{\geqslant 0}$, satisfying the commutation relations

$$[L_m, L_n] = (m - n)L_{m+n}.$$

Suppose that V is finite-dimensional and the operator L_0 is diagonalizable with integer eigenvalues. Then, there exists a natural way to construct a vector bundle on an arbitrary smooth algebraic curve starting from the representation V. One can find this construction in [2, Chapter 6]. Let us describe it briefly.

Consider the algebra of formal power series $\mathbb{C}[[z]]$. This algebra can be equipped with the topology whose neighbourhood basis of zero consists of the subspaces $z^N\mathbb{C}[z]$, $N \in \mathbb{Z}$.

It is easy to see that this turns $\mathbb{C}[[z]]$ into a complete topological algebra. Denote by $\mathrm{Aut}\mathcal{O}$ the group of continuous automorphisms of $\mathbb{C}[[z]]$. Note that any continuous automorphism of $\mathbb{C}[[z]]$ is uniquely determined by its action on $z \in \mathbb{C}[[z]]$. The map $z \mapsto f(z)$ gives an automorphism $\mathbb{C}[[z]]$ if and only if $f(z) = a_1 z + a_2 z^2 + a_3 z^3 + \ldots$ and $a_1 \neq 0$.

It can be checked via direct computation that the Lie algebra of the group $\mathrm{Aut}\mathcal{O}$ is isomorphic to the «non-negative part» Vir_+ of the Virasoro algebra. The isomorphism is given by $L_n \mapsto -z^{n+1}\partial_z$.

If V is a finite-dimensional representation of Vir_+ such that L_0 is diagonalizable with integer eigenvalues, then the action of Vir_+ on V can be exponentiated to an action of the group $\mathrm{Aut}\mathcal{O}$ on V thus turning V into a left $\mathrm{Aut}\mathcal{O}$-module.

Now, let X be a smooth algebraic curve. Denote by $\mathcal{O}_x$ the local ring of a point $x \in X$. Recall that a local parameter at x is any element $t_x \in \mathfrak{m}_x$, where $\mathfrak{m}_x \subset \mathcal{O}_x$ is the unique maximal ideal of the ring $\mathcal{O}_x$. We define Aut_X as the set consisting of the elements (x, t_x), where $x \in X$ and t_x is a local parameter at x.

The vector bundle associated to the representation V is defined as follows:

$$\mathcal{V}_X = \mathfrak{Aut}_X \underset{\mathrm{Aut}\mathcal{O}}{\times} V = \mathfrak{Aut}_X \times V / \{((x, t_x), f \cdot v) \sim ((x, f(t_x)), v)\}.$$

It is natural to ask how to describe this vector bundle more explicitly in some particular cases. In this paper, we consider the case when X is the complex projective line $\mathbb{CP}^1$. It is well-known that any finite-dimensional vector bundle on the projective line is isomorphic to a direct sum of some line bundles which are all of the form $\mathcal{O}(k)$, $k \in \mathbb{Z}$. So, the question we want to answer is how to decompose the vector bundles on $\mathbb{CP}^1$ emerging from the abstract construction into the direct some of line bundles.

In Section 3, we work with an arbitrary finite-dimensional representation of Vir_+ and show that the corresponding vector bundle on $\mathbb{P}^1$ is uniquely determined by the action of the operators L_0 and L_1. Moreover, we see how to compute this bundle explicitly knowing the Jordan decomposition of L_1.

In Section 4, we consider a concrete example of representation of Vir_+, namely the so-called Fock modules defined in terms of certain semi-infinite monomials. We describe these modules explicitly and compute the corresponding vector bundles on $\mathbb{P}^1$.

In Sections 5 and 6, we consider Demazure modules of level 1 corresponding to the affine Kac-Moody algebra $\widehat{\mathfrak{sl}}(2, \mathbb{C})$. Again, we describe these modules explicitly and compute the corresponding vector bundles.

In Sections 7-9, we work with Demazure modules of level 1 for $\widehat{\mathfrak{sl}}(n, \mathbb{C})$, where $n \geqslant 2$ is arbitrary. In this case, we compute the vector bundles corresponding to Demazure modules with highest weight equal to a multiple of the first fundamental weight of the root lattice of $\mathfrak{sl}(n, \mathbb{C})$. In the case $n = 3$ considered in Section 8 the answer is given in a particularly compact form.

2 Preliminaries

We start with describing the main construction that we are going to study. The details can be found in [2, Chapter 6].

Denote by $\mathcal{O}$ the $\mathbb{C}$-algebra of formal power series $\mathbb{C}[[z]]$. We introduce a topology on $\mathcal{O}$ letting the set of subspaces $z^N\mathbb{C}[z]$ be a neighbourhood basis of zero. This turns $\mathcal{O}$ into a complete topological algebra. We define $\mathrm{Aut}\mathcal{O}$ as the group of continuous automorphisms

of $\mathcal{O}$. Any such automorphism is determined uniquely by its action on the topological generator z of $\mathcal{O}$, so any element $\rho \in \mathrm{Aut}\mathcal{O}$ is given by a formal power series $\rho(z) \in \mathbb{C}[[z]]$. Note that ρ is a continuous automorphism of $\mathcal{O}$ if and only if $\rho(z) = a_1 z + a_2 z^2 + a_3 z^3 + \dots$ and $a_1 \neq 0$.

Also, we need the following abstract construction. Suppose that G is a group and S is a set. The set S is called a *G-torsor* if S is equipped with a free transitive right action of G, i. e. for any $x, y \in S$ there exists a unique element $g \in G$ such that $y = x \cdot g$. If we fix any element $x \in S$, we can identify S with G via the map $x \cdot g \mapsto g$.

If V is a left G-module and S is a G-torsor, we define the S-twist of V as the set

$$V_S = S \underset{G}{\times} V = S \times V / \{(s \cdot g, v) \sim (s, gv)\}.$$

Choosing $x \in S$, we can identify V with V_S via the map $v \mapsto (x, v)$. This identification depends on x. Nevertheless, since the action of G on V is linear, the vector space structure on V_S is independent of the choice of $x \in S$. Note, however, that V_S does not have a left G-module structure (unless the group G is abelian).

Now, let X be a smooth algebraic curve. For any point $x \in X$ denote by $\mathcal{O}_x$ the local ring at x. Recall that a local parameter (a local coordinate) at $x \in X$ is defined as any element $t_x \in \mathfrak{m}_x \subset \mathcal{O}_x$, where $\mathfrak{m}_x$ is the unique maximal ideal of the ring $\mathcal{O}_x$. A choice of a local parameter $t_x \in \mathcal{O}_x$ defines a (non-canonical) isomorphism $\mathcal{O}_x \cong \mathcal{O} = \mathbb{C}[[z]]$, $t_x \mapsto z$.

Let $\mathfrak{Aut}_x$ be the set of all local coordinates of $\mathcal{O}_x$. The group $\mathrm{Aut}\mathcal{O}$ acts naturally on $\mathfrak{Aut}_x$: if $t_x \in \mathfrak{Aut}_x$ and $\rho \in \mathrm{Aut}\mathcal{O}$, then $\rho(t_x) \in \mathfrak{Aut}_x$. Also, $(\rho \star \mu)(t) = \mu(\rho(t))$, so it is indeed a right action of $\mathrm{Aut}\mathcal{O}$ on $\mathfrak{Aut}_x$. It is clear that this action is free and transitive. So, $\mathfrak{Aut}_x$ is a $\mathrm{Aut}\mathcal{O}$-torsor.

Let V be a $\mathrm{Aut}\mathcal{O}$-module. Then, we define the $\mathfrak{Aut}_x$-twist of V in the following way:

$$\mathcal{V}_x = \mathfrak{Aut}_x \underset{\mathrm{Aut}\mathcal{O}}{\times} V.$$

We wish to combine all torsors $\mathfrak{Aut}_x$ into one bundle on X. For this purpose, define Aut_X as the set of pairs $\{(x, t_x) \mid x \in X, t_x \text{ is a local parameter at x}\}$. Note that Aut_X is also an $\mathrm{Aut}\mathcal{O}$-torsor since $\rho \in \mathrm{Aut}\mathcal{O}$ maps (x, t_x) to $(x, \rho(t_x))$.

We are ready to give the main definition.

Definition 1. *If V is a finite-dimensional $\mathrm{Aut}\mathcal{O}$-module, then*

$$\mathcal{V}_X = \mathfrak{Aut}_X \underset{\mathrm{Aut}\mathcal{O}}{\times} V$$

is the vector bundle associated with V and $\mathfrak{Aut}_X$.

Note that the fibre of $\mathcal{V}_X$ over $x \in X$ is identified with $\mathcal{V}_x$.

A natural way to construct left $\mathrm{Aut}\mathcal{O}$-modules is to exponentiate modules over the corresponding Lie algebra. Namely, consider the Lie algebra $\mathrm{Der}_0\mathcal{O} = z\mathbb{C}[z]\partial_z$. It is not hard to check that $\mathrm{Lie}(\mathrm{Aut}\mathcal{O}) = \mathrm{Der}_0\mathcal{O}$. So, if V is a $\mathrm{Der}_0\mathcal{O}$-module, it is natural to try to exponentiate the action of $\mathrm{Der}_0\mathcal{O}$ on V and obtain the corresponding action of $\mathrm{Aut}\mathcal{O}$ on V. However, this is possible if the action of Der_0 on V satisfies certain conditions. We are going to find these conditions.

A representation of the Lie algebra $\mathrm{Lie}(\mathbb{G}_m) \cong \mathbb{C} \cdot z\partial_z$ on a vector space V can be exponentiated to a representation of the multiplicative group $\mathbb{G}_m$ if and only if the action

of the element $z\partial_z$ on V is diagonalizable with integer eigenvalues. In this case, $a \in \mathbb{G}_m$ acts on the eigenvectors of the operator $z\partial_z$ with eigenvalues n multiplying them by a^n. A short way to write this is the following: $a \mapsto a^{z\partial_z}$.

Next, there is the exponential map

$$\mathrm{Der}_+\mathcal{O} = z^2\mathbb{C}[z]\partial_z \to \mathrm{Aut}_+\mathcal{O} = \{z + a_2z^2 + a_3z^3 + \ldots \mid a_i \in \mathbb{C}\}$$

which is an isomorphism between $\mathrm{Der}_+\mathcal{O}$ and $\mathrm{Aut}_+\mathcal{O}$.

A Der_+-action on a vector space M can be exponentiated to an $\mathrm{Aut}_+\mathcal{O}$-action if it is locally nilpotent, i.e. for any $v \in M$ and $x \in \mathrm{Der}_+\mathcal{O}$ we have $x^N \cdot v = 0$ for sufficiently large N.

So, we arrive at the following proposition.

Proposition 1. *An action of* $\mathrm{Der}_0\mathcal{O}$ *on a vector space* M *can be exponentiated to an action of* $\mathrm{Aut}\mathcal{O}$ *if the following condions hold:*

- *the action of the vector field* $z\partial_z$ *is diagonalizable with integer eigenvalues;*
- *the action of* $\mathrm{Der}_+\mathcal{O}$ *is locally nilpotent.*

From now on, we assume that all representations we consider satisfy these two conditions.

Recall the definition of the Virasoro algebra.

Definition 2. *The Virasoro algebra* Vir *is a complex Lie algebra generated as a vector space by* L_n, $n \in \mathbb{Z}$, *and* the central element c, *satisfying the commutation relations:*

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}, \quad [c, L_n] = 0.$$

It can be checked directly that the elements $-z^{n+1}\partial_z \in \mathrm{Der}_0\mathcal{O}$, $n \geq 0$, satisfy the same commutation relations as $L_n \in \mathrm{Vir}$. This means that the Lie algebra $\mathrm{Der}_0\mathcal{O}$ can be identifed with the Lie subalgebra $\mathrm{Vir}_+ \subset \mathrm{Vir}$ generated by L_n, $n \geq 0$. For brevity, we will often write simply $L_n = -z^{n+1}\partial_z$.

Let V be a representation of $\mathrm{Der}_0\mathcal{O}$ satisfying these two conditions. One can ask how to write the action of $\mathrm{Aut}\mathcal{O}$ on V explicitly. Recall that any element of the group $\mathrm{Aut}\mathcal{O}$ can be viewed as a power series

$$z \mapsto f(z) = a_1z + a_2z^2 + a_3z^3 + \ldots, a_1 \neq 0.$$

We need to find $v_i \in \mathbb{C}$ such that

$$f(z) = \exp\left(\sum_{i>0} v_i z^{i+1}\partial_z\right) v_0^{z\partial_z} \cdot z. \tag{1}$$

Since $z\partial_z \cdot z = z$, we have $v_0^{z\partial_z} \cdot z = v_0 z$. This means that

$$\exp\left(\sum_{i>0} v_i z^{i+1}\partial_z\right) v_0 z = v_0 z + \sum_{i>0} v_i v_0 z^{i+1} + \frac{1}{2}\left(\sum_{i>0} v_i z^{i+1}\partial_z\right) \cdot \sum_{i>0} v_i v_0 z^{i+1} + \ldots,$$

which implies consecutively

$$v_0 = f'(0),$$

$$v_1 = \frac{1}{2}\frac{f''(0)}{f'(0)},$$

$$v_2 = \frac{1}{6}\frac{f'''(0)}{f'(0)} - \frac{1}{4}\left(\frac{f''(0)}{f'(0)}\right)^2, \text{ etc.}$$

The linear operator $R(f)\colon V \to V$ is given by the formula

$$R(f) = \exp\left(-\sum_{j>0} v_j L_j\right) v_0^{-L_0} \tag{2}$$

.

3 Vector bundles on the projective line

Let V be a finite-dimensional representation of Vir_+ satisfying the conditions of Proposition 1. In this section, we prove that the corresponding vector bundle on the projective line depends only on the action of the operators L_0 and L_1 and show a way to compute the bundle explicitly (i.e. to find its decomposition into the direct sum of line bundles) knowing the Jordan decomposition of L_1.

Recall that the vector bundle we are interested in is defined by the formula

$$\mathcal{V}_{\mathbb{P}^1} = \mathrm{Aut}_{\mathbb{P}^1} \underset{\mathrm{Aut}\mathcal{O}}{\times} V,$$

where

$$\mathrm{Aut}_{\mathbb{P}^1} = \{(x, t_x) \mid x \in \mathbb{P}^1,\ t_x \text{ is a local parameter at } x\}.$$

The identification of pairs (x, t_x) is done as follows:

$$((x, f(t_x)), v) \sim ((x, t_x), R(f)v),$$

where

$$f = f(z) = a_1 z + a_2 z^2 + a_3 z^3 + \dots, \quad R(f) = \exp(-\sum_{j>0} v_j L_j) v_0^{-L_0},$$

and the numbers v_j, $j \geqslant 0$, are defined uniquely by the formula

$$f(z) = \exp(-\sum_{j>0} v_j z^{j+1}\partial_z) v_0 z.$$

Consider homogeneous coordinates $(z_0 : z_1)$ on $\mathbb{P}^1$. Then, $\mathbb{P}^1$ is covered by the two standard affine charts $U_0 = \{(z_0 : z_1) \mid z_0 \neq 0\}$, $U_1 = \{(z_0 : z_1) \mid z_1 \neq 0\}$. A vector bundle on $\mathbb{P}^1$ is defined uniquely by the gluing cocycle φ_{01} corresponding to these two charts.

Let $(a_0 : a_1)$ be any point of $\mathbb{P}^1$. The chart U_0 defines the local parameter at $(a_0 : a_1)$ equal to $\frac{z_1}{z_0} - \frac{a_1}{a_0} = t - a$, while the chart U_1 defines the local parameter $\frac{z_0}{z_1} - \frac{a_0}{a_1} = \frac{1}{t} - \frac{1}{a}$ (here $t = \frac{z_1}{z_0}$, $a = \frac{a_1}{a_0}$). The element $f \in \mathrm{Aut}\mathcal{O}$ mapping the first local parameter to the second one must satisfy the equality

$$f(t - a) = \frac{1}{t} - \frac{1}{a}, \text{ so}$$

$$f(z) = \frac{1}{z + a} - \frac{1}{a} = -\frac{z}{a^2} + \frac{z^2}{a^3} - \frac{z^3}{a^4} + \dots$$

The following proposition gives an explicit formula for the gluing cocycle $\varphi_{01} = R(f)$.

Proposition 2. $\varphi_{01} = R(f) = \exp(\frac{L_1}{a})(-\frac{1}{a^2})^{-L_0}$.

Proof. By definition, $R(f) = \exp(-\sum_{j>0} v_j L_j) v_0^{-L_0}$, where $v_0, v_1, v_2, \ldots$ are determined uniquely by (1). So, it suffices to check that the values $v_0 = -\frac{1}{a^2}$, $v_1 = -\frac{1}{a}$, $v_2 = v_3 = \ldots = 0$ satisfy (1).

Denote $T = z^2\partial_z$. We are going to prove by induction that $T^n z = n! z^{n+1}$. The base $n = 1$ is clear. The induction step is done as follows:

$$T^{n+1}z = T(T^n z) = T(n!z^{n+1}) = z^2(n+1)n!z^n = (n+1)!z^{n+2}.$$

This implies that

$$-\frac{1}{a^2}\exp(-\frac{z^2\partial_z}{a})z = -\frac{1}{a^2}(1 - \frac{T}{a} + \frac{T^2}{2!a^2} - \frac{T^3}{3!a^3} + \ldots)z =$$

$$= -\frac{1}{a^2}(z - \frac{z^2}{a} + \frac{z^3}{a^2} - \ldots) = -\frac{z}{a^2} + \frac{z^2}{a^3} - \frac{z^3}{a^4} + \ldots = f(z).$$

This finishes the proof of the proposition. □

Example 1. *Consider a one-dimensional vector space $V = \langle v \rangle$ such that $L_0(v) = Nv$, $N \in \mathbb{Z}$ (and $L_i(v) = 0$ if $i > 0$). In this case,*

$$R(f)v = (-\frac{a_0^2}{a_1^2})^{-N} = (-1)^N(\frac{a_1}{a_0})^{2N},$$

so the gluing cocycle $\varphi_{01}(z)$ is equal to $(-1)^N(\frac{z_1}{z_0})^{2N}$. The line bundle corresponding to this cocycle is $\mathcal{O}(2N)$.

Example 1 is generalized in the following proposition.

Proposition 3. *Let V be an s-dimensional representation of the algebra Vir_+ such that the action of L_1 on V is one Jordan block (the corresponding eigenvalue is automatically 0) and v_1, v_2, $\ldots$, v_n are L_0-homogeneous vectors of V forming a Jordan basis of L_1. Then, if $L_0(v_1) = lv_1$, the corresponding vector bundle on $\mathbb{P}^1$ is isomorphic to $\mathcal{O}(2l - s + 1)^{\oplus s}$. In other words, this bundle is isomorphic to $\mathcal{O}(l_{max} + l_{min})^{\oplus s}$, where $l_{max} = l$ and l_{min} are the maximal and minimal L_0-degrees of vectors in V.*

Proof. The commutation relation $[L_1, L_0] = L_1$ implies that $L_0(v_i) = (l - i + 1)v_i$ for all $i = 1, 2, \ldots, s$.

The matrix of the operator $\exp(\frac{L_1}{a})$ relative to the given basis is the following:

$$\begin{pmatrix} 1 & \frac{1}{a} & \frac{1}{2!a^2} & \frac{1}{3!a^3} & \cdots & \frac{1}{(s-1)!a^{s-1}} \\ 0 & 1 & \frac{1}{a} & \frac{1}{2!a^2} & \cdots & \frac{1}{(s-2)!a^{s-2}} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \ldots & 1 \end{pmatrix},$$

so the gluing cocycle is given by the matrix

$$\begin{pmatrix} a^{2l-2s+2} & a^{2l-2s+3} & \frac{1}{2!}a^{2l-2s+4} & \frac{1}{3!}a^{2l-2s+5} & \cdots & \frac{1}{(s-1)!}a^{2l-s+1} \\ 0 & a^{2l-2s+4} & a^{2l-2s+5} & \frac{1}{2!}a^{2l-2s+6} & \cdots & \frac{1}{(s-2)!}a^{2l-s+2} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \ldots & a^{2l} \end{pmatrix}.$$

For convenience, we factor out the factor $a^{2l-2s+2}$ from all elements of the matrix. In geometric terms, this corresponds to the «twisting», i.e. tensor multiplication by $\mathcal{O}(-2l+2s-2)$. We obtain the matrix

$$\begin{pmatrix} 1 & a & \frac{1}{2!}a^2 & \frac{1}{3!}a^3 & \dots & \frac{1}{(s-1)!}a^{s-1} \\ 0 & a^2 & a^3 & \frac{1}{2!}a^4 & \dots & \frac{1}{(s-2)!}a^{s} \\ 0 & 0 & a^4 & a^5 & \dots & \frac{1}{(s-3)!}a^{s+1} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & a^{2s-2} \end{pmatrix}.$$

We can perform the following actions with this matrix:

- add a linear combination of any set of columns with coefficients in $\mathbb{C}[a^{-1}]$ to another column (this corresponds to a regular change of trivialization over the chart U_1);
- add a linear combination of any set of rows with coefficients in $\mathbb{C}[a]$ to another row (this corresponds to a regular change of trivialization over the chart U_0).

We can also permute the columns and the rows and multiply them by arbitrary non-zero numbers.

First, we add the rightmost column multiplied by suitable coefficients to all other columns in such a way that all elements in the first row vanish (except for the last one). This gives us the matrix

$$\begin{pmatrix} 0 & 0 & 0 & 0 & \dots & \frac{1}{(s-1)!}a^{s-1} \\ a & a^2 & \frac{1}{2!}a^3 & \frac{1}{3!}a^4 & \dots & \frac{1}{(s-2)!}a^{s} \\ 0 & 2a^3 & 2a^4 & \frac{2}{2!}a^5 & \dots & \frac{1}{(s-3)!}a^{s+1} \\ 0 & 0 & 3a^5 & 3a^6 & \dots & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & a^{2s-2} \end{pmatrix}.$$

Now, we add the first row multiplied by suitable coefficients to the remaining ones. After this operation, all elements in the rightmost column become zero (except for the first one). Finally, we move the last column to the leftmost position, divide the i-th row by $i-1$ ($i = 2, 3, 4, \dots, s$) and multiply the first row by $(s-1)!$. The matrix we obtain is the following:

$$\begin{pmatrix} a^{s-1} & 0 & 0 & 0 & \dots & 0 \\ 0 & a & a^2 & \frac{1}{2!}a^3 & \dots & \frac{1}{(s-2)!}a^{s-2} \\ 0 & 0 & a^3 & a^4 & \dots & \frac{1}{(s-3)!}a^{s-3} \\ 0 & 0 & 0 & a^5 & \dots & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & a^{2s-3} \end{pmatrix}.$$

Note that right bottom $(s-1)\times(s-1)$ minor of this matrix is of the same form as the initial matrix of the cocycle. Now, we can apply the procedure we just described to this minor. Continuing similarly with the minors of sizes $(s-2)\times(s-2)$, $(s-3)\times(s-3)$ etc., we eventually obtain the scalar matrix with all diagonal elements equal to a^{s-1}. The cocycle given by this matrix corresponds to the bundle $\mathcal{O}(s-1)^{\oplus s}$. It remains to «twist» back this bundle by $\mathcal{O}(2l-2s+2)$ obtaining the bundle $\mathcal{O}(2l-s+1)^{\oplus s}$. $\square$

So, we can speak about vector bundles corresponding to spaces equipped with the action of only two operators L_0, L_1 such that $[L_1, L_0] = L_1$. We will use it later computing vector bundles corresponding to subspaces of Vir_+-modules that are invariant under L_0 and L_1 but not necessarily under the whole algebra Vir_+.

4 The representations of Vir_+ on the space of semi-infinite products

In this section, we compute the vector bundles associated with certain representations of Vir_+ called the Fock modules. First, we define these representations.

Denote by $\mathcal{F}$ the space with a basis consisting of all semi-infinite monomials of the form

$$\psi_{i_1} \wedge \psi_{i_2} \wedge \psi_{i_3} \wedge \ldots,$$

where $i_1 < i_2 < i_3 < \ldots$ and $i_{n+1} - i_n = 1$ for $n \gg 0$.

The space $\mathcal{F}$ is the direct sum

$$\mathcal{F} = \bigoplus_{m \in \mathbb{Z}} \mathcal{F}^{(m)},$$

where $\mathcal{F}^{(m)}$ is the linear span of all semi-infinite monomials of charge m, i.e. the monomials coinciding with $|m\rangle = \psi_m \wedge \psi_{m+1} \wedge \psi_{m+2} \wedge \ldots$ for all sufficiently large indices. The monomial $|m\rangle$ is called *the vacuum monomial of charge* m.

Introduce *the free fermions* ψ_j, ψ_j^*, $j \in \mathbb{Z}$ acting on the space $\mathcal{F}$. By definition,

$$\psi_j(\psi_{i_1} \wedge \psi_{i_2} \wedge \ldots) = \begin{cases} 0, \text{if } j = i_s \text{ for some } s, \\ (-1)^s \psi_{i_1} \wedge \ldots \wedge \psi_{i_s} \wedge \psi_j \wedge \psi_{i_{s+1}} \wedge \ldots, \text{ if } i_s < j < i_{s+1}, \end{cases}.$$

$$\psi_j^*(\psi_{i_1} \wedge \psi_{i_2} \wedge \ldots) = \begin{cases} 0, \text{if } j \neq i_s \text{ for all } s, \\ (-1)^{s+1} \psi_{i_1} \wedge \ldots \wedge \psi_{i_{s-1}} \wedge \psi_{i_{s+1}} \wedge \ldots, \text{ if } i_s = j. \end{cases}.$$

It is easy to check that the operators ψ_j, ψ_j^* satisfy the following relations:

$$\psi_i \psi_j^* + \psi_j^* \psi_i = \delta_{ij},\ \psi_i \psi_j + \psi_j \psi_i = 0,\ \psi_i^* \psi_j^* + \psi_j^* \psi_i^* = 0.$$

It follows from the definitions that $\psi_j(\mathcal{F}^{(m)}) \subset \mathcal{F}^{(m+1)}$, $\psi_j^*(\mathcal{F}^{(m)}) \subset \mathcal{F}^{(m-1)}$.

Next, we introduce the so-called *free bosons*. They are defined by the formulas

$$b_n = \sum_{k \in \mathbb{Z}} \psi_k^* \psi_{n+k},\ n \neq 0;\ \ b_0 = \sum_{k \geqslant 0} \psi_k^* \psi_k - \sum_{k<0} \psi_k \psi_k^*. \tag{3}$$

It follows from the definitions that $b_n(\mathcal{F}^{(m)}) \subset \mathcal{F}^{(m)}$. It can also be checked that b_0 acts on $\mathcal{F}^{(m)}$ by the scalar m.

The free bosons satisfy the following commutation relations:

$$[b_n, b_m] = n\delta_{n,m}.$$

and act on *the Fock space* $\mathcal{B} = \mathbb{C}[x_1, x_2, \ldots; q, q^{-1}]$ by the formulas

$$r(b_m) = m\frac{\partial}{\partial x_m},\ r(b_{-m}) = x_m,\ r(b_0) = q\frac{\partial}{\partial q}.$$

Recall the definition of the Heisenberg algebra.

Definition 3. *The Heisenberg algebra $\mathcal{H}$ is a Lie algebra with the basis $\{a_n, n \in \mathbb{Z}; h\}$ satisfying the commutation relations $[h, a_n] = 0$, $[a_m, a_n] = m\delta_{m,-n}h$.*

It follows that the operators $r(b_m)$ define a representation of the Heisenberg algebra on the Fock space $\mathcal{B}$ (the element h acts as the identity). In the definition, we used the notation a_n in order not to confuse with the operators b_n introduced previously. From now on, we use b_n instead of a_n as in [2].

The *boson-fermion* correspondence (see [2, 3]) identifies the spaces $\mathcal{B}$ and $\mathcal{F}$. Any vacuum vector $|m\rangle$ is mapped to q^m under this identification.

The Virasoro algebra acts on $\mathcal{B}$ and hence on $\mathcal{F}$ by the following formulas:

$$L_0 = \frac{1}{2}b_0^2 + b_{-1}b_1 + b_{-2}b_2 + \ldots, \; L_k = \frac{1}{2}\sum_s b_s b_{k-s} \text{ при } k \neq 0.$$

Now, we are ready to define the Fock modules $V_{n,k}$. Consider the vector

$$v_{n,k} = \psi_{-n} \wedge \psi_{-n+1} \wedge \ldots \wedge \psi_{-n+k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \ldots$$

Definition 4. *The Fock module $V_{n,k}$ is the linear span of all vectors obtained from the vector $v_{n,k}$ under the action of all polynomials in b_n, $n \geqslant 0$.*

We are going to do the following:

1. Describe the spaces $V_{n,k}$ explicitly and find their dimensions.
2. Show that $V_{n,k}$ are invariant under the action of Vir_+.
3. Compute explicitly the vector bundles on $\mathbb{P}^1$ associated with the representations of Vir_+ on the spaces $V_{n,k}$.

Proposition 4. *The space $V_{n,k}$ is the linear span of all semi-infinite monomials of the form $\psi_{i_0} \wedge \psi_{i_1} \wedge \ldots \wedge \psi_{i_k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \ldots$, где $-n \leqslant i_0 < i_1 < \ldots < i_k \leqslant k$.*

To prove this proposition, we need the following lemma.

Lemma 1. *Suppose that*

$$b_{-j_s}b_{-j_{s-1}}\ldots b_{-j_1}(|0\rangle) = \sum_{t=(t_0,t_1,\ldots,t_k)} a_t\psi_{t_0}\wedge\psi_{t_1}\wedge\ldots\wedge\psi_{t_k}\wedge\psi_{k+1}\wedge\psi_{k+2}\wedge\ldots + \text{«extra» monomials},$$

where the sum is taken over all t such that $-n \leqslant t_0 < t_1 < \ldots < t_k \leqslant k$ and all monomials not of this form are called «extra». Then

$$b_{j_s}b_{j_{s-1}}\ldots b_{j_1}(v_{n,k}) = \sum_{t=(t_0,t_1,\ldots,t_k)} a_t\psi_{k-n-t_k} \wedge \psi_{k-n-t_{k-1}} \wedge \ldots \wedge \psi_{k-n-t_0} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \ldots$$

Proof. We prove the lemma by induction on s. The base for $s = 0$ is trivial. Let us prove the induction step from s to $s + 1$.

First, we note that formulas (3) imply $V_{n,k} \subset \mathcal{F}^{(0)}$ and $b_0(V_{n,k}) = 0$. The explicit formula for the action of b_m on the semi-infinite monomials is the following:

$$b_m(\psi_{i_1}\wedge\psi_{i_2}\wedge\psi_{i_3}\wedge\ldots) = -(\psi_{i_1+m}\wedge\psi_{i_2}\wedge\psi_{i_3}\wedge\ldots+\psi_{i_1}\wedge\psi_{i_2+m}\wedge\psi_{i_3}\wedge\ldots+\psi_{i_1}\wedge\psi_{i_2}\wedge\psi_{i_3+m}\wedge\ldots) \tag{4}$$

(we assume here that the elements ψ_{i_s} anti-commute). So, we have

$$b_{-j_{s+1}} b_{-j_s} \dots b_{-j_1}(\psi_0 \wedge \psi_1 \wedge \psi_2 \wedge \dots) =$$

$$= b_{-j_{s+1}}(\sum_{t=(t_0,t_1,\dots,t_k)} a_t \psi_{t_0} \wedge \psi_{t_1} \wedge \dots \wedge \psi_{t_k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots + (\text{«extra» monomials})) =$$

$$= -\sum_t a_t(\psi_{t_0-j_{s+1}} \wedge \psi_{t_1} \wedge \dots \wedge \psi_{t_k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots + \psi_{t_0} \wedge \psi_{t_1-j_s+1} \wedge \dots \wedge \psi_{t_k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots + \dots$$

$$\dots + \psi_{t_0} \wedge \psi_{t_1} \wedge \dots \wedge \psi_{t_k-j_s+1} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots) + \text{«extra» monomials.}$$

We remark that some of the summands in the last sum might be «extra».

At the same time,

$$b_{j_{s+1}} b_{j_s} \dots b_{j_1}(\psi_{-n} \wedge \psi_{-n+1} \wedge \dots \wedge \psi_{-n+k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots) =$$

$$= b_{j_{s+1}}(\sum_t a_t \psi_{k-n-t_k} \wedge \psi_{k-n-t_{k-1}} \wedge \dots \wedge \psi_{k-n-t_0} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots) =$$

$$= -\sum_t a_t(\psi_{k-n-t_k+j_{s+1}} \wedge \psi_{k-n-t_{k-1}} \wedge \dots \wedge \psi_{k-n-t_0} \wedge \psi_{k+1} \wedge \dots +$$

$$+\psi_{k-n-t_k} \wedge \psi_{k-n-t_{k-1}+j_{s+1}} \wedge \dots \wedge \psi_{k-n-t_0} \wedge \dots + \dots + \psi_{k-n-t_k} \wedge \psi_{k-n-t_{k-1}} \wedge \dots \wedge \psi_{k-n-t_0+j_{s+1}} \wedge \dots).$$

□

We are ready to prove Proposition 4

Proof of Proposition 4. The explicit formula for the action of the operators b_i immediately implies that each element of the space $V_{n,k}$ is a linear combination of monomials of the form $\psi_{i_0} \wedge \psi_{i_1} \wedge \dots \wedge \psi_{i_k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots$, where $-n \leqslant i_0 < i_1 < \dots < i_k \leqslant k$. Our purpose is to prove that all such monomials lie in $V_{n,k}$.

Any semi-infinite monomial $\psi_{i_0} \wedge \psi_{i_1} \wedge \dots \wedge \psi_{i_k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots$ of this form is an element of the space $\mathcal{F}^{(0)}$, so the corresponding under the boson-fermion correspondence polynomial is indepenedent of q, q^{-1}. It follows that

$$\psi_{i_0} \wedge \psi_{i_1} \wedge \dots \wedge \psi_{i_k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots = P(b_{-1}, b_{-2}, \dots)(\psi_0 \wedge \psi_1 \wedge \psi_2 \wedge \dots)$$

for some polynomial P. We are going to prove that

$$P(b_1, b_2, \dots)v_{n,k} = \psi_{k-n-i_k} \wedge \psi_{k-n-i_{k-1}} \wedge \dots \wedge \psi_{k-n-i_0} \wedge \psi_{k+1} \wedge \psi_{k+2} \dots$$

But this follows easily from Lemma 1. Indeed, since

$$\psi_{i_0} \wedge \psi_{i_1} \wedge \dots \wedge \psi_{i_k} \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \dots = P(b_{-1}, b_{-2}, \dots)(\psi_0 \wedge \psi_1 \wedge \psi_2 \wedge \dots),$$

application of Lemma 1 to each of the monomials of the polynomial P gives us the equality

$$P(b_1, b_2, \dots)v_{n,k} = \psi_{k-n-i_k} \wedge \psi_{k-n-i_{k-1}} \wedge \dots \wedge \psi_{k-n-i_0} \wedge \psi_{k+1} \wedge \psi_{k+2} \dots$$

This concludes the proof of Proposition 4. □

Corollary 1. $\dim V_{n,k} = \binom{n+k+1}{n}$.

Next, we are going to prove that $V_{n,k}$ is a representation of Vir_+.

Proposition 5. *The space $V_{n,k}$ is invariant under the action of* Vir_+.

Proof. Recall that L_n is given by the following formulas:

$$L_0 = \frac{1}{2}b_0^2 + b_{-1}b_1 + b_{-2}b_2 + \dots, \; L_k = \frac{1}{2}\sum_s b_s b_{k-s} \text{ for } k \neq 0.$$

One can check (see [2, Chapters 2,3]) that the action of L_0 on any homogeneous polynomial $f \in \mathbb{C}[x_1, x_2, \dots]$ consists in multiplying it by $\deg f$. Under the boson-fermion correspondence, any semi-infinite monomial is identified with a homogeneous in $x_1, x_2, \dots$ polynomial in the Fock space $\mathcal{B} = \mathbb{C}[x_1, x_2, \dots; q, q^{-1}]$ (the L_0-degree of any semi-infinite monomial $\psi_{i_1} \wedge \psi_{i_2} \wedge \psi_{i_3} \wedge \dots \in \mathcal{F}^{(0)}$ is equal to $-i_0 + (1 - i_1) + (2 - i_2) + \dots$). So, the space $V_{n,k}$ is L_0-invariant.

We need to check that $L_m(V_{n,k}) \subset V_{n,k}$ if $m > 0$. Consider the action of L_m on an arbitrary semi-infinite monomial $v = \psi_{i_1} \wedge \psi_{i_2} \wedge \psi_{i_3} \wedge \dots$, $i_1 < i_2 < i_3 < \dots$. It follows from the formulas for L_m that $L_m(v)$ is a linear combination of the monomials $\psi_{j_1} \wedge \psi_{j_2} \wedge \psi_{j_3} \dots$ such that not more than two indices j_s differ from i_s (we do not assume here that $j_1 < j_2 < j_3 < \dots$).

Suppose that $j_s \neq i_s$ and $j_r \neq i_r$. Then, the monomial $\psi_{j_1} \wedge \psi_{j_2} \wedge \psi_{j_3} \dots$ either does not appear or appears twice in the expression for $L_m(v)$. In the second case, it appears after the applicatoin of $b_{j_r - i_r} b_{j_s - i_s}$ and $b_{j_r - i_s} b_{j_s - i_r}$ to the monomial v. The application of $b_{j_r - i_r} b_{j_s - i_s}$ gives us this monomial with coefficient $+1$ while the application of $b_{j_r - i_s} b_{j_s - i_r}$ gives us this monomial with coefficient -1. So, these two monomials cancel each other.

If exactly one index j_s does not equal i_s, then $j_s > i_s$ for L_0-degree reasons since L_m changes the L_0-degree of an L_0-homogeneous monomial by m.

So, we proved that the expression for $L_m(v)$ $(m > 0)$ contains only the monomials $\psi_{j_1} \wedge \psi_{j_2} \wedge \psi_{j_3} \dots$ such that $j_s \geq i_s$ for all s (moreover, the equality holds for all s except one). This implies Vir_+-invariance of $V_{n,k}$. □

Example 2. *Before we proceed to the general situation, let us compute the vector bundle associated with $V_{n,k}$ for $k = 0$. Denote by v_k the semi-infinite monomial $v_k = \psi_{-k} \wedge \psi_1 \wedge \psi_2 \wedge \dots$. Then, the vectors $v_n, v_{n-1}, \dots, v_1, v_0$ form a basis of the space $V_{n,0}$. It is easy to see from the formula for L_1 that $L_1(v_s) = (s-1)v_{s-1}$, $s = 1, 2, \dots, n$, $L_1(v_0) = 0$. So, the space $V_{n,0}$ is the direct sum of two Jordan blocks*

$$V_{n,0} = \langle v_n, v_{n-1}, \dots, v_1 \rangle \oplus \langle v_0 \rangle.$$

Moreover, $v_1, v_2, \dots, v_n$ is a Jordan basis of the first block up to multiplication of the vectors by non-zero coefficients. Since $L_0(v_n) = nv_n$, $L_0(v_0) = 0$, it follows from Proposition 3 that the first and the second blocks give us the bundles $\mathcal{O}(n+1)^{\oplus n}$ and $\mathcal{O}$ respectively. So, the whole vector bundle is isomorphic to

$$\mathcal{O}(n+1)^{\oplus n} \oplus \mathcal{O}.$$

In order to work with the case of general k, we need an explicit formula for the action of L_1 on the semi-infinite monomials. This formula is given in the following lemma.

Lemma 2.

$$L_1(\psi_{-i_0} \wedge \psi_{-i_1} \wedge \dots \psi_{-i_k} \wedge \psi_{k+1} \wedge \dots) = (i_0 - 1)\psi_{-i_0+1} \wedge \psi_{-i_1} \wedge \dots \psi_{-i_k} \wedge \psi_{k+1} \wedge \dots +$$

$$+(i_1 - 1)\psi_{-i_0} \wedge \psi_{-i_1+1} \wedge \dots \psi_{-i_k} \wedge \psi_{k+1} \wedge \dots + \dots + (i_k - 1)\psi_{-i_0} \wedge \psi_{-i_1} \wedge \dots \psi_{-i_k+1} \wedge \psi_{k+1} \wedge \dots$$

Proof. It follows from the proof of Proposition 5 that the expression for $L_1(\psi_{-i_0} \wedge \psi_{-i_1} \wedge \ldots \psi_{-i_k} \wedge \psi_{k+1} \wedge \ldots)$ can contain only the monomials from the right hand side of the formula we prove (they are obtained from the initial monomial by increasing one index by 1). It remains to find the coefficient of each of these monomials.

Rewrite the formula for L_1 in the following more convenient form:

$$L_1 = b_0 b_1 + b_{-1} b_2 + b_{-2} b_3 + \ldots$$

It follows from this formula that the result of the action of L_1 on the monomial $\psi_{-i_0} \wedge \psi_{-i_1} \wedge \ldots \psi_{-i_k} \wedge \psi_{k+1} \wedge \ldots$ is the sum of monomials obtained by the following operation: we choose two (probably coinciding) indices $-i_r$, $-i_s$, replace one of them (say, $-i_r$) with $-i_r + m + 1$, $m \geq 1$ and, assuming that the monomial we obtained is not zero (i.e., $-i_r + m + 1$ is not equal to any of i_p), replace $-i_s$ with $-i_s - m$.

Consider, for example, the monomial $\psi_{-i_0+1} \wedge \psi_{-i_1} \wedge \ldots \psi_{-i_k} \wedge \psi_{k+1} \wedge \ldots$ It can be obtained from the initial monomial by shifting i_0 to an arbitrary index from $-i_0 + 2$ to k except for $-i_1, -i_2, \ldots, -i_k$ and then returning the new index to the position $-i_0 + 1$. This gives us the coefficient $k + i_0 - 1 - k = i_0 - 1$.

A similar argument for the monomial $\psi_{-i_0} \wedge \psi_{-i_1+1} \wedge \ldots \psi_{-i_k} \wedge \psi_{k+1} \wedge \ldots$ gives the coefficient $k + i_1 - 1 - (k - 1) = i_1$. However, this monomial can also be obtained in the following way: we can replace $-i_0$ by $-i_1 + 1$ and then replace $-i_1$ with $-i_0$. The result of this operation is the monomial

$$\psi_{-i_1+1} \wedge \psi_{-i_0} \wedge \ldots \wedge \psi_{-i_k} \wedge \psi_{k+1} \wedge \ldots$$

equal to $-\psi_{-i_0} \wedge \psi_{-i_1+1} \wedge \ldots \psi_{-i_k} \wedge \psi_{k+1} \wedge \ldots$ due to the anticommutation relations. The resulting coefficient is equal to $i_1 - 1$.

For the other monomials, the argument is similar. □

It is easy to deduce the following proposition from Lemma 2.

Proposition 6. *The Fock module $V_{n,k}$ is the direct sum of the L_1-invariant subspaces $V^r_{n,k}$, $r = 0, 1, \ldots, k, k+1$, where $V^r_{n,k}$ is the linear span of all semi-infinite monomials from $V_{n,k}$ with exactly r negative indices.*

Denote by V_n the linear span of all monomials of the form $\psi_i \wedge \psi_0 \wedge \psi_1 \wedge \ldots$ such that $-n \leqslant i \leqslant -1$. Also, denote by W_k the linear span of the monomials of the form $\psi_i \wedge \psi_{k+1} \wedge \psi_{k+2} \wedge \ldots$ such that $0 \leqslant i \leqslant k$. Note that each space $V^r_{n,k}$ is isomorphic to the tensor product $V_n^{\otimes r} \otimes W_k^{\otimes k-r+1}$ with the relations of anticommutation of the first k and the last $k - r + 1$ tensor factors. Moreover, this identification agrees with the L_1-action on $V^r_{n,k}$, V_n and W_k: the action of L_1 on $V^r_{n,k}$ is identified with the action of the operator

$$L_1 \otimes \mathrm{Id} \otimes \mathrm{Id} \otimes \ldots \otimes \mathrm{Id} + \mathrm{Id} \otimes L_1 \otimes \mathrm{Id} \otimes \ldots \otimes \mathrm{Id} + \ldots + \mathrm{Id} \otimes \mathrm{Id} \otimes \mathrm{Id} \otimes \ldots \otimes L_1$$

on $V_n^{\otimes r} \otimes W_k^{\otimes k-r+1}$. Also, this identification agrees with the L_0-grading in the following sense: $L_0(v_1 \otimes v_2 \otimes \ldots \otimes v_{k+1}) = \sum_i L_0(v_i)$ up to adding a constant.

So, the problem is reduced to the following question. Suppose that we have L_0-graded finite-dimensional vector spaces $V_1, V_2, \ldots, V_m$ equipped with an action of nilpotent operators $A_1, A_2, \ldots, A_m$ such that each A_i acts by one Jordan block and has a Jordan basis consisting of L_0-homogeneous vectors. Consider the space $V_1 \otimes V_2 \otimes \ldots \otimes V_n$ equipped with the action of the operator

$$A = A_1 \otimes \mathrm{Id} \otimes \ldots \otimes \mathrm{Id} + \mathrm{Id} \otimes A_2 \otimes \ldots \otimes \mathrm{Id} + \ldots + \mathrm{Id} \otimes \mathrm{Id} \otimes \ldots \otimes A_m.$$

Let V be the vector space obtained from $V_1 \otimes V_2 \otimes \ldots \otimes V_n$ under factorization by some relations agreeing with the A-action. The question is how to find the Jordan decomposition of the operator on V induced by A and also to compute the L_0-degrees of the vectors in each Jordan block.

We start with the case when there are no relations, i.e. $V = V_1 \otimes V_2 \otimes \ldots \otimes V_n$. It turns out that the most important case is $n = 2$. In this case, we have L_0-graded finite-dimensional vector spaces V, W with the action of nilpotent operators A, B respectively such that both operators act by one Jordan block. Also, suppose that A, B are L_0-homogeneous of degree 1, i.e. these operators reduce the L_0-grading of any L_0-homogeneous vector by 1. Let $v_1, v_2, \ldots, v_n$ and $w_1, w_2, \ldots, w_m$ be L_0-homogeneous Jordan basis of V and W respectively, and $\deg_{L_0}(v_n) = l_1$, $\deg_{L_0}(w_m) = l_2$. Then, the following proposition holds.

Proposition 7. *The space $V \otimes W$ equipped with the action of the operator $A \otimes \mathrm{Id} + \mathrm{Id} \otimes B$ is the direct sum of Jordan blocks of sizes $n+m-1, n+m-3, n+m-5, \ldots$. Each of these blocks has an L_0-homogeneous Jordan basis and the maximal L_0-degrees of vectors in these bases are equal to $l_1 + l_2$, $l_1 + l_2 - 1$, $l_1 + l_2 - 2$, $\ldots$ respectively.*

Proof. The vector spaces V, W can be equipped with a structure of representations of the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ in such a way that the action of the element $f \in \mathfrak{sl}(2, \mathbb{C})$ on V and W coincide with the action of the operators A and B respectively and the action of $h \in \mathfrak{sl}(2, \mathbb{C})$ is diagonalizable in the given bases (the h-grading is obtained from the L_0-grading after a shift and multiplication by 2). This turns V and W into irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$ with highest weights $n-1$ and $m-1$ respectively (since $\dim V = n$ and $\dim W = m$). It is known that the $\mathfrak{sl}(2, \mathbb{C})$-representation $V \otimes W$ is the direct sum of irreducible subrepresentations with highest weights $n+m-2$, $n+m-4$, $n+m-6$ etc. These subrepresentations are Jordan blocks of the operator $A \otimes \mathrm{Id} + \mathrm{Id} \otimes B$. $\square$

Denote by $(A \otimes B)_d$ the subspace of $A \otimes B$ consisting of all L_0-homogeneous vector of degree d. Proposition 7 implies the following useful observation.

Corollary 2. *Denote by T the operator $A \otimes \mathrm{Id} + \mathrm{Id} \otimes B$. Then*

$$\dim(T^k(A \otimes B)_d) = \min(\dim(A \otimes B)_d, \dim(A \otimes B)_{d-k})$$

for all d and k.

In other words, the kernel of the operator $T^k \colon (A \otimes B)_d \to (A \otimes B)_{d-k}$ is of the minimal dimension allowed by the dimensions of the spaces $(A \otimes B)_d$ and $(A \otimes B)_{d-k}$. In particular, if $\dim(A \otimes B)_d \geqslant \dim(A \otimes B)_{d-k}$, then this operator is surjective.

Next, we need the following observation. Let V_1, V_2, $\ldots$, V_n be L_0-graded finite-dimensional vector spaces equipped with the action of nilpotent operators A_1, A_2, $\ldots$, A_n such that each of these operators is one Jordan block. Also, suppose that these spaces satisfy the conditions from Proposition 7, i.e. the operators A_i are L_0-homogeneous of degree 1 and each V_i has an L_0-homogeneous Jordan basis. Denote by

$$T = A_1 \otimes \mathrm{Id} \otimes \ldots \otimes \mathrm{Id} + \mathrm{Id} \otimes A_2 \otimes \ldots \otimes \mathrm{Id} + \ldots + \mathrm{Id} \otimes \mathrm{Id} \otimes \ldots \otimes A_n$$

the operator acting on $V = V_1 \otimes V_2 \otimes \ldots \otimes V_n$. Denote by V_d the L_0-homogeneous component of degree d of the space V. Let D, d be the maximal and the minimal numbers (respectively) such that the spaces V_D and V_d are non-zero. Then, it is easy to see that

$\dim V_{D-s} = \dim V_{d+s}$ for all integer s. Informally, this can be viewed as the symmetry of the decomposition of V into the direct sum of the homogeneous components. From now on, we sometimes refer to this property as «the symmetry of L_0-decomposition». Now, we state the following proposition.

Proposition 8. *$T^k(V_{D-s}) = V_{d+s}$ for all s.*

Proof. We proceed by induction on the number n of the spaces. The base of induction $n = 2$ is Corollary 2. Now, we are going to prove the induction step from n to $n + 1$.

By the induction hypothesis, $V_1 \otimes V_2 \otimes \ldots \otimes V_n = J_1 \oplus J_2 \oplus \ldots \oplus J_r$ is the direct sum of L_0-homogeneous Jordan subspaces such that $D_i + d_i$ is the same for all i, where D_i and d_i are the maximal and the minimal L_0-degrees respectively of vectors in J_i.

The operator A_{n+1} acts on the space V_{n+1} by one Jordan block. Denote by D', d' the maximal and the minimal L_0-degrees respectively of the vectors in V_{n+1}. Now,

$$V_1 \otimes V_2 \otimes \ldots \otimes V_n \otimes V_{n+1} \cong (J_1 \otimes V_{n+1}) \oplus (J_2 \otimes V_{n+1}) \oplus \ldots \oplus (J_r \otimes V_{n+1}).$$

Applying the statement of the induction base to each of the spaces $J_k \otimes V_{n+1}$, we obtain that $J_k \otimes V_{n+1}$ is the direct sum of Jordan blocks such that the maximal and the minimal L_0-degrees of vectors in such blocks are equal to $D_k + D'$ and $d_k + d'$, $D_k + D' - 1$ and $d_k + d' + 1$ etc. respectively. The sum of the numbers in these pairs are equal to $D_k + d_k + D' + d'$, so they do not depend on k due to the induction hypothesis. This proves the proposition. □

Now, we return to the spaces $V^r_{n,k}$. Recall that $V^r_{n,k}$ is isomorphic to the space ${V_n}^{\otimes r} \otimes W_k^{\otimes k-r+1}$ with the relations of anticommutation of the first k and the last $k-r+1$ tensor factors. Note that this factorization does not affect the symmetry of the decomposition of ${V_n}^{\otimes r} \otimes W_k^{\otimes k-r+1}$ into the direct sum of L_0-homogeneous components (i.e., the L_0-homogeneous components of degrees $d + s$ и $D - s$ ($s \in \mathbb{Z}$) have the same dimension also after the factorization). It is also clear that the factorization preserves surjectivity of the map $L_1^{D-d-2s} \colon (V^r_{n,k})_{D-s} \to (V^r_{n,k})_{d+s}$, so this map is an isomorphism of the corresponding spaces due to the equality of their dimensions. Informally, the decomposition of $V^r_{n,k}$ into the direct sum of L_0-homogeneous components if symmetric. This implies that the decomposition of $V^r_{n,k}$ into the Jordan subspaces of the operator L_1 is also symmetric. More formally, this result can be stated as follows.

Proposition 9. *Let D and d be, respectively, the maximal and the minimal number such that the spaces $(V^r_{n,k})_D$ and $(V^r_{n,k})_d$ are non-zero. Then, each Jordan subspace of the operator L_1 acting on $V^r_{n,k}$ has an L_0-homogeneous Jordan basis such that the maximal and the minimal L_0-degree of vectors in this Jordan subspace are equal to $D - s$ and $d + s$ respectively for some $s \in \mathbb{Z}_{\geqslant 0}$.*

Proof. Choose a non-zero homogeneous vector $u_1 \in V^r_{n,k}$ of the maximal possible degree D. We consider the vectors $L_1(v)$, $L_1^2(v)$, ... until we obtain zero for the first time. Denote by v_1 the last non-zero vector of this form. Our observations imply that the map $L_1^{D-d} \colon (V^r_{n,k})_D \to (V^r_{n,k})_d$ is an isomorphism, so $\deg_{L_0}(v_1) = d$. The vectors u_1, $L_1(u_1)$, $L_1^2(u_1)$,..., v_1 span the first Jordan subspace which we denote by U_1.

Next, let s be the minimal integer number such that $(V^r_{n,k})_{D-s}$ is not contained in U_1. Choose a vector $u_2 \in (V^r_{n,k})_{D-s}$ not contained in U_1. Since $L_1^{D-d-2s} \colon (V^r_{n,k})_{D-s} \to (V^r_{n,k})_{d+s}$ is an isomorphism, we see that the vector $v_2 = L_1^{D-d-2s}(u_2)$ is also not contained in U_1.

At the same time, $L_1(v_2) \in U_1$ due to the symmetry of the L_0-decomposition of our space. Now, we can replace u_2 with a vector $\tilde{u}_2$ such that $L_1^{D-d-2s+1}(\tilde{u}_2) = 0$ (we obtain $\tilde{u}_2$ by adding a vector from U_1 to u_2). The linear span of the vectors $L_1^p(\tilde{u}_2)$, $p = 0, 1, \ldots, D - d - 2s$, is the second Jordan subspace of the operator L_1. Denote by U_2 the sum of this subspace and U_1.

Proceeding similarly, we construct a desired decomposition into Jordan blocks. The symmetry with respect to the L_0-grading is clear from the construction. □

Now, it is easy to compute the vector bundle associated with the space $V^r_{n,k}$.

Proposition 10. *The vector bundle associated with $V^r_{n,k}$ is isomorphic to*

$$\mathcal{O}((k+1)(n+k+1))^{\oplus\binom{n}{k+1}}.$$

Proof. Let us choose a Jordan subspace of maximal dimension in $V^r_{n,k}$. Let l and s be the greatest L_0-degree of a vector in this subspace and its dimension respectively. We know from Proposition 3 that the vector bundle associated with this block is isomorphic to $\mathcal{O}(2l - s + 1)^{\oplus s}$.

Next, Proposition 9 implies the following: if $l - x$ is the maximal L_0-degree of a vector in another Jordan subspace, then its dimension is equal to $s - 2x$. Thus, the vector bundle associated with this Jordan subspace is isomorphic to

$$\mathcal{O}(2(l-x) - (s-2x) + 1)^{\oplus(s-2x)} = \mathcal{O}(2l - s + 1)^{\oplus(s-2x)}.$$

It is clear now that the vector bundle associated with the whole space $V^r_{n,k}$ is isomorphic to $\mathcal{O}(2l - s + 1)^{\oplus \dim V^r_{n,k}}$. It remains to compute l and s.

The Jordan subspace of the maximal dimension in $V^r_{n,k}$ is generated by the monomial

$$\psi_{-n} \wedge \psi_{-n+1} \wedge \ldots \wedge \psi_{-n+r-1} \wedge \psi_0 \wedge \psi_1 \wedge \ldots \wedge \psi_{k-r} \wedge \psi_{k+1} \wedge \ldots,$$

while the monomial of the least L_0-degree in this subspace is equal to

$$\psi_{-r} \wedge \psi_{-r+1} \wedge \ldots \wedge \psi_{-1} \wedge \psi_r \wedge \psi_{r+1} \wedge \ldots \wedge \psi_k \wedge \psi_{k+1} \wedge \ldots$$

The L_0-degrees of these monomials are equal to $nr + r(k-r+1) = r(n+k-r+1)$ and r^2 respectively, so the dimension of this Jordan subspace is equal to

$$r(n+k-r+1) - r^2 + 1 = r(n+k-2r+1) + 1.$$

Now,

$$2l - s + 1 = 2r(n+k-r+1) - r(n+k-2r+1) + 1 = r(n+k+1),$$

and $\dim V^r_{n,k} = \binom{n}{r} \cdot \binom{k+1}{k-r+1}$ since a semi-infinite monomial in $V^r_{n,k}$ is determined by a choice of r negative indices from the set $\{-1, -2, \ldots, -n\}$ and $k - r + 1$ non-negative indices from the set $\{0, 1, \ldots, k\}$. The computation we did shows that the vector bundle we want to compute is isomorphic to $\mathcal{O}((k+1)(n+k+1))^{\oplus\binom{n}{k+1}}$ as stated. □

The vector bundle associated with the whole Fock module $V_{n,k}$ is the direct sum of the bundles associated with the subspaces $V^r_{n,k}$, $r = 0, 1, \ldots, k+1$. This implies the main result of this section.

Proposition 11. *The vector bundle on $\mathbb{CP}^1$ associated with the representation of* Vir_+ *on the Fock module $V_{n,k}$ is isomorphic to the direct sum*

$$\bigoplus_{r=0}^{k+1} \mathcal{O}(r(n+k+1))^{\oplus\binom{n}{r}\binom{k+1}{k-r+1}}.$$

5 The representation of Vir_+ on the principal subspace of $\pi_{\sqrt{2}\mathbb{Z}}$ for the bosonic vertex operator $V_{\sqrt{2}}(z)$

In this section, we compute explicitly the bundles on $\mathbb{P}^1$ corresponding to another class of representations of Vir_+. The details of the following construction can be found in [2, Section 5.2].

For $\lambda \in \mathbb{C}$, define π_λ as the $\mathcal{H}$-module generated by the vacuum vector $|\lambda\rangle$ such that

$$b_n|\lambda\rangle = 0, n \geqslant 0, \;\; b_0|\lambda\rangle = \lambda|\lambda\rangle.$$

In other words, π_λ is the vector space $\mathbb{C}[b_n]_{n<0}|\lambda\rangle$ where b_n acts as:

- the multiplication by b_n if $n < 0$;
- the derivation $n\partial/\partial b_{-n}$ if $n > 0$;
- the scalar multiplication by λ if $n = 0$,

while h acts identically. Define a $\mathbb{Z}$-grading on π_λ by the rule $\deg|\lambda\rangle = 0$, $\deg b_n = -n$. Each π_λ is equipped with the following action of the Virasoro algebra:

$$L_0 = \frac{1}{2}b_0^2 + b_{-1}b_1 + b_{-2}b_2 + \ldots, \; L_k = \frac{1}{2}\sum_s b_s b_{k-s} \text{ при } \; k \neq 0.$$

For any positive integer N, define the space

$$V_{\sqrt{N}\mathbb{Z}} = \bigoplus \pi_{m\sqrt{N}}.$$

The algebra Vir acts on $V_{\sqrt{N}\mathbb{Z}}$ by the same formulas as above.

One can define the so-called *bosonic vertex operators*

$$V_\lambda(z) = \sum_{n\in\mathbb{Z}} V_\lambda[n] z^{-n-\lambda^2/2},$$

where $V_\lambda[n]$ are certain linear operators on $\pi_{\sqrt{N}\mathbb{Z}}$, $\lambda \in \sqrt{N}\mathbb{Z}$. The explicit formula for $V_\lambda(z)$ is the following:

$$V_\lambda(z) = S_\lambda z^{\lambda b_0} \exp\left(-\lambda \sum_{n<0} \frac{b_n}{n} z^{-n}\right) \exp\left(-\lambda \sum_{n>0} \frac{b_n}{n} z^{-n}\right), \tag{5}$$

where $S_\lambda \colon \pi_\mu \to \pi_{\lambda+\mu}$ is the shift operator defined by the conditions

$$S_\lambda|\nu\rangle = |\nu+\lambda\rangle; \quad [S_\lambda, b_n] = 0, \; n \neq 0.$$

The introduced bosonic vertex operators define a structure of vertex algebra on $V_{\sqrt{N}\mathbb{Z}}$ if N is even (for odd N, it is a vertex superalgebra). The other vertex operators are determined by the standard reconstruction theorem (see, for example, [2, Chapter 2.3]). Moreover, it is a conformal vertex algebra with the conformal vector $\frac{1}{2}b_{-1}^2$.

The vertex operators $V_\lambda(z)$ satisfy the following relations:

$$[b_n, V_\lambda(z)] = \lambda z^n V_\lambda(z). \tag{6}$$

We will work with the subspace $W_{2n} \subset \pi_{\sqrt{N}\mathbb{Z}}$ obtained by the application of all polynomials in the operators $V_{\sqrt{N}}[k]$, $k \geqslant 0$, to the vector $|-n\sqrt{N}\rangle$, $n \geqslant 0$. In this paper, we deal only with the case $N = 2$.

We are going to follow the same plan as in the previous section.

1. First, we describe the space W_{2n} more explicitly. In particular, we prove that this space is finite-dimensional and compute its dimension.

2. The, we show that W_{2n} is invariant under the action of Vir_+.

3. Finally, we compute explicitly the vector bundles on $\mathbb{P}^1$ corresponding to the representation of Vir_+ on W_{2n}.

The structure of W_{2n} is described by the following proposition.

Proposition 12. *The subspaces $\pi_{k\sqrt{2}}$ having a non-trivial intersection with W_{2n} are precisely the ones for which $-n \leqslant k \leqslant n$. Each space $W_{2n} \cap \pi_{k\sqrt{2}}$ is obtained by application of all possible polynomials in b_m, $m \geqslant 0$, to one vector $v_k \in \pi_{k\sqrt{2}}$. Moreover, the polynomial in b_m, $m < 0$, corresponding to the vector v_k is obtained from the polynomial corresponding to the highest weight vector of the space $V_{n+k,n-k-1}$ from the previous section after the change $b_{-m} \mapsto \sqrt{2}b_{-m}$.*

The following corollary follows immediately from Proposition 12.

Corollary 3. *1. $\dim W_{2n} = 2^{2n}$;*

2. W_{2n} is invariant under the action of Vir_+.

The proof of Corollary 3. 1. An explicit calculation:

$$\dim W_{2n} = \sum_{k=-n}^{n} \dim(W_{2n} \cap \pi_{k\sqrt{2}}) = \sum_{k=-n}^{n} \binom{2n}{n+k} = \sum_{i=0}^{2n} \binom{2n}{i} = 2^{2n}.$$

2. Let $Q(b_{-1}, b_{-2}, \ldots) \in W_{2n} \cap \pi_{k\sqrt{2}}$. It follows easily from Proposition 12 that

$$Q(b_{-1}, b_{-2}, \ldots) = P(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots),$$

where the polynomial $P(b_{-1}, b_{-2}, \ldots)$ corresponds to some vector in the space $V_{n+k,n-k-1}$ under the boson-fermion correspondence. Vice versa, if $P \in V_{n+k,n-k-1}$ (here, we identify polynomials in $b_{-1}, b_{-2}, \ldots$ and the corresponding vectors in $V_{n+k,n-k-1}$), then

$$P(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots) \in W_{2n} \cap \pi_{k\sqrt{2}}.$$

Indeed, this is true for the generating vector $v_k \in W_{2n} \cap \pi_{k\sqrt{2}}$ due to Proposition 12. All other vectors are obtained from v_k by application of all possible polynomials in $n \geqslant 0$. Up to multiplication by a constant, each b_n acts as the derivation $\partial/\partial b_{-n}$, so the space we consider is obtained from the polynomial

$$v_k = Q_0(b_{-1}, b_{-2}, \ldots) = P_0(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots)$$

under all possible derivations. The space $V_{n+k,n-k-1}$ (under the boson-fermion correspondence) is also obtained from the polynomial P_0 under all possible derivations. But

$$\partial/\partial b_{-m}(P(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots)) = \sqrt{2}\frac{\partial P}{\partial b_{-m}}(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots),$$

which implies our assertion.

Next, the formula for L_m, $m \geqslant 0$, can be rewritten in the following way:

$$L_m = R(b_0, b_1, b_2, \ldots) + b_{-1}b_{m+1} + b_{-2}b_{m+2} + \ldots,$$

where R is some polynomial. The space $V_{n+k,n-k-1}$ is invariant under the action of b_m and L_m, $m \geqslant 0$, so

$$((m+1)b_{-1}\frac{\partial}{\partial b_{-m-1}}+(m+2)b_{-2}\frac{\partial}{\partial b_{-m-2}}+\ldots)P(b_{-1}, b_{-2}, \ldots) = P'(b_{-1}, b_{-2}, \ldots) \in V_{n+k,n-k-1}.$$

At the same time,

$$((m+1)b_{-1}\frac{\partial}{\partial b_{-m-1}} + (m+2)b_{-2}\frac{\partial}{\partial b_{-m-2}} + \ldots)P(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots) =$$

$$= (m+1)\sqrt{2}b_{-1}\frac{\partial P}{\partial b_{-m-1}}(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots)+(m+2)b_{-2}\frac{\partial P}{\partial b_{-m-2}}(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots)+\ldots =$$

$$= P'(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \ldots).$$

Due to invariance of W_{2n} under the action of $b_0, b_1, \ldots$, this implies invariance of W_{2n} under L_m.

□

Now, we are going to prove Proposition 12. In order to do this, we first need to consider the case $N = 1$. In this case, the bosonic vertex operator $V_1(z)$ is given by the formula

$$V_1(z) = S_1 z^{\lambda b_0} \exp(b_{-1}z + \frac{b_{-2}}{2}z^2 + \frac{b_{-3}}{3}z^3 + \ldots) \exp(-(\partial b_{-1}z^{-1} + \partial b_{-2}z^{-2} + \ldots)).$$

The boson-fermion correspondence gives us the equality $V_1(z) = \sum\limits_{m\in\mathbb{Z}} \psi_m^* z^m$.

Consider the power series

$$\exp(b_{-1}z + \frac{b_{-2}}{2}z^2 + \ldots) = P_0 + P_1 z + P_2 z^2 + \ldots \tag{7}$$

Here, $P_0, P_1, P_2, \ldots$ are polynomials in $b_{-1}, b_{-2}, \ldots$ (in particular, $P_0 = 1$). We also write $P_m = 0$ for $m < 0$. Then, the action of $V_1(z)$ on the vector $|n\rangle$ is given by the formula

$$V_1(z)|n\rangle = (P_0 z^n + P_1 z^{n+1} + P_2 z^{n+2} + \ldots)|n-1\rangle.$$

Denote by $\widetilde{V}_1(z)$ the expression

$$\widetilde{V}_1(z) = \exp(b_{-1}z + \frac{b_{-2}}{2}z^2 + \frac{b_{-3}}{3}z^3 + \ldots) \exp(-(\partial b_{-1}z^{-1} + \partial b_{-2}z^{-2} + \ldots)).$$

Lemma 3. $\widetilde{V}_1(z)(P_n) = (P_n - z^{-1}P_{n-1})\widetilde{V}_1(z)$.

Proof. We have

$$\exp(-\partial b_{-1}z^{-1} - \partial b_{-2}z^{-2} - \ldots) \exp(b_{-1}w + \frac{b_{-2}}{2}w^2 + \ldots) =$$

$$= \exp(b_{-1}w + \frac{b_{-2}}{2}w^2 + \ldots) \exp(-\partial b_{-1}z^{-1} - \partial b_{-2}z^{-2} - \ldots) \exp(-\frac{w}{z} - \frac{w^2}{2z^2} - \frac{w^3}{3z^3} - \ldots) =$$

$$= (1-\frac{w}{z})\exp(b_{-1}w+\frac{b_{-2}}{2}w^2+\ldots) = (1-\frac{w}{z})(1+P_1w+P_2w^2+\ldots) =$$
$$= 1+w(P_1-z^{-1})+w^2(P_2-P_1z^{-1})+\ldots$$

This implies

$$\widetilde{V}_1(z)(P_n) = \exp(b_{-1}z+\frac{b_{-2}}{2}z^2+\frac{b_{-3}}{3}z^3+\ldots)(P_n-z^{-1}P_{n-1}) =$$

$$= (P_n-z^{-1}P_{n-1})\exp(b_{-1}z+\frac{b_{-2}}{2}z^2+\frac{b_{-3}}{3}z^3+\ldots) = (P_n-z^{-1}P_{n-1})\widetilde{V}_1(z).$$

□

Now, let us describe explicitly the polynomials obtained from $|-n\rangle$ by applying the operators $V_1[m]=\psi^*_m$, $m\geqslant 0$.

Proposition 13. *Up to the sign, the result of application of the operator $V_1[m_k]\ldots V_1[m_2]V_1[m_1]$ to the vector $|-n\rangle$ is the polynomial*

$$\det\begin{pmatrix} P_{m_1-k+1} & P_{m_1-k+2} & \ldots & P_{m_1} \\ P_{m_2-k+1} & P_{m_2-k+2} & \ldots & P_{m_2} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m_k-k+1} & P_{m_k-k+2} & \ldots & P_{m_k} \end{pmatrix}. \tag{8}$$

Proof. We proceed by induction on k. The base for $k=1$ is clear: the coefficient of z^{m_1} in the expression for $V_1(z)$ is equal precisely to P_{m_1}.

To prove the induction step, we use Lemma 3.

$$\widetilde{V}_1(z)(P_m) = (P_m-z^{-1}P_{m-1})(P_0+P_1z+P_2z^2+\ldots),$$

so $V_1[m_k]\ldots V_1[m_2]V_1[m_1]|n\rangle$ is equal to the coefficient of $z^{m_{k+1}+k}$ in the expression

$$\det\begin{pmatrix} P_{m_1-k+1}-z^{-1}P_{m_1-k} & P_{m_1-k+2}-z^{-1}P_{m_1-k+1} & \ldots & P_{m_1}-z^{-1}P_{m_1-1} \\ P_{m_2-k+1}-z^{-1}P_{m_2-k} & P_{m_2-k+2}-z^{-1}P_{m_2-k+1} & \ldots & P_{m_2}-z^{-1}P_{m_2-1} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m_k-k+1}-z^{-1}P_{m_k-k} & P_{m_k-k+2}-z^{-1}P_{m_k-k+1} & \ldots & P_{m_k}-z^{-1}P_{m_k-1} \end{pmatrix}.$$
$$\cdot(P_0+P_1z+P_2z^2+\ldots).$$

Using multilinearity of determinant with respect to the columns, we obtain the sum

$$P_{m_{k+1}-k}\det\begin{pmatrix} P_{m_1-k+1} & P_{m_1-k+2} & \ldots & P_{m_1} \\ P_{m_2-k+1} & P_{m_2-k+2} & \ldots & P_{m_2} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m_k-k+1} & P_{m_k-k+2} & \ldots & P_{m_k} \end{pmatrix} - P_{m_{k+1}-k+1}\det\begin{pmatrix} P_{m_1-k} & P_{m_1-k+2} & \ldots & P_{m_1} \\ P_{m_2-k} & P_{m_2-k+2} & \ldots & P_{m_2} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m_k-k} & P_{m_k-k+2} & \ldots & P_{m_k} \end{pmatrix} + \ldots$$

$$\ldots + (-1)^k P_{m_{k+1}}\det\begin{pmatrix} P_{m_1-k} & P_{m_1-k+1} & \ldots & P_{m_1-1} \\ P_{m_2-k} & P_{m_2-k+1} & \ldots & P_{m_2-1} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m_k-k} & P_{m_k-k+1} & \ldots & P_{m_k-1} \end{pmatrix} = \pm\det\begin{pmatrix} P_{m_1-k} & P_{m_1-k+1} & \ldots & P_{m_1} \\ P_{m_2-k} & P_{m_2-k+1} & \ldots & P_{m_2} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m_{k+1}-k} & P_{m_{k+1}-k+1} & \ldots & P_{m_{k+1}} \end{pmatrix},$$

which concludes the proof of the induction step. □

It is easy to compute the semi-infinite monomials corresponding to the polynomials from Proposition 13. Indeed, $V_1[m] = \psi_m^*$, so $V_1[m_k] \dots V_1[m_2]V_1[m_1]|0\rangle$ is the semi-infinite monomial

$$\psi_0 \wedge \dots \wedge \widehat{\psi}_{m_1} \wedge \dots \wedge \widehat{\psi}_{m_2} \wedge \dots \wedge \widehat{\psi}_{m_k} \wedge \dots$$

This gives us the following corollary.

Corollary 4. *Up to the sign, the semi-infinite monomial*

$$\psi_0 \wedge \dots \wedge \widehat{\psi}_{m_1} \wedge \dots \wedge \widehat{\psi}_{m_2} \wedge \dots \wedge \widehat{\psi}_{m_k} \wedge \dots$$

corresponds to the polynomial

$$\det \begin{pmatrix} P_{m_1-k+1} & P_{m_1-k+2} & \dots & P_{m_1} \\ P_{m_2-k+1} & P_{m_2-k+2} & \dots & P_{m_2} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m_k-k+1} & P_{m_k-k+2} & \dots & P_{m_k} \end{pmatrix}. \tag{9}$$

Now, we are ready to return to the case $N = 2$. Recall that

$$V_{\sqrt{2}}(z) = S_{\sqrt{2}} z^{\sqrt{2}b_0} \exp(\sqrt{2}(b_{-1}z + \frac{b_{-2}}{2}z^2 + \dots)) \exp(-\sqrt{2}(\partial b_{-1}z^{-1} + \partial b_{-2}z^{-2} + \dots)).$$

Introduce the following notation:

$$\exp(\sqrt{2}(b_{-1}z^{-1} + \frac{b_{-2}}{2}z^{-2} + \frac{b_{-3}}{3}z^3 + \dots) = Q_0 + Q_1 z + Q_2 z^2 + \dots$$

(in particular, $Q_0 = 1$). Assume also that $Q_m = 0$, $m < 0$. It is clear that

$$Q_m(b_{-1}, b_{-2}, \dots) = P_m(\sqrt{2}b_{-1}, \sqrt{2}b_{-2}, \dots).$$

Denote by $\widetilde{V}_{\sqrt{2}}(z)$ the expression defined by the same formula as $V_{\sqrt{2}}(z)$ but without the operator $S_{\sqrt{2}} z^{\sqrt{2}b_0}$. Let us state an assertion similar to Lemma 3.

Lemma 4. $\widetilde{V}_{\sqrt{2}}(z)(Q_m) = (Q_m - 2z^{-1}Q_{m-1} + z^{-2}Q_{m-2})\widetilde{V}_{\sqrt{2}}(z)$.

Proof. Similar to the proof of Lemma 3. □

Now, we prove the following proposition.

Proposition 14. *Each polynomial obtained from $|n\rangle$ by application of the operators $V_{\sqrt{2}}[m]$, $m \geqslant 0$, is a linear combinations of the polynomials of the following form:*

$$\det \begin{pmatrix} Q_{m_1-k+1} & Q_{m_1-k+2} & \dots & Q_{m_1} \\ Q_{m_2-k+1} & Q_{m_2-k+2} & \dots & Q_{m_2} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{m_k-k+1} & Q_{m_k-k+2} & \dots & Q_{m_k} \end{pmatrix}, \tag{10}$$

where $m_1, m_2, \dots, m_k \leqslant 2n - 1$.

Proof. First, let us ignore the restriction $m \geqslant 0$. Proceeding by induction, as in the proof of Proposition 13, we are going to compute the coefficients of z^s in the expression

$$\det \begin{pmatrix} Q_{m_1-k+1} - 2z^{-1}Q_{m_1-k} + z^{-2}Q_{m_1-k-1} & \dots & Q_{m_1} - 2z^{-1}Q_{m_1-1} + z^{-2}Q_{m_1-2} \\ Q_{m_2-k+1} - 2z^{-1}Q_{m_2-k} + z^{-2}Q_{m_2-k-1} & \dots & Q_{m_2} - 2z^{-1}Q_{m_2-1} + z^{-2}Q_{m_2-2} \\ \vdots & \ddots & \vdots \\ Q_{m_k-k+1} - 2z^{-1}Q_{m_k-k} + z^{-2}Q_{m_k-k-1} & \dots & Q_{m_k} - 2z^{-1}Q_{m_k-1} + z^{-2}Q_{m_k-2} \end{pmatrix} \cdot$$

$$\cdot(Q_0 + Q_1 z + Q_2 z^2 + \dots).$$

Due to the equality

$$Q_m - 2z^{-1}Q_{m-1} + z^{-2}Q_{m-2} = (Q_m - z^{-1}Q_{m-1}) - z^{-1}(Q_{m-1} - z^{-1}Q_{m-2}),$$

the last determinant can be expanded along the rows as the following sum of determinants:

$$\det \begin{pmatrix} Q_{m_1-k+1} - z^{-1}Q_{m_1-k} & \dots & Q_{m_1} - z^{-1}Q_{m_1-1} \\ Q_{m_2-k+1} - z^{-1}Q_{m_2-k} & \dots & Q_{m_2} - z^{-1}Q_{m_2-1} \\ \vdots & \ddots & \vdots \\ Q_{m_k-k+1} - z^{-1}Q_{m_k-k} & \dots & Q_{m_k} - z^{-1}Q_{m_k-1} \end{pmatrix} -$$

$$-z^{-1} \det \begin{pmatrix} Q_{m_1-k} - z^{-1}Q_{m_1-k-1} & \dots & Q_{m_1-1} - z^{-1}Q_{m_1-2} \\ Q_{m_2-k+1} - z^{-1}Q_{m_2-k} & \dots & Q_{m_2} - z^{-1}Q_{m_2-1} \\ \vdots & \ddots & \vdots \\ Q_{m_k-k+1} - z^{-1}Q_{m_k-k} & \dots & Q_{m_k} - z^{-1}Q_{m_k-1} \end{pmatrix} + \dots$$

So, the statement follows from the proof of Proposition 13.

To prove that $m_1, m_2, \dots, m_k \leqslant 2n-1$, it is enough to check that the coefficients of z^s, $s < 0$, do not contain the polynomials Q_m such that $m \geqslant 2n$. We check it by induction on k. We have

$$V_{\sqrt{2}}(z)|-n\sqrt{2}\rangle = (Q_0 z^{-2n} + Q_1 z^{-2n+1} + \dots + Q_{2n} + Q_{2n+1}z + \dots)|-(n-1)\sqrt{2}\rangle.$$

The coefficients of the negative powers of z in this expression are only Q_m, $m \leqslant 2n-1$.

Next, we proceed by induction. At the k-th step, we are interested in the coefficients of the negative powers of z in the expression

$$\det \begin{pmatrix} Q_{n_1-k+1} - 2z^{-1}Q_{n_1-k} + z^{-2}Q_{n_1-k-1} & \dots & Q_{n_1} - 2z^{-1}Q_{n_1-1} + z^{-2}Q_{n_1-2} \\ \vdots & \ddots & \vdots \\ Q_{n_k-k+1} - 2z^{-1}Q_{n_k-k} + z^{-2}Q_{n_k-k-1} & \dots & Q_{n_k} - 2z^{-1}Q_{n_k-1} + z^{-2}Q_{n_k-2} \end{pmatrix} \cdot$$

$$\cdot(Q_0 z^{-2n+2k} + Q_1 z^{-2n+2k+1} + \dots + Q_{2n-1}z^{2k-1} + Q_{2n}z^{2k} + \dots)$$

After the full expansion of this determinant, the minimal power of z is $-2k$, so, after expanding all brackets, the polynomials Q_m, $m \geqslant 2k$, cannot appear in the expressions for the coefficients of the negative powers of z. □

It is also easy to prove the following statement.

Proposition 15. *The space $W_{2n} \cap \pi_{(-n+k)\sqrt{2}}$, $k \geqslant 0$, is invariant under the action of the operators b_s, $s \geqslant 0$.*

Proof. Induction on k. The base for $k = 0$ is clear. To prove the induction step, we use the commutation relation (6) equivalent to the equalities

$$[b_n, V_{\sqrt{2}}[i]] = \sqrt{2}V_{\sqrt{2}}[i+n].$$

Now, if $v \in W_{2n} \cap \pi_{(-n+k)\sqrt{2}}$, then

$$b_n(V_{\sqrt{2}}[i](v)) = V_{\sqrt{2}}[i](b_n(v)) + \sqrt{2}V_{\sqrt{2}}[i+n](v) \in W_{2n} \cap \pi_{(-n+k+1)\sqrt{2}}$$

since $b_n(v) \in W_{2n} \cap \pi_{(-n+k)\sqrt{2}}$ by the induction hypothesis. Since each vector in the space $W_{2n} \cap \pi_{(-n+k+1)\sqrt{2}}$ is of the form $V_{\sqrt{2}}[n](v)$ for some $v \in W_{2n} \cap \pi_{(-n+k)\sqrt{2}}$ (by the definition of W_{2n}), the step is proved. □

We showed that the space $W_{2n} \cap \pi_{(-n+k)\sqrt{2}}$ is contained in the linear span of polynomials of the form

$$\det \begin{pmatrix} Q_{m_1-k+1} & Q_{m_1-k+2} & \dots & Q_{m_1} \\ Q_{m_2-k+1} & Q_{m_2-k+2} & \dots & Q_{m_2} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{m_k-k+1} & Q_{m_k-k+2} & \dots & Q_{m_k} \end{pmatrix} \cdot |-(n+k)\sqrt{2}\rangle,$$

where $m_1, m_2, \dots, m_k \leqslant 2n-1$. It remains to show that each such polynomial is in $W_{2n} \cap \pi_{(-n+k)\sqrt{2}}$. For this purpose, it is sufficient to show that the polynomial

$$\mathcal{Q}_{n,k} = \det \begin{pmatrix} Q_{2n-k} & Q_{2n-k+1} & \dots & Q_{2n-1} \\ Q_{2n-k-1} & Q_{2n-k} & \dots & Q_{2n-2} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{2n-2k+1} & Q_{2n-2k+2} & \dots & Q_{2n-k} \end{pmatrix} \cdot |-(n+k)\sqrt{2}\rangle$$

is contained in $W_n \cap \pi_{(-n+k)\sqrt{2}}$. Indeed, as we know from Corollary 4, this polynomial corresponds to the semi-infinite monomial

$$\psi_0 \wedge \psi_1 \wedge \ldots \wedge \psi_{2n-k-1} \wedge \psi_{2n} \wedge \psi_{2n+1} \wedge \ldots$$

In Section 4, we proved that all semi-infinite monomials of the form

$$\psi_{m_0} \wedge \psi_{m_1} \wedge \ldots \wedge \psi_{m_{2n-k}} \wedge \psi_{2n} \wedge \psi_{2n+1} \wedge \ldots,$$

where $0 \leqslant m_0 < m_1 < \ldots < m_{2n-k} < 2n$, can be obtained from this monomial by applying polynomials in b_s, $s \geqslant 0$. Up to the sign, such semi-infinite monomials correspond to the polynomials that we consider. But we proved in Proposition 15 that W_{2n} is invariant under the action of b_s, $s \geqslant 0$. This will imply that all such polynomials lie in $W_{2n} \cap \pi_{\sqrt{2}(-n+k)}$.

In order to check that $\mathcal{Q}_{n,k}$ lies in W_{2n}, it is sufficient to ensure that the vector $V^k_{\sqrt{2}}[0]|-n\sqrt{2}\rangle$ is non-zero. Indeed, the operator $V_{\sqrt{2}}[m]$ is of degree m with respect to the grading

$$\deg(Q(b_{-1}, \ldots)|k\rangle) = r^2 + \deg Q$$

(see [2, Chapter 5.2]). So,

$$\deg V_{\sqrt{2}}[m_k] \ldots V_{\sqrt{2}}[m_2] V_{\sqrt{2}}[m_1]|-n\sqrt{2}\rangle \leqslant n^2,$$

and equality holds if and only if $m_1 = m_2 = \ldots = m_k = 0$. In this case, we have the equality

$$V^k_{\sqrt{2}}[0]|-n\sqrt{2}\rangle = Q(b_{-1}, b_{-2}, \ldots)|(-n+k)\sqrt{2}\rangle$$

for some polynomial Q that is either zero or of degree $n^2-(n-k)^2=2nk-k^2$. At the same time, the maximal possible degree of a polynomial of form (10) is equal to $k(2n-k)$. This degree is equal to $k(2n-k)$ if and only if $m_1,m_2,\dots,m_k$ are equal to $2n-1,2n-2,\dots,2n-k$ (in some order). So, if the vector $V^k_{\sqrt{2}}[0]|-n\sqrt{2}\rangle$ is non-zero, it is automatically proportional to $\mathcal{Q}_{n,k}$.

In order to prove that $V^k_{\sqrt{2}}[0]|-n\sqrt{2}\rangle$ is non-zero, we use the fact that the three operators $V_{\sqrt{2}}[0]$, $V_{-\sqrt{2}}[0]$ and $\sqrt{2}b_0$ span a Lie algebra isomorphic to $\mathfrak{sl}(2,\mathbb{C})$, see [4] ($\sqrt{2}b_0$ corresponds to the Cartan element h). So, the space

$$\{V^k_{\sqrt{2}}[0]|-n\sqrt{2}\rangle \mid 0\leqslant k\leqslant 2n\}$$

is a representation of this Lie algebra. Since $V^k_{-\sqrt{2}}[0]|-n\sqrt{2}\rangle=0$ and $\sqrt{2}b_0|-n\sqrt{2}\rangle=-2n|\sqrt{2}\rangle$, the vector $|-n\sqrt{2}\rangle$ is a highest vector of this representation. Moreover, its eigenvalue with respect to $\sqrt{2}b_0$ is equal to $-2n$, so the irreducible representation generated by $|-n\sqrt{2}\rangle$ has dimension $2n+1$. This implies that the whole representation $\{V^k_{\sqrt{2}}[0]|-n\sqrt{2}\rangle \mid 0\leqslant k\leqslant 2n\}$ is an irreducible representation of $\mathfrak{sl}(2,\mathbb{C})$ and all vectors of the form $V^k_{\sqrt{2}}[0]|-n\sqrt{2}\rangle, 0\leqslant k\leqslant 2n$, are non-zero.

Now, we are ready to compute explicitly the vector bundles on $\mathbb{P}^1$ corresponding to the representations of Vir_+ on the spaces W_{2n}. To this aim, we compute separately the bundles corresponding to each subspace $W_{2n}\cap\pi_{\sqrt{2}(-n+k)}$, $0\leqslant k\leqslant 2n$. For these subspaces, this problem was virtually solved in Section 4. Indeed, each of them is obtained from the polynomial corresponding to the semi-infinite monoial

$$\psi_{-k}\wedge\psi_{-k+1}\wedge\ldots\wedge\psi_{2n-2k-1}\wedge\psi_{2n-k}\wedge\psi_{2n-k+1}\wedge\ldots$$

by application of the operators b_s, $s\geqslant 0$. The only difference is that we now have to take into account the summand b_0b_1 in the expression for L_1. This summand acts by 0 on the spaces that we considered in Section 4. However, the operator b_0 acts on $W_{2n}\cap\pi_{\sqrt{2}(-n+k)}$ by the scalar $\sqrt{2}(-n+k)$. So, the summand b_0b_1 acts on the polynomial $P(\sqrt{2}b_{-1},\sqrt{2}b_{-2},\ldots)$ by the formula

$$b_0b_1(P(\sqrt{2}b_{-1},\sqrt{2}b_{-2},\ldots))=2\frac{\partial P}{\partial b_{-1}}(\sqrt{2}b_{-1},\sqrt{2}b_{-2},\ldots).$$

So, the result of the action of L_1 on the monomial $\psi_{i_1}\wedge\psi_{i_2}\wedge\ldots$ is

$$(i_0-1+2k-2n)\psi_{-i_0+1}\wedge\psi_{-i_1}\wedge\ldots+(i_1-1+2k-2n)\psi_{-i_0}\wedge\psi_{-i_1+1}\wedge\ldots+\ldots$$

It is convenient for us to add $2k-2n$ to all indices in the monomials we consider. After this change, we will work with the space equal to the linear span of all semi-infinite monomials of the form

$$\psi_{i_1}\wedge\psi_{i_2}\wedge\ldots\wedge\psi_{i_{2n-k}}\wedge\psi_k\wedge\psi_{k+1}\ldots,$$

where $-2n+k\leqslant i_1<i_2<\ldots<i_{2n-k}\leqslant k-1$. The operator L_1 acts by the formula from Lemma 2 on these monomials. This subspace can be split into the direct sum of subspaces U_r, $r=0,1,\dots,k$, where each of these subspaces U_r is defined as the linear span of semi-infinite monomials having exactly r non-negative indices from 0 to $k-1$. The bundle on $\mathbb{P}^1$ associated with U_r can be computed in the same way as in Section 4. The vector of maximal degree in such subspace is equal to

$$\psi_{-2n+k}\wedge\psi_{-2n+k+1}\wedge\ldots\psi_{-r-1}\wedge\psi_0\wedge\psi_1\wedge\ldots\wedge\psi_{r-1}\wedge\psi_k\wedge\psi_{k+1}\wedge\ldots,$$

while the vector of minimal degree is

$$\psi_{-2n+r+k} \wedge \psi_{-2n+r+k+1} \wedge \ldots \psi_{-1} \wedge \psi_{k-r} \wedge \psi_{k-r+1} \wedge \ldots \wedge \psi_{k-1} \wedge \psi_k \wedge \psi_{k+1} \wedge \ldots$$

Their degrees are equal to

$$l_{\max} = (n-k)^2 + r(k-r) + k(2n-k-r)$$

and

$$l_{\min} = (n-k)^2 + (k-r)(2n-k-r)$$

respectively, while the dimension of U_r is $\binom{2n-k}{r}\binom{k}{r}$.

So, the vector bundle corresponding to U_r is isomorphic to

$$\mathcal{O}(l_{\max} + l_{\min})^{\oplus\binom{2n-k}{r}\binom{k}{r}} = \mathcal{O}(2n(n-r))^{\oplus\binom{2n-k}{r}\binom{k}{r}},$$

while the vector bundle associated with the whole space U_r is

$$\bigoplus_{r=0}^{k} \mathcal{O}(2n(n-r))^{\oplus\binom{2n-k}{r}\binom{k}{r}}.$$

It remains to sum these bundles over all k, $0 \leqslant k \leqslant 2n$. The summand $\mathcal{O}(2n(n-r))$ is contained with multiplicity

$$\sum_{k=0}^{2n} \binom{2n-k}{r}\binom{k}{r} = \binom{2n+1}{2r+1}$$

in this direct sum. The last equality can be proved in a combinatorial way. Indeed, its right-hand side is equal to the number of choices of $2r+1$ elements from $2n+1$ elements (without repetitions). At the same time, if we place these elements in a line, then the number of choices of $2r+1$ elements in such a way that the $(r+1)$-th chosen element is the $(k+1)$-th in this line is equal to $\binom{2n-k}{r}\binom{k}{r}$. This concludes the proof of this equality.

Now, we state the final result.

Proposition 16. *The vector bundle on $\mathbb{CP}^1$ associated with the representation of* Vir_+ *on the space W_{2n} is isomorphic to the direct sum*

$$\bigoplus_{r=0}^{n} \mathcal{O}(2n(n-r))^{\oplus\binom{2n+1}{2r+1}}.$$

6 Demazure modules of level 1: the case of $\mathfrak{sl}(2,\mathbb{C})$

In this section, we describe Demazure modules of level 1 corresponding to the affine Kac-Moody algebra $\widehat{\mathfrak{sl}}(2,\mathbb{C})$ and compute the corresponding vector bundles on $\mathbb{P}^1$. We start with a general construction of the conformal vertex algebra associated to an even lattice (in the previous section, we did it for 1-dimensional lattices). The details can be found in [2, Chapter 5.4]. Let L be a lattice of finite rank equipped with a symmetric bilinear form $(\cdot,\cdot)\colon L \times L \to \mathbb{Z}$ such that $(\lambda,\lambda) > 0$ for all $\lambda \in L \setminus \{0\}$. We assume that L is an even lattice, i.e. (λ,λ) is even for any $\lambda \in L$ (without this restriction, the construction we describe gives a vertex superalgebra).

Let us set $\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$. The bilinear form on L induces a bilinear form on $\mathfrak{h}$. Denote by $\hat{\mathfrak{h}}$ the central extension of $\mathfrak{h}((t))$,

$$0 \to \mathbb{C}\mathbf{1} \to \hat{\mathfrak{h}} \to \mathfrak{h}((t)),$$

with the commutation relations

$$[A \otimes f(t), B \otimes g(t)] = -(A, B)(\mathrm{Res} f(t) g'(t) dt)\mathbf{1}.$$

This is the Heisenberg Lie algebra associated to $\mathfrak{h}$ equipped with the bilinear form $(\cdot, \cdot)$.

The Weyl algebra $\widetilde{\mathcal{H}}_L$ is defined as the (completed) enveloping algebra of $\hat{\mathfrak{h}}$ modulo the relation $\mathbf{1} = 1$. It has (topological) generators $h_n = h \otimes t^n$, $h \in \mathfrak{h}$, $n \in \mathbb{Z}$, and relations

$$[h_n, g_m] = n(h, g)\delta_{n,-m}.$$

For $\lambda \in \mathfrak{h}$, we define the Fock representation π_λ of $\widetilde{\mathcal{H}}_L$ generated by the vector $|\lambda\rangle$ such that

$$h_n|\lambda\rangle = 0, n > 0; h_0|\lambda\rangle = (\lambda, h)|\lambda\rangle.$$

Now, denote by V_L the space

$$V_L = \bigoplus_{\lambda \in L} \pi_\lambda.$$

The space V_L is equipped with the following $\mathbb{Z}$-gradation: $\deg h_n = -n$, $\deg |\lambda\rangle = \frac{(\lambda,\lambda)}{2}$. The canonical structure of vertex algebra on V_L can be defined by the formulas similar to (5). Namely, for any $\lambda \in L$, we define the bosonic vertex operator V_λ by the following formula:

$$V_\lambda(z) = S_\lambda z^\lambda \exp\left(\sum_{n<0} \frac{\lambda_n}{n} z^{-n}\right) \exp\left(\sum_{n>0} \frac{\lambda_n}{n} z^{-n}\right), \tag{11}$$

where z^λ is a shorthand for the linear map that takes any element of the α-Fock space V_α to the monomial $z^{(\lambda,\alpha)}$, while S_λ is, up to a non-zero constant, the shift by λ. The vertex operators for other elements of the Fock space are, as in the previous section, determined by reconstruction.

Finally, V_L has a conformal vector given by the formula

$$\omega = \frac{1}{2} \sum_{i=1}^{\dim \mathfrak{h}} (h_i)_{-1} (h^i)_{-1}, \tag{12}$$

where $\{h_i\}$ and $\{h^i\}$ are dual bases of $\mathfrak{h}$ with respect to the given bilinear form.

It is possible to classify all simple modules over the vertex algebra V_L. Let

$$L' = \{\lambda \in L \otimes_{\mathbb{Z}} \mathbb{Q} \mid (\lambda, \mu) \in \mathbb{Z} \ \forall \mu \in L\}$$

be the dual lattice to L. For each $\gamma \in L'/L$, consider the vector space

$$V_L^\gamma = \bigoplus_{\lambda \in \gamma + L} \pi_\lambda.$$

Note that the vertex operators $V_\lambda(z)$, $\lambda \in L$, give rise to well-defined fields on each V_L^γ since $(\lambda, \mu) \in \mathbb{Z}$ for all $\lambda \in \gamma + L$ and $\mu \in L$. Thus, for any $A \in V_L$, the vertex operator $Y(A, z)$ is a field on V_L^γ. It can be checked that these fields define a structure of conformal

V_L-module on V_L^γ. Moreover, it is known (see, for example, [1]) that V_L^γ, $\gamma \in L'/L$, are all simple V_L-modules up to isomorphism.

Let us consider the 1-dimensional lattice $L = \sqrt{2}\mathbb{Z}$ from the previous section. In this case, $L' = \frac{1}{2}L$. So, $L'/L \cong \mathbb{Z}/2\mathbb{Z}$ and any element of L' is of the form $k\omega$, $k \in \mathbb{Z}$, where $\omega = \frac{\sqrt{2}}{2}$. Note that L is the root lattice of the simple Lie algebra $\mathfrak{sl}(2,\mathbb{C})$ while L' is its weight lattice.

Now, we define the so-called Demazure modules of level 1 for the affine Kac-Moody algebra $\hat{\mathfrak{sl}}(2,\mathbb{C})$.

Definition 5. *For any $\lambda = k\omega \in L'$, the Demazure module $W^\lambda = W_k$ of level 1 is defined as the subspace of V_L or V_L^ω obtained from the vector $|k\omega\rangle$ by application of all possible operators $V_\mu[m]$, $\mu \in L$, $m \geq 0$.*

It is easy to see that $W_n = W_{-n}$ for symmetry reasons. So, it is sufficient to consider only Demazure modules W_{-n}, $n \geq 0$. Moreover, it can be shown that the module W_n, $n \geqslant 0$, can be obtained from the vector $|-n\omega\rangle$ by applying only the operators $V_{\sqrt{2}}[m]$, $m \geqslant 0$. So, for even n, the spaces W_n are precisely the spaces from the previous section. For odd n, the structure of W_n is very similar to that in the case of even n. We state the following proposition generalizing Proposition 12.

Proposition 17. *Let n be any positive integer. The subspaces $\pi_{-n\omega+k\sqrt{2}}$ having a non-trivial intersection with W_n are precisely the ones for which $0 \leqslant k \leqslant n$. Each space $W_n \cap \pi_{-n\omega+k\sqrt{2}}$ is obtained by application of all possible polynomials in b_m, $m \geqslant 0$, to one vector $v_k \in \pi_{-n\omega+k\sqrt{2}}$. Moreover, the polynomial in b_m, $m < 0$, corresponding to the vector v_k is obtained from the polynomial corresponding to the highest weight vector of the space $V_{k,n-k-1}$ from the Section 4 after the change $b_{-m} \mapsto \sqrt{2}b_{-m}$.*

Proof. The proof is similar to the proof of Proposition 12, so we only give its sketch. First, as in the proof Proposition 14, one shows that any polynomial obtained from $|-n\omega\rangle$ by application of the operators $V_{\sqrt{2}}[m]$, $m \geq 0$ is a linear combination of determinants of form (10) such that $m_1, m_2, \ldots, m_k \leqslant n-1$. Invariance of W_n under the action of b_s, $s \geqslant 0$, is proved exactly as in the proof of Proposition 15. Finally, one shows that the vector $V_{\sqrt{2}}^k[0]|-n\omega\rangle$ is non-zero if $0 \leqslant k \leqslant n$. This completes the proof of the proposition. □

Similarly to Corollary 3, the following corollary follows from Proposition 17.

Corollary 5. *1.* $\dim W_n = 2^n$*;*

2. W_n is invariant under the action of Vir_+.

Proof. Similar to the proof of Corollary 3. □

We want to compute the vector bundles on $\mathbb{P}^1$ corresponding to W_n. However, the situation is now different from the one in the case of even n. The problem is that the operator L_0 acts on $\pi_{k\sqrt{2}/2}$ by the scalar $\frac{k^2}{4}$ which is not integer for odd k. In order to fix this, we can shift the action of L_0 by a scalar operator $\alpha\mathrm{Id}$, i.e. replace the operator L_0 with the operator $L_0 + \alpha\mathrm{Id}$. Here, α is chosen in such a way that all eigenvalues of the operator $L_0 + \alpha\mathrm{Id}$ are integer. Note that this would not be possible if we worked with representations of the whole Virasoro algebra because of the commutation relations

$$[L_m, L_{-m}] = 2mL_0 + \frac{c}{12}(m^3 - m).$$

But it is easy to see that the operators $L_0+\alpha\mathrm{Id}$ and L_m, $m \geqslant$ still give a representation of the algebra Vir_+. However, there is no canonical choice of the constant α. So, we obtain a family of representations of Vir_+ and hence a family of the corresponding vector bundles on $\mathbb{P}^1$. Any two such bundles can be obtained from each other by tensor multiplication by $\mathcal{O}(2N)$, $N \in \mathbb{Z}$.

Now, the computation of the vector bundles corresponding to W_n, $n \geqslant 0$, can be done as in the case of even n. Again, we compute separately the bundles corresponding to each subspace $W_n \cap \pi_{-n\omega+k\sqrt{2}}$, $0 \leqslant k \leqslant n$. As before, this space is isomorphic to the linear span of all semi-infinite monomials of the form

$$\psi_{i_1} \wedge \psi_{i_2} \wedge \ldots \wedge \psi_{i_{n-k}} \wedge \psi_k \wedge \psi_{k+1} \ldots,$$

where $-n+k \leqslant i_1 < i_2 < \ldots < i_{n-k} \leqslant k-1$. This space can be split into the direct sum of the subspaces U_r, $r = 0, 1, \ldots, k$, where U_r is the linear span of semi-infinite monomials having exactly r non-negative indices from 0 to $k-1$. The vector of maximal degree in such subspace is equal to

$$\psi_{-n+k} \wedge \psi_{-n+k+1} \wedge \ldots \psi_{-r-1} \wedge \psi_0 \wedge \psi_1 \wedge \ldots \wedge \psi_{r-1} \wedge \psi_k \wedge \psi_{k+1} \wedge \ldots,$$

while the vector of minimal degree is

$$\psi_{-n+r+k} \wedge \psi_{-n+r+k+1} \wedge \ldots \psi_{-1} \wedge \psi_{k-i} \wedge \psi_{k-i+1} \wedge \ldots \wedge \psi_{k-1} \wedge \psi_k \wedge \psi_{k+1} \wedge \ldots$$

Their degrees are equal to

$$l_{\max} = \frac{1}{4}(n-2k)^2 + r(k-r) + k(n-k-r) + C =$$

and

$$l_{\min} = \frac{1}{4}(n-2k)^2 + (k-r)(n-k-r) + C =$$

respectively, where C is a constant chosen in such a way that both numbers $l_{\max}$ and $l_{\min}$ are integer. Also, the dimension of U_r is $\binom{n-k}{r}\binom{k}{r}$. This implies that the vector bundle on $\mathbb{P}^1$ corresponding to the space U_r is isomorphic to

$$\bigoplus_{r=0}^{k} \mathcal{O}(l_{\max}+l_{\min})^{\dim U_r} = \bigoplus_{r=0}^{k} \mathcal{O}(\frac{1}{2}n^2 - rn + 2C)^{\binom{n-k}{r}\binom{k}{r}} = \bigoplus_{r=0}^{k} \mathcal{O}(2N - rn)^{\binom{n-k}{r}\binom{k}{r}},$$

where $N = \frac{1}{4}n^2 + C$. Note that N can be any integer number.

In the sum over all k, $0 \leq k \leq n$, the summand $\mathcal{O}(\frac{1}{2}n^2 - rn + 2C)$ is contained with multiplicity

$$\sum_{k=0}^{n} \binom{n-k}{r}\binom{k}{r} = \binom{n+1}{2r+1}.$$

So, the final result is the following.

Proposition 18. *The family of vector bundles on $\mathbb{CP}^1$ corresponding to the representations of Vir_+ on the space W_n is the direct sum*

$$\bigoplus_{r=0}^{n} \mathcal{O}(2N - nr)^{\oplus\binom{n+1}{2r+1}},$$

where N is any integer.

7 Demazure modules of level 1: the case of $\mathfrak{sl}(n,\mathbb{C})$

In this section, we discuss a more general situation. Let L_n be the root lattice of the simple Lie algebra $\mathfrak{sl}(n,\mathbb{C})$, i.e. $\mathrm{rk}(L_n) = n-1$ and L_n has a basis α^1, α^2, $\ldots$, α^{n-1} of simple roots such that $(\alpha^i,\alpha^i) = 2$, $(\alpha^i,\alpha^{i+1}) = -1$ and $(\alpha_i,\alpha_j) = 0$ if $|i-j| > 1$ for all $i,j = 1,2,\ldots,n-1$.

The dual lattice L'_n is the weight lattice of $\mathfrak{sl}(n,\mathbb{C})$. Its basis is formed by the so-called *fundamental weights* ω^1, ω^2, $\ldots$, ω^{n-1} such that $\langle\omega^i,\alpha^j\rangle = \delta_{ij}$ for all $i,j = 1,2,\ldots,n-1$. It is known that the group L'_n/L_n is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ via the following isomorphism: $k \in \mathbb{Z}/n\mathbb{Z} \mapsto [\omega^k] \in L'_n Ł_n$, $1 \leqslant k \leqslant n-1$. This implies that there exist n pairwise non-isomorphic simple modules over the conformal vertex algebra V_{L_n} and these modules are V_{L_n} and $V_{L_n}^{\omega^k}$, $1 \le k \le n-1$ (as before, we denote by $V_{L_n}^{\lambda}$ the module over V_{L_n} corresponding to the weight λ).

As in the case $n = 2$, we introduce the notion of Demazure module of level 1.

Definition 6. *For any $\lambda \in L'_n$, the Demazure module W^λ of level 1 is defined as the subspace of $V_{L_n}^{\lambda}$ obtained from the vector $|\lambda\rangle$ by application of all possible operators $V_\mu[m]$, $\mu \in L$, $m \ge 0$.*

As in the case $n = 2$, it is clear that $W^\lambda = W^{\lambda'}$ if λ and λ' are equivalent with respect to the action of the Weyl group.

In this paper, we will work only with the case $\lambda = k\omega^1$, $k \geqslant 0$. For convenience, we denote ω^1 simply by ω.

Similarly to the case $n = 2$, in order to obtain the Demazure module $W^{k\omega}$, it suffices to apply to $k\omega$ only the operators $V_{-\alpha^i}[m]$, $m \geqslant 0$. This implies that any element of $W^{k\omega}$ can be represented as a polynomial in the variables α^i_{-s}, where $i = 1,2,\ldots,n-1$, $s > 0$. Our first task is to describe the space $W^{k\omega}$ explicitly. To this aim, it is convenient to consider another basis of the lattice L_n, namely $a^1, a^2, \ldots, a^{n-1}$, where $a^i = \alpha^1 + \alpha^2 + \ldots + \alpha^i$. It can be checked by a direct computation that $(a^i,a^i) = 2$, $(a^i,a^j) = 1$ if $i \neq j$ for all $i,j = 1,2,\ldots,n-1$. Again, it can be shown that $W^{k\omega}$ can be obtained from $|k\omega\rangle$ by applying only the operators $V_{-a^i}[m]$, $m \geqslant 0$.

As we know, $[h_n, g_m] = n(h,g)\delta_{n,-m}$, so the elements of any «vertical» component π_μ of $V_{L_n}^{\lambda}$ can be represented in the form $P|\mu\rangle$, where P is a polynomial in a^i_{-s}, $1 \leqslant i \leqslant n-1$, $s > 0$, and the elements a^i_s, $s > 0$, act as the operators

$$s\sum_{j=1}^{n-1}\frac{\partial}{\partial a^s_{-j}} + s\frac{\partial}{\partial a_{-s^i}}, \tag{13}$$

while a^i_0 act as multiplication by (a^i,μ).

Denote by P_i, $i \geqslant 0$, the polynomials defined by (7). As before, we assume additionally that $P_i = 0$ if $i < 0$.

Let r_1, r_2, $\ldots$, r_{n-1} be any non-negative integers and $r = r_1 + r_2 + \ldots + r_{n-1}$. The following proposition describes the structure of the intersection of $W^{k\omega}$ with the «vertical» component $\pi_{k\omega - r_1 a_1 - \ldots - r_{n-1}a_{n-1}}$ (i.e. the elements of $V_{L_n}^{k\omega}$ obtained from $|k\omega\rangle$ by applying the operators $V_{-a^1}[m]$ r_1 times, the operators $V_{-a^2}[m]$ r_2 times, $\ldots$, the operators $V_{-a^{n-1}}[m]$ r_{n-1} times, where $m \ge 0$).

Proposition 19. *The intersection $W^{k\omega} \cap \pi_{k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}}$ is the linear span of all polynomials of the following form:*

$$P_{t_1,t_2,\ldots,t_r} = \det \begin{pmatrix} P_{t_1-r+1} & P_{t_1-r+2} & \ldots & P_{t_1} \\ P_{t_2-r+1} & P_{t_2-r+2} & \ldots & P_{t_2} \\ \vdots & \vdots & \ddots & \vdots \\ P_{t_r-r+1} & P_{t_r-r+2} & \ldots & P_{t_r} \end{pmatrix}, \tag{14}$$

where the polynomials P_i in the rows from 1 to r_1, from r_1+1 to r_1+r_2, ..., from $r_1+\ldots+r_{n-2}+1$ to $r_1+\ldots+r_{n-1}$ depend on the variables a^1_{-s}, a^2_{-s}, ..., a^{n-1}_{-s} respectively and $t_1, t_2, \ldots, t_r \leq k-1$.

The proof of Proposition 19 is similar to the one in the case $n = 2$. We start with the following analogue of Lemmas 3 and 4. For simplicity, from now on we denote by $P_j(a^i)$ the polynomial P_j in variables a^j_{-s}, $s < 0$. Again, by $\widetilde{V}_\lambda$ we denote the expression defined by the same formula as (11) but without $S_\lambda z^\lambda$.

Lemma 5. *For any $i = 1, 2, \ldots, n-1$*

$$\widetilde{V}_{-a^i}(z)(P_t(a^i)) = (P_t(a^i) - 2z^{-1}P_{t-1}(a^i) + z^{-2}P_{t-2}(a^i))\widetilde{V}_{-a^i}(z)$$

and for any $i, j = 1, 2, \ldots, n-1$, $i \neq j$

$$\widetilde{V}_{-a^i}(z)(P_t(a^j)) = (P_t(a^j) - z^{-1}P_{t-1}(a^j)\widetilde{V}_{-a^i}(z).$$

Proof. The first equality is just the statement of Lemma 4. The proof of the second equality is the same as the proof of Lemma 3. □

It follows from Lemma 5that the following statement weaker than Proposition 19 holds.

Proposition 20. *Any element in the intersection $W^{k\omega} \cap \pi_{k\omega - r_1 a_1 - \ldots - r_{n-1} a_{n-1}}$ is a linear combination of the polynomials from Proposition 19.*

Proof. If we consider the space obtained from the vector $|k\omega\rangle$ by applying all possible operators $V_{-a^i}[m]$ (without the restriction $m \geq 0$) then the statement is proved in the same way as in the Propositions 13 and 14. So, it remains to show that the restriction $m \geq 0$ implies the restriction $t_1, t_2, \ldots, t_r \leq k-1$. The proof of this fact is also similar to the proof of the second part of Proposition 14 although it requires more complicated calculations.

We are going to prove this by induction on $r = r_1 + r_2 + \ldots + r_{n-1}$. The base $r = 0$ is trivial. To prove the induction step, we start with computing the degrees of $|k\omega\rangle$ and $|k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}\rangle$.

$$\deg |k\omega\rangle = \frac{(k\omega, k\omega)}{2} = \frac{k^2}{2} \cdot \frac{n-1}{n},$$

$$\deg |k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}\rangle = \frac{1}{2}(k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}, k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}) =$$

$$\frac{1}{2}((k\omega, k\omega) - 2(k\omega, r_1 a^1 + \ldots + r_{n-1} a^{n-1}) + (r_1 a^1 + \ldots + r_{n-1} a^{n-1}, r_1 a^1 + \ldots + r_{n-1} a^{n-1})) =$$

$$= \frac{k^2}{2} \cdot \frac{n-1}{n} - k(r_1 + \ldots + r_{n-1}) + (r_1^2 + \ldots + r_{n-1}^2) + \sum_{i<j} r_i r_j.$$

If we consider, for example, the «vertical» component $|k\omega\rangle - (r_1+1)a^1 - r_2a^2 - \ldots - r_{n-1}a^{n-1}\rangle$ obtained from the previous one via the shift by a^1, i.e. corresponding to the tuple $(r_1+1, r_2, \ldots, r_{n-1})$, then the degree we are interested in is equal to

$$\deg|k\omega - (r_1+1)a^1 - r_2a^2 - \ldots - r_{n-1}a^{n-1}\rangle =$$

$$= \frac{k^2}{2}\cdot\frac{n-1}{n} - k(r_1+\ldots+r_{n-1}) + (r_1^2+\ldots+r_{n-1}^2) + \sum_{i<j} r_ir_j + 2r_1 + 1 + (r_2+\ldots+r_{n-1}) - k =$$

$$= \deg|k\omega - r_1a^1 - \ldots - r_{n-1}a^{n-1}\rangle + 2r_1 + 1 + (r_2+\ldots+r_{n-1}) - k.$$

Consider a polynomial of form 14. It follows from Lemma 5 that the result of application of the bosonic vertex operator $V_{-a^1}(z)$ to this polynomial is the following expression:

$$\det\begin{pmatrix} P_{t_1-r+1} - 2z^{-1}P_{t_1-r} + z^{-2}P_{t_1-r-1} & \ldots & P_{t_1} - 2z^{-1}P_{t_1-1} + z^{-2}P_{t_1-2} \\ \vdots & \ddots & \vdots \\ P_{t_r-r+1} - z^{-1}P_{t_r-r} & \ldots & P_{t_r} - z^{-1}P_{t_r-1} \end{pmatrix}\cdot$$

$$\cdot(P_0 + P_1z + P_2z^2 + \ldots)$$

(the elements in the first r_1 and in the remaining rows of the matrix are of the form

$$P_t - 2z^{-1}P_{t-1} + z^{-2}P_{t-2} \text{ and } P_t - z^{-1}P_{t-1}$$

respectively).

Recall that the operator $V_{-a^i}[m]$ is of degree $-m$ for any $m \in \mathbb{Z}$ while the degrees of the vacuum vectors $|k\omega-(r_1+1)a^1-r_2a^2-\ldots-r_{n-1}a^{n-1}\rangle$ and $|k\omega-r_1a^1-r_2a^2-\ldots-r_{n-1}a^{n-1}\rangle$ differ by $2r_1+1+(r_2+\ldots+r_{n-1})-k$. So, the polynomials in the intersection of $W^{k\omega}$ with the «vertical» component $\pi_{k\omega-(r_1+1)a^1-r_2a^2-\ldots r_{n-1}a^{n-1}}$ appear in the last expression as the coefficients of z^s such that $s \leq -2r_1-1-(r_2+\ldots+r_{n-1})+k$. But it is easy to see that the minimal power of z in the full expansion of the determinant is equal to $-2r_1-r_2-\ldots-r_{n-1}$. It follows that the powers of z taken from the bracket $(P_0 + P_1z + P_2z^2 + \ldots)$ cannot be greater than $k-1$. This concludes the proof of the proposition.

□

Proof of Proposition 19. To prove that all polynomials of form (14) lie in the space $W^{k\omega} \cap \pi_{k\omega-r_1a^1-\ldots-r_{n-1}a^{n-1}}$, consider the polynomials of the maximal degree in this space. On the one hand, this degree cannot be greater than

$$\deg|k\omega\rangle - \deg|k\omega - r_1a^1 - \ldots - r_{n-1}a^{n-1}\rangle = k(r_1+\ldots+r_{n-1}) - (r_1^2+\ldots+r_{n-1}^2) - \sum_{i<j} r_ir_j.$$

On the other hand, one can easily compute that this number is the maximal possible degree of a polynomial of form 14 and such a polynomial has this degree if and only if the numbers $t_1, t_2, \ldots, t_{r_1}$ are equal to $k-1, k-2, \ldots, k-r_1$ (in some order), $t_{r_1+1}, t_{r_1+2}, \ldots, t_{r_1+r_2}$ are equal to $k-1, k-2, \ldots, k-r_2$, etc. So, up to multiplication by a constant, there is only one polynomial of form 14 of this degree (of course, provided that this number is non-negative). We can prove that this polynomial is in $W^{k\omega} \cap \pi_{k\omega-r_1a^1-\ldots-r_{n-1}a^{n-1}}$ using the same technique as in the case $n=2$. Namely, we use the fact that the operators $V_{\pm a^i}[0]$ and a_0^i $(i = 1, 2, \ldots, n-1)$ span a Lie algebra isomorphic to $\mathfrak{sl}(n, \mathbb{C})$. So, the vectors $V_{-a^1}[0]^{r_1} \ldots V_{-a^{n-1}}[0]^{r_{n-1}}|k\omega\rangle$ (for all possible tuples $r_1, r_2, \ldots, r_{n-1}$) form a representation of $\mathfrak{sl}(n, \mathbb{C})$ with the highest vector $k\omega$. It is known that the irreducible representation of

$\mathfrak{sl}(n,\mathbb{C})$ with the highest weight $k\omega$ is isomorphic to the k-th symmetric power of the n-dimensional vector representation. So, this irreducible representation is exactly the representation spanned by the vectors $V_{-a^1}[0]^{r_1}\dots V_{-a^{n-1}}[0]^{r_{n-1}}|k\omega\rangle$ and all these vectors are non-zero.

It is not difficult to prove that each space $W^{k\omega}\cap\pi_{k\omega-r_1a^1-\dots-r_{n-1}a^{n-1}}$ is invariant under the action of a^i_s, $s\geq 0$. This statement can be proved similarly to Proposition 15 using an obvious analogue of relations (6). It follows from 13 that this is equivalent to invariance under all derivations $\frac{\partial}{\partial a^i_{-s}}$, $s>0$. But the action of $\frac{\partial}{\partial a^i_{-s}}$, $s>0$ on the polynomial $P_{t_1,t_2,\dots,t_r}$ is given by the following formula (for brevity, we denote by R_j the sum $r_1+r_2+\dots+r_j$):

$$\frac{\partial}{\partial a^i_{-s}}(P_{t_1,t_2,\dots,t_r})=\sum_{j=1}^{r_i}P_{t_1,\dots,t_{R_{i-1}},t_{R_{i-1}+1},\dots,t_{R_{i-1}+j}-s,\dots t_{R_i},\dots,t_r} \tag{15}$$

In words, we reduce by s each index from $R_{i-1}+1$ to R_i, i.e. each index corresponding to the variables a^i_{-q}, and take the sum over all such polynomials (compare this with (4)). Now, the fact that all polynomials of form (14) can be obtained from the one of the maximal degree follows from the similar statement of Proposition 4. □

Let us also prove the following lemma.

Lemma 6. *The space $W^{k\omega}\cap\pi_{k\omega-r_1a^1-\dots-r_{n-1}a^{n-1}}$ contains all polynomials of the form*

$$\det\begin{pmatrix} P_{t_1-j_{r-1}} & P_{t_1-j_{r-2}} & \dots & P_{t_1-j_1} & P_{t_1}\\ P_{t_2-j_{r-1}} & P_{t_2-j_{r-2}} & \dots & P_{t_2-j_1} & P_{t_2}\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ P_{t_r-j_{r-1}} & P_{t_r-j_{r-2}} & \dots & P_{t_r-j_1} & P_{t_r}\end{pmatrix}, \tag{16}$$

where $j_1>j_2>\dots>j_{r-1}$ are any positive integers and $t_1,t_2,\dots,t_{r-1}\leq k-1$.

Proof. We need to prove that each polynomial (16) can be expressed as a linear combination of polynomials (14). If we replace each polynomial P_i in the j-th row of this determinant with x^i_j (x_j, $j=1,2,\dots,r$, are formal variables), we obtain the following determinant:

$$\det\begin{pmatrix} x_1^{t_1-j_{r-1}} & x_1^{t_1-j_{r-2}} & \dots & x_1^{t_1}\\ x_2^{t_2-j_{r-1}} & x_2^{t_2-j_{r-2}} & \dots & x_2^{t_2}\\ \vdots & \vdots & \ddots & \vdots\\ x_r^{t_r-j_{r-1}} & x_r^{t_r-j_{r-2}} & \dots & x_r^{t_r}\end{pmatrix}=x_1^{t_1-j_{r-1}}\dots x_r^{t_r-j_{r-1}}\det\begin{pmatrix} 1 & x_1^{j_{r-1}-j_{r-2}} & \dots & x_1^{j_{r-1}}\\ 1 & x_2^{j_{r-1}-j_{r-2}} & \dots & x_2^{j_{r-1}}\\ \vdots & \vdots & \ddots & \vdots\\ 1 & x_r^{j_{r-1}-j_{r-2}} & \dots & x_r^{j_{r-1}}\end{pmatrix}.$$

The last determinant is a skew-symmetric polynomial in x_1, x_2, …, x_r, so it is a symmetric polynomial (in fact, a Schur polynomial) multiplied by $\prod_{i<j}(x_j-x_i)$. Writing the corresponding Schur polynomial as a sum of monomials, we obtain that the last expression is a linear combination of expressions of the following form:

$$x_1^{t_1-r+1}x_2^{t_2-r+1}\dots x_r^{t_r-r+1}\det\begin{pmatrix} 1 & x_1 & \dots & x_1^{r-2} & x_1^{r-1}\\ 1 & x_2 & \dots & x_2^{r-2} & x_2^{r-1}\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ 1 & x_r & \dots & x_r^{r-2} & x_r^{r-1}\end{pmatrix}=\det\begin{pmatrix} x_1^{t_1-r+1} & x_1^{t_1-r+2} & \dots & x_1^{t_1}\\ x_2^{t_2-r+1} & x_2^{t_2-r+2} & \dots & x_2t_2\\ \vdots & \vdots & \ddots & \vdots\\ x_r^{t_r-r+1} & x_r^{t_r-r+2} & \dots & x_r^{t_r}\end{pmatrix}.$$

Replacing back each x^i_j with P_i (P_i depends on the corresponding set of variables a^m_{-s}), we obtain the desired expression of polynomial (16) as a linear combination of polynomials (14). □

We proved that $W^{k\omega} \cap \pi_{k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}}$ is linearly spanned by the elements $P_{t_1,\ldots,t_r}$ which can be numbered by tuples $(t_1, t_2, \ldots, t_{r_1})$, $(t_{r_1+1}, \ldots t_{r_1+r_2})$, etc. of non-negative integers not exceeding $k-1$. For this numbering, the anti-commutation relations hold, i.e. if we permute the numbers in any of these tuples, then the corresponding polynomial is multiplied by the sign of the permutation. This is similar to the case $n = 2$ when we have only one tuple. However, the main difficulty for general n is that the polynomials corresponding to different sets of such tuples are not linearly independent. For example, for $r_1 = r_2 = 1$ and $r_i = 0$, $i \geq 3$, the following linear combination vanishes:

$$\begin{vmatrix} 0 & P_0(a^1) \\ P_{t-1}(a^2) & P_t(a^2) \end{vmatrix} + \begin{vmatrix} P_0(a^1) & P_1(a^1) \\ P_{t-2}(a^2) & P_{t-1}(a^2) \end{vmatrix} + \cdots + \begin{vmatrix} P_{t-1}(a^1) & P_t(a^1) \\ 0 & P_0(a^2) \end{vmatrix} = 0.$$

So, it is natural to try to find a linearly independent subset of the set of all polynomials (14) that spans linearly the whole space $W^{k\omega} \cap \pi_{k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}}$. The following proposition describes a way to do that.

Proposition 21. *The polynomials (14) such that*

1. *the indices corresponding to each of $n-1$ set of variables a^i_{-s} $(i = 1, 2, \ldots, n-1)$ are given in descending order (i.e., $t_1 > \ldots > t_{r_1}$, $t_{r_1+1} > \ldots > t_{r_1+r_2}$, etc.);*
2. *the indices of P_i on the main diagonal (i. e., $t_1 - r + 1$, $t_2 - r + 2$, $\ldots$, t_r) are non-negative*

form a basis of the space $W^{k\omega} \cap \pi_{k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}}$.

Proof. First, we prove by induction on n that such polynomials are linearly independent. The base $n = 2$ was proved in Section 5. To prove the induction step from $n-1$ to n, suppose that there is a non-trivial linear combination of such determinants equal to 0. Expand all determinants in this linear combinations with respect to the last r_{n-1} rows. Each term of this expansion is, up to the sign, the product of an $(r_{n-1} \times r_{n-1})$-minor of the initial matrix taken from the last r_{n-1} rows and the complement $(R_{n-2} \times R_{n-2})$-minor. Consider all terms in this expansion with the maximal degree in the variables a^{n-1}_{-s}. The sum of these terms must be zero. But the $(r_{n-1} \times r_{n-1})$-minor in each of these terms is the rightmost minor of the initial matrix which implies that this minor is of form (8). By our assumption, all indices on the main diagonal in the initial $(r \times r)$-matrices are non-negative. So, the rightmost $(r_{n-1} \times r_{n-1})$-minors are non-zero. Since these minors are linearly-independent by the induction base, the coefficient of each such minor must be zero. But all these coefficients are sums (with coefficients) of $(R_{n-2} \times R_{n-2})$-minors and each of these minors are leftmost ones in the initial $(r \times r)$-matrices, so they are of form (14) with $r_{n-1} = 0$. By the induction hypothesis, these minors are linearly independent. So, each coefficient of the corresponding $r \times r$-matrices in the initial linear combination must be 0. Proceeding similarly with the terms of smaller degree in a^{n-1}_{-s}, we eventually prove that the initial linear combination is trivial.

Now, we need to prove that each polynomial (14) can be expressed as a linear combination of polynomials satisfying the conditions of the proposition. It suffices to prove this in the case when all r_i are equal to 1, i.e. the polynomials in each row depend on their own set of variables. For brevity, denote this space by U^k_r. It suffices to prove that $\dim U^k_r$ is equal to the number of polynomials satisfying the conditions of the proposition, i.e., as it is easy to see, $k(k-1)\ldots(k-r+1)$. To prove this, we use induction on r. The base $r = 1$ is clear. To do the induction step from U^k_{r-1} to U^k_r, consider a new space $\widehat{U}^k_r \supset U^k_r \oplus V^k_r$,

where V_r^k is spanned by all polynomials (14) such that $t_1 + t_2 + \ldots + t_r \leqslant r(k-1)$ but without the restriction $t_1 \leq k-1$. Using expansion along the first row and Lemma 6, we can show by induction on t_1 that the space $\widehat{U}_r^k$ has a basis consisting of polynomials

$$P_{t_1}^1 \cdot \det \begin{pmatrix} P_{t_2-r+2}^2 & P_{t_2-r+3}^2 & \ldots & P_{t_2}^2 \\ \vdots & \vdots & \ddots & \vdots \\ P_{t_r-r+2}^r & P_{t_r-r+3}^r & \ldots & P_{t_r}^r \end{pmatrix}$$

such that $t_1 + t_2 + \ldots + t_r \leq r(k-1) - r + 1$ and the determinant satisfies the conditions of our proposition (P_t^i are the usual polynomials P_t; we assume that polynomials P_t^i and P_t^j depend on different sets of variables for $i \neq j$), while V_k^r has a basis consisting of polynomials of the same form but with the additional restriction $t_1 \geq k - r + 1$. So, the dimension of U_r^k is equal to the dimension of the space with a basis consisting of polynomials of the same form but with the additional restriction $t_1 \leq k - r$. Applying the induction hypothesis, we see that this dimension is equal to

$$(k-r+1) \cdot k(k-1)\ldots(k-r+2) = k(k-1)\ldots(k-r+1)$$

as desired.

□

Corollary 6. *The dimension of $W^{k\omega}$ is equal to k^n.*

Proof. For any $(r_1, r_2, \ldots, r_{n-1})$, the basis of the space $W^{k\omega} \cap \pi_{k\omega - r_1 a^1 - \ldots - r_{n-1} a^{n-1}}$ consists of all elements of form (14) such that $k-1 \geq t_1 > t_2 > \ldots > t_{r_1} \geq 0$, $k - r_1 - 1 \geq t_{r_1+1} > \ldots > t_{r_1+r_2} \geq 0$, etc. So, the dimension of this space is equal to

$$\binom{k}{r_1} \cdot \binom{k-r_1}{r_2} \cdot \binom{k-r_1-\ldots r_{n-2}}{r_{n-1}} = \frac{k!}{r_1! r_2! \ldots r_{n-1}! (k - r_1 - \ldots - r_{n-1})!},$$

which is the multinomial coefficient $\binom{k}{r_1, r_2, \ldots r_{n-1}, k-r_1-\ldots-r_{n-1}}$. These are the coefficients in the expansion of the expression $(x_1 + x_2 + \ldots + x_n)^k$. So, their sum is equal to n^k. □

In Section 8, we will use the following result describing explicitly linear relations for the polynomials $P_{t_1,\ldots,t_r}$.

Proposition 22. *If $t_m - m + r < 0$, i.e., the m-th diagonal element of matrix (14) is zero, then we have*

$$\sum_{\substack{j_{m+1},\ldots,j_r \geqslant 0 \\ J = j_{m+1} + \ldots + j_r}} P_{t_1,\ldots,t_{m-1},t_m+J,t_{m+1}-j_{m+1},\ldots,t_r-j_r} = 0.$$

Proof. Denote by $\widetilde{P}_{t_1,\ldots,t_r}$ the determinant obtained from (14) after replacing each P_s^i with x_i^s. We have

$$\widetilde{P}_{t_1,\ldots,t_r} = \prod_{i=1}^{r} x_i^{t_i - r + 1} \cdot \prod_{i>j} (x_i - x_j)$$

(the second product is the Vandermonde determinant). Take the left-hand side of the relation we are proving and replace each P with $\widetilde{P}$. We obtain

$$\sum_{\substack{j_{m+1},\ldots,j_r \geqslant 0 \\ J = j_{m+1} + \ldots + j_r}} \widetilde{P}_{t_1,\ldots,t_{m-1},t_m+J,t_{m+1}-j_{m+1},\ldots,t_r-j_r} =$$

$$= \sum_{\substack{j_{m+1},\dots,j_r\geqslant 0\\ J=j_{m+1}+\dots+j_r}} \prod_{i=1}^{r} x_i^{t_i-r+1}\cdot x_m^J\cdot \prod_{i=m+1}^{r} x_i^{-j_i}\cdot \prod_{i>j}(x_i-x_j)=$$

$$=\prod_{i=1}^{r} x_i^{t_i-r+1}\prod_{i>j\neq m}(x_i-x_j)\sum_{\substack{j_{m+1},\dots,j_r\geqslant 0\\ J=j_{m+1}+\dots+j_r}} x_m^J x_{m+1}^{-j_1}\dots x_r^{-j_r}\prod_{i>m}(x_i-x_m)=$$

$$=\prod_{i=1}^{r} x_i^{t_i-r+1}\prod_{i>j\neq m}(x_i-x_j)\sum_{\substack{j_{m+1},\dots,j_r\geqslant 0\\ J=j_{m+1}+\dots+j_r}} x_m^J x_{m+1}^{-j_1+1}\dots x_r^{-j_r+1}\prod_{i>m}(1+x_m x_i^{-1}+x_m^2x_i^{-2}+\dots)^{-1}=$$

$$=\prod_{i=1}^{r} x_i^{t_i-r+1}\prod_{i>j\neq m}(x_i-x_j)\Big(\sum_{\substack{j_{m+1},\dots,j_r\geqslant 0\\ J=j_{m+1}+\dots+j_r}} x_m^J x_{m+1}^{-j_1+1}\dots x_r^{-j_r+1}\Big)\Big(\sum_{\substack{j_{m+1},\dots,j_r\geqslant 0\\ J=j_{m+1}+\dots+j_r}} x_m^J x_{m+1}^{-j_1+1}\dots x_r^{-j_r+1}\Big)^{-1}=$$

$$=\prod_{i=1}^{r} x_i^{t_i-r+1}\prod_{i>j\neq m}(x_i-x_j).$$

After expanding the brackets in the last product, we obtain an expression containing only negative powers of x_m (because $t_m-m+r<0$). So, after we replace back x_i^s with P_s^i, this expression becomes zero. □

Next, we are going to find an explicit formula for the action of L_1 on $W^{k\omega}$. Generally, for an arbitrary conformal vertex algebra V, the action of Vir is given by the formula

$$Y(\omega,z)=\sum_{n\in\mathbb{Z}} L_n z^{-n-2},$$

where ω is the conformal vector and $Y(v,z)\in \mathrm{End}V[[z^{\pm 1}]]$ is the vertex operator corresponding to $v\in V$. In our case, for the vertex algebra π_{L_n}, the conformal vector ω is given by formula (12). The basis of L dual to $\{a^i \mid 1\leq i\leq n-1\}$ is $\{-\frac{a^1+a^2+\dots+a^{n-1}}{n}+a^i \mid 1\leq i\leq n-1\}$. So, we have

$$\omega=\frac{1}{2n}\Big((n-1)\sum_{i=1}^{n-1}(a^i)^2-\sum_{i\neq j}a^ia^j\Big).$$

The operator L_1, i.e. the coefficient of z^{-3} in $Y(\omega,z)$, is thus given by

$$\frac{1}{2n}\Big((n-1)\sum_{\substack{1\leq i\leq n-1\\ k\in\mathbb{Z}}} a_k^i a_{-k+1}^i-\sum_{\substack{i\neq j\\ k\in\mathbb{Z}}} a_k^i a_{-k+1}^j\Big)=$$

$$=\frac{1}{n}\Big((n-1)\sum_{i=1}^{n-1}(a_0^ia_1^i+a_{-1}^ia_2^i+\dots)-\sum_{i\neq j}(a_0^ia_1^j+a_{-1}^ia_2^j+\dots\Big).$$

Consider the sum of all coefficients $a_k^ia_{-k+1}^j$ in the last expression for fixed k. Using formula (13), we obtain that this sum is equal to

$$\frac{n-1}{n}\sum_{i=1}^{n-1}a_{-k+1}^ia_k^i-\frac{1}{n}\sum_{i\neq j}a_{-k+1}^ia_k^j=$$

$$= \frac{n-1}{n}\sum_{i=1}^{n-1} ka_{-k+1}(\sum_{r=1}^{n-1}\partial a^r_{-k} + \partial a^i_{-k}) - \frac{1}{n}\sum_{i\neq j} ka^i_{-k+1}(\sum_{r=1}^{n-1}\partial a^r_{-k} + \partial a^j_{-k}).$$

In the sum of these expressions for all possible j, the coefficient of $a^i_{-k+1}\partial a^r_{-k}$ such that $i \neq r$ is equal to $\frac{1}{n}(k(n-1) - k(n-3+2)) = 0$, while the coefficient of $a^i_{-k+1}\partial a^i_{-k}$ is $\frac{1}{n}(2k(n-1) - k(n-2)) = k$. So, we arrive at the following formula:

$$L_1 = \sum_{i=1}^{n-1}(a^i_0\partial a^i_{-1} + 2a^i_{-1}\partial a^i_{-2} + 3a^i_{-2}\partial a^i_{-3} + \ldots).$$

As in the case $n = 2$, the operator $\widetilde{L}_1 = \sum_{i=1}^{n-1}(2a^i_{-1}\partial a^i_{-2} + 3a^i_{-2}\partial a^i_{-3} + \ldots)$ acts on P_m by the rule $\widetilde{L}_1(P_m) = (m-1)P_{m-1}$. Using this, we can prove the following proposition.

Proposition 23. *The action of $\widetilde{L}_1$ on any polynomial $P_{t_1,t_2,\ldots,t_r}$ (using the notation as in (14)) is given by the formula*

$$\widetilde{L}_1(P_{t_1,t_2,\ldots,t_r}) = \sum_{i=1}^{r}(t_i - r)P_{t_1,\ldots,t_i-1t_i-1,t_{i+1},\ldots,t_r}$$

.

Proof. Replace each element P^i_s of determinant (14) with x_i^{s-1}. The resulting polynomial is equal to

$$\widehat{P}_{t_1,\ldots,t_r} = \prod_{i=1}^{r} x_i^{t_i-r+1}\cdot\prod_{i>j}(x_i - x_j).$$

In terms of $x_1,\ldots,x_r$, the action of $\widetilde{L}_1$ is the derivation $\frac{\partial}{\partial x_1} + \frac{\partial}{\partial x_2} + \ldots + \frac{\partial}{\partial x_r}$. So,

$$\widetilde{L}_1(\widehat{P}_{t_1,\ldots,t_r}) = \sum(t_i - r)\prod_{i=1}^{r} x^{t_i-r-1}\prod_{i>j}(x_i - x_j) + \prod_{i=1}^{r} x_i^{t_i-r}(\frac{\partial}{\partial x_1} + \frac{\partial}{\partial x_2} + \ldots + \frac{\partial}{\partial x_r})(\prod_{i>j}(x_i - x_j)).$$

The last term of this sum is zero since $(\frac{\partial}{\partial x_1} + \frac{\partial}{\partial x_2} + \ldots + \frac{\partial}{\partial x_r})(\prod_{i>j}(x_i - x_j))$ is skew-symmetric and hence is divisible by $\prod_{i>j}(x_i - x_j)$ but has a smaller degree. Replacing back x_i^{s-1} with P^i_s, we obtain the initial formula. □

It remains to compute the action of $a^i_0\partial a^i_{-1}$, $1 \leq i \leq n-1$. The operator ∂a^i_{-1} acts on $P_{t_1,t_2,\ldots,t_r}$ by the formula

$$\partial a^i_{-1}(P_{t_1,t_2,\ldots,t_r}) = \sum_{j=1}^{r_i} P_{t_1,\ldots,t_{R_{i-1}},t_{R_{i-1}+1},\ldots t_{R_{i-1}+j}-1,\ldots,t_{R_i},\ldots t_r},$$

while the action a^i_0 on $\pi_{k\omega - r_1a^1 - \ldots - r_{n-1}a^{n-1}}$ is the scalar multiplication by

$$(a^i, k\omega + r_1a^1 + \ldots r_{n-1}a^{n-1}) = -(k - (r_1 + \ldots + r_{n-1}) - r_i) = -(k - r - r_i).$$

This shows that the action of L_1 on $W^{k\omega} \cap \pi_{k\omega - r_1a^1 + \ldots - r_{n-1}a^{n-1}}$ is given by

$$L_1(P_{t_1,t_2,\ldots,t_r}) = \sum_{i=1}^{r}(t_i - k + r_j)P_{t_1,\ldots,t_i-1t_i-1,t_{i+1},\ldots,t_r}, \tag{17}$$

where we assume that the polynomials in the i-th row of determinant (14) depend on the variables a^j_{-s}. It follows, in particular, that the polynomial of the maximal degree in $W^{k\omega} \cap \pi_{k\omega - r_1a^1 - \ldots - r_{n-1}a^{n-1}}$ is sent to 0 by the operator L_1.

8 Explicit computation of vector bundles corresponding to Demazure modules of level 1 for $\mathfrak{sl}(3,\mathbb{C})$

In this section, we compute explicitly the vector bundles on $\mathbb{CP}^1$ corresponding to the Demazure modules $W^{k\omega}$ for the Lie algebra $\mathfrak{sl}(3,\mathbb{C})$. For convenience, we slightly change the notation from the previous section. Namely, we replace a^1 and a^2 with a and b respectively. Also, we will denote the polynomials P_m depending on b_{-s} by Q_m. As we know from the previous section, the space $W^{k\omega} \cap \pi_{k\omega-(k-r)a-sb}$ is the linear span of all polynomials of the form

$$\widetilde{R}_{\mu,\lambda} = \widetilde{R}_{\mu_1,\ldots,\mu_{k-r};\lambda_1,\ldots,\lambda_s} = \det \begin{pmatrix} P_{\lambda_1-s+1} & \ldots & P_{\lambda_1} & P_{\lambda_1+1} & \ldots & P_{\lambda_1+k-r} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ P_{\lambda_s-s+k-r} & \ldots & P_{\lambda_s} & P_{\lambda_s+1} & \ldots & P_{\lambda_s+r} \\ \ldots & \ldots & \ldots & Q_{\mu_1-k+r+1} & \ldots & Q_{\mu_1} \\ \vdots & \ddots & \ldots & \vdots & \ddots & \vdots \\ \ldots & \ldots & \ldots & Q_{\mu_{k-r}-k+r+1} & \ldots & Q_{\mu_{k-r}} \end{pmatrix},$$

where μ and λ are tuples $(\mu_1,\ldots,\mu_{k-r})$ and $(\lambda_1,\ldots,\lambda_s)$ respectively such that $\mu_t \leqslant k-1$ and $\lambda_t \leqslant r-1$ for all t. Here, we do not assume that all λ_t and μ_t are non-negative. However, we have the following relation for any $\mu_1,\ldots,\mu_{k-r}$, $\lambda_1,\ldots,\lambda_{s-1}$ and $p>0$ (see Proposition 22):

$$\widetilde{R}_{\mu_1,\ldots,\mu_{k-r};\lambda_1,\ldots,\lambda_{s-1},-p} = -\sum_{\substack{i_1,\ldots,i_{k-r}\geq 0 \\ I=i_1+\ldots+i_{k-r}>0}} \widetilde{R}_{\mu_1-i_1,\ldots,\mu_{k-r}-i_{k-r};\lambda_1,\ldots,\lambda_{s-1},I-p} \tag{18}$$

Let us introduce the following change of notation. We replace the tuple $(\mu_1,\ldots\mu_{k-r})$ with the tuple $(\widetilde{\mu}_1,\ldots,\widetilde{\mu}_r)$ such that $\{\mu_1,\ldots,\mu_{k-r}\}\bigsqcup\{\widetilde{\mu}_1,\ldots,\widetilde{\mu}_r\} = \{0,1,\ldots,k-1\}$. Now, denote by $\widehat{R}_{\mu_1,\ldots,\mu_r;\lambda_1,\ldots,\lambda_s}$ the polynomial $\widetilde{R}_{k-1-\widetilde{\mu}_{k-r},\ldots,k-1-\widetilde{\mu}_1;\lambda_1,\ldots,\lambda_s}$.

Lemma 7. *We have the following relations:*

$$\widehat{R}_{\mu_1,\ldots,\mu_r;\lambda_1,\ldots,\lambda_{s-1},-1} = -\sum_{\substack{\delta_1,\ldots,\delta_r=0,1 \\ \Delta=\delta_1+\ldots+\delta_r>0}} \widehat{R}_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda_1,\ldots,\lambda_{s-1},\Delta-1} \tag{19}$$

Proof. We are going to show that relations (18) imply relations (19). In formula (18), the sum is taken over all tuples $(i_1,\ldots,i_{k-r})$ of non-negative integers with positive sum. However, it suffices to take the sum over all tuples $(i_1,\ldots,i_{k-r})$ such that $\mu_1 \geqslant \mu_1 - i_1 \geqslant \mu_2 \geqslant \mu_2 - i_2 \geqslant \ldots \geqslant \mu_{k-r} \geqslant \mu_{k-r} - i_{k-r}$ (assuming that $\mu_1 > \mu_2 > \ldots > \mu_{k-r}$). Indeed, it is not difficult to see that the summands corresponding to other tuples mutually cancel. But any such shift of the indices $\mu_1,\mu_2,\ldots,\mu_{k-r}$ is equivalent to shifting some of the indices $\widetilde{\mu}_1,\widetilde{\mu}_2,\ldots,\widetilde{\mu}_r$ by 1, and vice versa. This completes the proof. □

The operator L_1 acts on the new polynomials $\widehat{R}_{\mu,\lambda}$ as follows:

$$L_1(\widehat{R}_{\mu,\lambda}) = -\sum_{t=1}^{r}(\mu_t - k + r)\widehat{R}_{\mu_1,\ldots,\mu_t-1,\ldots,\mu_r;\lambda} + \sum_{t=1}^{s}(\lambda_t - r + s)\widehat{R}_{\mu;\lambda_1,\ldots,\lambda_t-1,\ldots,\lambda_s}.$$

Finally, if we define $R_{\mu,\lambda}$ as $R_{\mu,\lambda} = (-1)^{\mu_1+\ldots+\mu_r}\widehat{R}_{\mu;\lambda}$, then we have

$$L_1(R_{\mu,\lambda}) = \sum_{t=1}^{r}(\mu_t - k + r)R_{\mu_1,\ldots,\mu_t-1,\ldots,\mu_r;\lambda} + \sum_{t=1}^{s}(\lambda_t - r + s)R_{\mu;\lambda_1,\ldots,\lambda_t-1,\ldots,\lambda_s} \tag{20}$$

and

$$R_{u;\lambda_1,\ldots,\lambda_{s-1},-p} = \sum_{\substack{\delta_1,\ldots,\delta_r=0,1\\ \Delta=\delta_1+\ldots+\delta_r>0}} (-1)^{\Delta-1}R_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda_1,\ldots,\lambda_{s-1},\Delta-p} \tag{21}$$

for any $p < 0$.

It is clear that the polynomials $R_{\mu_1,\ldots,\mu_r;\lambda_1,\ldots,\lambda_s}$ such that $k-1 \geqslant \mu_1 > \ldots > \mu_r \geqslant 0$ and $r-1 \geqslant \lambda_1 > \ldots > \lambda_s \geqslant 0$ form a basis of the space $W^{k\omega} \cap \pi_{k\omega-(k-r)a-sb}$. In order to compute the vector bundles corresponding to this space, we are going to do a certain change of basis. We start with a description of this change in the case $s = 1$. Denote by $S_{\mu;\lambda}$ (in the particular case $s = 1$, λ is just one number) the polynomials defined by the following formulas:

$$\begin{aligned} R_{\mu;\lambda} &= S_{\mu;\lambda} \text{ if } \lambda = r-1,\\ R_{\mu;\lambda} &= \sum_{\substack{j_1,\ldots,j_r\geqslant 0\\ J=j_1+\ldots+j_r\leqslant r-2-\lambda}} S_{\mu_1-j_1,\ldots,\mu_r-j_r;\lambda+J} \text{ if } \lambda \leqslant r-2. \end{aligned} \tag{22}$$

These formulas clearly define an invertible change of basis of the space $W^{k\omega}\cap\pi_{k\omega-(k-r)a-sb}$.

Lemma 8. *The action of L_1 on $S_{\mu,\lambda}$ is given by*

$$L_1(S_{\mu_1,\ldots,\mu_r;\lambda}) = \sum_{i=1}^{r}(\mu_i - k + r)S_{\mu_1,\ldots,\mu_i-1,\ldots,\mu_r;\lambda} \text{ if } \lambda = r-1$$

$$L_1(S_{\mu_1,\ldots,\mu_r;\lambda}) = \sum_{i=1}^{r}(\mu_i - k + r - 1)S_{\mu_1,\ldots,\mu_i-1,\ldots,\mu_r;\lambda} - (r-\lambda-1)S_{\mu;\lambda-1} \text{ if } \lambda \leqslant r-2.$$

Proof. We use descending induction on λ. The base $\lambda = r-1$ is clear since the corresponding polynomials are the «old» polynomials $R_{\mu;\lambda}$. The step is an explicit computation:

$$L_1(S_{\mu;r-l}) = L_1(R_{\mu;r-l} - \sum_{0<J=j_1+\ldots+j_r\leqslant l-2} S_{\mu_1-j_1,\ldots,\mu_r-j_r;r-l+J}) =$$

$$= \sum_{i=1}^{r}(\mu_i - k + r)S_{\mu_1,\ldots,\mu_i-1,\ldots,\mu_r;r-l}+$$

$$+ \sum_{\substack{0<J\leqslant l-2\\ 1\leqslant i\leqslant r}}(\mu_i - k + r)S_{\mu_1-j_1,\ldots,\mu_i-j_i-1,\ldots,\mu_r-j_r;r-l+J} - (l-1)\sum_{0\leqslant J\leqslant l-1} S_{\mu_1-j_1,\ldots,\mu_r-j_r;r-l+J-1}-$$

$$-\sum_{0<J\leqslant l-2}\left(\sum_{i=1}^{r}(\mu_i-j_i-k+r-1)S_{\mu_1-j_1,\ldots,\mu_s-j_s-1,\ldots,\mu_r-j_r;r-l+J}+(J-l+1)S_{\mu_1-j_1,\ldots,\mu_r-j_r;r-l+J-1}\right)$$

We need to compute the coefficient of $S_{\mu_1-c_1,\ldots,\mu_r-c_r;r-l+c_1+\ldots+c_r-1}$ in this expression. First, consider the case $c_1+\ldots+c_r \geqslant 2$. Without loss of generality, assume that $c_1,\ldots,c_t > 0$ and $c_{t+1} = \ldots = c_r = 0$. Then, the coefficient of $S_{\mu_1-c_1,\ldots,\mu_r-c_r;r-\ldots}$ is equal to

$$\sum_{i=1}^{t}(\mu_i - k + r) - (l-1) - \sum_{i=1}^{t}(\mu_i - c_i - k + r) - (c_1+\ldots+c_t - l + 1) = 0.$$

If $c_1 + \ldots + c_r = 1$, then (when $c_i = 1$) this coefficient is equal to

$$\mu_i - k + r - (l-1) = \mu_i - k + r - 1.$$

Finally, if $c_1 = \ldots = c_r = 0$, then this coefficient is $-(l-1)$. This implies the stated formula. $\square$

Also, we need an expression of $S_{\mu;-1}$ in terms of $S_{\mu,\lambda}$ with $\lambda \geqslant 0$.

Lemma 9. *We have* $S_{\mu_1,\ldots,\mu_r;-1} = (-1)^{r-1} S_{\mu_1-1,\ldots,\mu_r-1;r-1}$

Proof. We have

$$S_{\mu;-1} = R_{\mu;-1} - \sum_{0<J\leqslant r-1} S_{\mu_1-j_1,\ldots,\mu_r-j_r;J-1} =$$

$$= \sum_{\substack{\delta_1,\ldots,\delta_r=0,1\\ \Delta=\delta_1+\ldots+\delta_r>0}} (-1)^{\Delta-1} R_{\mu_1-\delta_1,\ldots\mu_r-\delta_r;\Delta-1} - \sum_{0<J\leqslant r-1} S_{\mu_1-j_1,\ldots,\mu_r-j_r;J-1} =$$

$$= (-1)^{r-1} S_{\mu_1-1,\ldots,\mu_r-1;r-1} + \sum_{\substack{\Delta>0\\ r-1\geqslant J>0}} (-1)^{\Delta-1} S_{\mu_1-\delta_1-j_1,\ldots,\mu_r-\delta_r-j_r;\Delta+J-1} - \sum_{0<J\leqslant r-1} S_{\mu_1-j_1,\ldots,\mu_r-j_r;J-1}$$

If $c_1,\ldots,c_t > 0$ $(t>0)$ and $c_{t+1} = \ldots = c_r = 0$, then the number of occurrences of $S_{\mu_1-c_1,\ldots\mu_r-c_r}$ with signs $+$ and $-$ are both equal to $\frac{t!}{2}$. So, the only remaining summand is $(-1)^{r-1} S_{\mu_1-1,\ldots,\mu_r-1;r-1}$ and we are done. $\square$

In the case of general s, it is convenient to temporarily forget about anticommutativity of $R_{\mu_1,\ldots,\mu_r;\lambda_1,\ldots,\lambda_s}$ in $\mu_1,\ldots,\mu_r$ and $\lambda_1,\ldots,\lambda_s$ and consider these expressions formally. The change of basis we are going to do can be described in several steps. The first step is the following:

$$R_{\mu;\lambda_1,\ldots,\lambda_{s-1},\lambda_s} = R^s_{\mu;\lambda_1,\ldots,\lambda_{s-1},\lambda_s} \text{ if } \lambda_s \geqslant r-s,$$

$$R_{\mu;\lambda_1,\ldots,\lambda_{s-1},\lambda_s} = \sum_{\substack{j_1,\ldots,j_r\geqslant 0\\ J=j_1+\ldots+j_r\leqslant r-s-1-\lambda_s}} R^s_{\mu_1-j_1,\ldots,\mu_r-j_r;\lambda_1,\ldots,\lambda_{s-1},\lambda_s+J} \text{ if } \lambda_s \leqslant r-s-1.$$

Note that this change is similar to the one we did for $s=1$. Informally, we can call an index λ_i «big» if $\lambda_i \geqslant r-s$ and «small» otherwise. This terminology comes from formula (20): if λ_i is «big», then $L_1(R_{\mu;\lambda})$ is a linear combination of polynomials $R_{\mu';\lambda'}$ such that all indices λ'_i are also «big». The same is true for formulas (22). In these terms, the formula we just described is similar. Indeed, we call λ_s «big» if $\lambda_s \geqslant r-s$. Then, the polynomials $R_{\mu;\lambda}$ and $R^s_{\mu;\lambda}$ are equal if λ_s is «big». Otherwise, $R_{\mu;\lambda}$ is a sum of $R^s_{\mu';\lambda'}$ with $\lambda'_s \geq \lambda_s$ as in (22); the inequality $J \leqslant r-s-1-\lambda_s$ means that all λ'_s are «small». The indices $\lambda_1,\ldots,\lambda_{s-1}$ can be viewed as parameters. It can be shown in the same way as in the proof of Lemma 8 that the action of L_1 on $R^s_{\mu;\lambda}$ is given by the following formulas:

$$L_1(R^s_{\mu;\lambda}) = \sum_{i=1}^{r} (\mu_i - k + r) R^s_{\mu_1,\ldots,\mu_i-1,\ldots,\mu_r;\lambda} + \sum_{i=1}^{s} (\lambda_i - r + s) R^s_{\mu;\lambda_1,\ldots,\lambda_i-1,\ldots,\lambda_s} \text{ if } \lambda \geqslant r-s,$$

$$L_1(R^s_{\mu;\lambda}) = \sum_{i=1}^{r} (\mu_i - k + r - 1) R^s_{\mu_1,\ldots,\mu_i-1,\ldots,\mu_r;\lambda} + \sum_{i=1}^{s} (\lambda_i - r + s) R^s_{\mu;\lambda_1,\ldots,\lambda_i-1,\ldots,\lambda_s} \text{ if } \lambda < r-s.$$

Indeed, the operator L_1 can be written as the sum $L_1' + L_1''$, where

$$L_1'(R_{\mu,\lambda}) = \sum_{t=1}^{r}(\mu_t - k + r)R_{\mu_1,\ldots,\mu_t-1,\ldots,\mu_r;\lambda} + (\lambda_s - r + s)R_{\mu;\lambda_1,\ldots,\lambda_{s-1},\lambda_s-1}$$

and

$$L_1''(R_{\mu,\lambda}) = \sum_{t=1}^{s-1}(\lambda_t - r + s)R_{\mu;\lambda_1,\ldots,\lambda_t-1,\ldots,\lambda_s}.$$

The action of L_1' is similar to the action of L_1 for $s = 1$ if λ_i, $1 \leqslant i \leqslant s-1$ are viewed as parameters. So, after our change of basis, the action of L_1' is transformed in the same way as in Lemma 8 (this can be proved in the same way as Lemma 8). The fact that the action of L_1'' in the basis $(R^s_{\mu;\lambda})$ is given by precisely the same formulas as in the basis $(R_{\mu;\lambda})$ can be proved even easier by descending induction on λ_s. So, the action of L_1 on $R^s_{\mu;\lambda}$ is given by the formulas above.

Then, we move to polynomials $R^i_{\mu;\lambda}$, $i = s-1, s-2, \ldots, 1$ inductively. To be precise, if $R^i_{\mu;\lambda}$ are already defined, then

$$R^i_{\mu;\lambda} = R^{i-1}_{\mu;\lambda} \text{ if } \lambda_i \leqslant r - s,$$
$$R^i_{\mu;\lambda} = \sum_{\substack{j_1,\ldots,j_r \geqslant 0 \\ J=j_1+\ldots+j_r \leqslant r-s-1-\lambda_i}} R^{i-1}_{\mu_1-j_1,\ldots,\mu_r-j_r;\lambda_1,\ldots,\lambda_i+J,\ldots,\lambda_s} \text{ if } \lambda_i \leqslant r-s-1.$$

Denote by $S_{\mu,\lambda}$ the polynomials $R^1_{\mu,\lambda}$. It can be shown in the same way as above (for $i = s$) that the action of L_1 on $S_{\mu,\lambda}$ is given by the following formula:

$$L_1(S_{\mu;\lambda}) = \sum_{i=1}^{r}(\mu_i - k + r - p)S_{\mu_1,\ldots,\mu_i-1,\ldots,\mu_r;\lambda} + \sum_{i=1}^{s}(\lambda_i - r + s)S_{\mu;\lambda_1,\ldots,\lambda_i-1,\ldots,\lambda_s}\ , \quad (23)$$

where p is the number of λ_i such that $\lambda_i < r - s$.

Also, we have the following relations:

$$S_{\mu;\lambda_1,\ldots,\lambda_{s-1},-1} = \sum_{\substack{\delta_1,\ldots,\delta_r=0,1 \\ \Delta \geqslant r-s+1}} (-1)^{\Delta-1} S_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda_1,\ldots,\lambda_{s-1},\Delta-1} \quad (24)$$

Indeed, one can show as in the proof of Lemma 9 that these relations hold if we replace S with R^s. After that, it is easy to see that these relations remain unchanged when we pass from R^i to R^{i-1}, $i = s, s-1, \ldots, 2$.

Assume that $\lambda_1 > \ldots > \lambda_{s-p} \geqslant r - s$ and $r - s > \lambda_{s-p+1} > \ldots > \lambda_s$. Then, one can write the following explicit transition formulas from $S_{\mu;\lambda}$ to $R_{\mu;\lambda}$:

$$R_{\mu;\lambda} = \sum_{\substack{\lambda_{s-p+i}+J_i<r-s \\ i=1,\ldots,p}} S_{\mu_1-J^1,\ldots,\mu_r-J^r;\lambda_1,\ldots,\lambda_{s-p},\lambda_{s-p+1}+J_1,\ldots,\lambda_s+J_p}. \quad (25)$$

The sum is taken over all possible matrices of non-negative integers of the form

$$\begin{pmatrix} j_1^1 & j_1^2 & \ldots & j_1^r \\ j_2^1 & j_2^2 & \ldots & j_2^r \\ \vdots & \vdots & \ddots & \vdots \\ j_p^1 & j_p^2 & \ldots & j_p^r \end{pmatrix},$$

$J_i = \sum_{t=1}^r j_i^t$ is the sum of all elements in the i-th row and $J^i = \sum_{t=1}^p j_t^i$ is the sum of all elements in the i-th column. Of course, similar formulas hold without the assumption $\lambda_1 > \ldots > \lambda_s$. We can see that this agrees with anticommutativity of the basis elements in μ and λ (although this is not true for $R^i_{\mu,\lambda}$, $i = 2, \ldots, s$). So, these formulas define a change of basis in $W^{k\omega} \cap \pi_{k\omega-(k-r)a-sb}$.

From now on, we assume again that $k-r \geqslant \mu_1 > \ldots > \mu_r \geqslant 0$ and $r-s \geqslant \lambda_1 > \ldots > \lambda_s \geqslant 0$. The space $W^{k\omega} \cap \pi_{k\omega-(k-r)a-sb}$ is the direct sum of its subspaces

$$U_{d_1,d_2} = \mathrm{Span}(S_{\mu;\lambda} \mid \mu_{r-d_1} \geqslant k-r+d_2 > \mu_{r-d_1+1}, \lambda_{s-d_2} \geqslant r-s > \lambda_{s-d_2+1}).$$

In other words, U_{d_1,d_2} is the linear span of $S_{\mu;\lambda}$ with $r-d_1$ «big» indices λ_i and $s-d_2$ «big» indices μ_i (again, the terminology «big» and «small» comes from formula (23)). Note that the subspaces U_{d_1,d_2} are not invariant under the L_1-action. Indeed, if $\lambda_s = 0$, then the expression for $L_1(S_{\mu;\lambda})$ contains the summand $S_{\mu;\lambda_1,\ldots,\lambda_{s-1},-1}$ which is a linear combination (24) of elements not contained in U_{d_1,d_2}. However, we can introduce the following lexicographical order on the set of pairs (d_1, d_2): we say that $(d_1, d_2) \preceq (d_1', d_2')$ if either $d_1 < d_1'$ or $d_1 = d_1'$ and $d_2 < d_2'$. Then, one can see from (23) and (24) that if $S_{\mu;\lambda} \in U_{d_1,d_2}$, then $L_1(S_{\mu;\lambda})$ is a linear combination of $S_{\mu';\lambda'}$ lying in $U_{d_1',d_2'}$ such that $(d_1', d_2') \preceq (d_1, d_2)$.

First, we are going to compute the vector bundle corresponding to the space $\widetilde{U}$ with a basis $\widetilde{S}_{\mu,\lambda}$, $k-r \geqslant \mu_1 > \ldots > \mu_r \geqslant 0$, $r-1 \geqslant \lambda_1 > \ldots > \lambda_s \geqslant 0$ equipped with the L_1-action given by (23) but with the assumption $\widetilde{S}_{\mu;\lambda_1,\ldots,\lambda_{s-1},-1} = 0$. After that, we will show that the answer for $W^{k\omega} \cap \pi_{k\omega-(k-r)a-sb}$ is the same.

The space $\widetilde{U}$ is the direct sum of its subspaces $\widetilde{U}_{d_1,d_2}$ defined similarly to U_{d_1,d_2}:

$$\widetilde{U}_{d_1,d_2} = \mathrm{Span}(\widetilde{S}_{\mu;\lambda} \mid \mu_{r-d_1} \geqslant k-r+d_2 > \mu_{r-d_1+1}, \lambda_{s-d_2} \geqslant r-s > \lambda_{s-d_2+1}).$$

Since $\widetilde{S}_{\mu;\lambda_1,\ldots,\lambda_{s-1},-1} = 0$, all subspaces $\widetilde{U}_{d_1,d_2}$ are invariant under the L_1-action. This means that the whole vector bundle is the direct sum of all vector bundles corresponding to subspaces $\widetilde{U}_{d_1,d_2}$. Moreover, the structure of these subspaces is the same as the structure of the spaces from Section 4 (i.e., they are isomorphic to tensor products of exterior powers of single Jordan blocks). So, we need to know only $\dim \widetilde{U}_{d_1,d_2}$ and the maximal and the minimal L_0-degrees of vectors in $\widetilde{U}_{d_1,d_2}$ to compute the corresponding vector bundle.

The vectors of maximal and minimal degree in $\widetilde{U}_{d_1,d_2}$ are

$$\widetilde{S}_{k-1,\ldots,k-r+d_1,k-r+d_2-1,\ldots,k-r+d_2-d_1;r-1,\ldots,r-s+d_2;r-s-1,\ldots,r-s-d_2}$$

and

$$\widetilde{S}_{k-1-d_1+d_2,\ldots,k-r+d_2,d_1-1,\ldots,0;r-1-d_2,\ldots,r-s;d_2-1,\ldots,0}$$

respectively. Up to adding $2N$, $N \in \mathbb{Z}$, their sum is equal to

$$\big((k-1)+\ldots+(k-r+d_1)\big) + \big((k-r+d_2-1)+\ldots+(k-r+d_2-d_1)\big) +$$
$$+\big((r-1)+\ldots+(r-s+d_2)\big) + \big((r-s-1)+\ldots+(r-s-d_2)\big) +$$
$$+\big((k-1-d_1+d_2)+\ldots+(k-r+d_2)\big) + \big((d_1-1)+\ldots+0\big) +$$
$$+\big((r-1-d_2)+\ldots+(r-s)\big) + \big((d_2-1)+\ldots+0\big) =$$
$$= (2k-r+d_2-1)(r-d_1) + (k-r+d_2-1)d_1 + (2r-s-1)(s-d_2) + (r-s-1)d_2 =$$

$$2kr - r^2 - r - s^2 - s + 2sr - kd_1 = C(k,r,s) - kd_1,$$

where $C(k,r,s) = 2kr - r^2 - r - s^2 - s + 2sr$ is independent of d_1.

The dimension of $\widetilde{U}_{d_1,d_2}$ is just the number of choices of $(\lambda;\mu)$ satisfying the definition of $\widetilde{U}_{d_1,d_2}$ and is equal to

$$\binom{r-d_2}{r-d_1}\binom{k-r+d_2}{d_1}\binom{s}{s-d_2}\binom{r-s}{d_2}.$$

To compute the vector bundle corresponding to the whole $\widetilde{U}$, we need to sum over all pairs (d_1,d_2). So, this bundle is isomorphic to

$$\bigoplus_{d_1\geqslant d_2\geqslant 0} \mathcal{O}(2N-d_1k)^{\oplus\binom{r-d_2}{r-d_1}\binom{k-r+d_2}{d_1}\binom{s}{s-d_2}\binom{r-s}{d_2}} \tag{26}$$

Now, we are going to show that U and $\widetilde{U}$ give rise to isomorphic vector bundles. To do this, consider a more general situation. Let V be a finite-dimensional vector space equipped with an L_1-action. Suppose that V is a direct sum of subspaces $V = \bigoplus_{i=1}^N V_i$ such that $L_1(V_i) \subset \bigoplus_{j=1}^i V_i$ for any $i = 1,\dots,N$. Write $L_1|_{V_i} = L_1'|_{V_i} + L_1''|_{V_i}$, where the operators $L_1'|_{V_i}$ and $L_1''|_{V_i}$ map V_i to V_i and $\bigoplus_{j=1}^{i-1} V_i$ respectively. In addition, suppose that the Jordan decomposition of each V_i with respect to $L_1'|_{V_i}$ is L_0-«symmetric» in the sense of Section 4. This means that the sum of the maximal and minimal L_0-degrees of vectors in each Jordan block of V_i is the same. Moreover, denoting this sum by D_i for V_i, suppose that $D_1 \geqslant D_2 \geqslant \ldots \geqslant D_N$. The, we have the following proposition.

Proposition 24. *The vector bundles corresponding to the action of L_1 and L_1' on V are isomorphic.*

Proof. We are going to repeat the proof of Proposition 9 with a slight modification. First, following the procedure from that proof, we can construct a Jordan decomposition of V_1 (note that L_1 maps V_1 to itself). Then, we start constructing a Jordan decomposition of V_2 with respect to L_1. For this, we take a vector $v \in V_2$ of maximal L_0-degree and consider the vectors $L_1(v)$, $L_1^2(v)$, ... Suppose that k is the minimal number such that $L_1^{k+1}(v)$ lies in the linear span of the Jordan subspaces we already constructed (in this case, this is V_1). Then, $L_1^{k+1}(v)$ is a linear combination of vectors lying in Jordan blocks J_1, J_2, ..., J_s. But, since $D_2 \leqslant D_1$, the maximal L_0-degrees of vectors in J_1, J_2, ..., J_s cannot be less than the L_0-degree of v. So, we can replace v with $v - v'$, $v' \in V_1$, $\deg_{L_0}(v) = \deg_{L_0}(v')$ such that $L_1^{k+1}(v - v') = 0$. After that, we again take a vector of maximal degree lying in V_2 and not lying in the sum of the already constructed Jordan subspaces, and so on. This way, after several iterations of the process, we construct a basis of $V_1 \oplus V_2$. Next, we repeat this procedure for V_3, V_4, ..., V_N. At the end, we obtain a decomposition of V into a direct sum of Jordan blocks with respect to L_1. The sizes and L_0-degrees of these Jordan blocks coincide with the ones for L_1'. So, the vector bundles corresponding to L_1 and L_1' are isomorphic. □

In our situation, as we already observed, L_1 sends each subspace U_{d_1,d_2} to the sum of subspaces $U_{d_1',d_2'}$ such that $(d_1',d_2') \preceq (d_1,d_2)$. Moreover, the sum of the maximal and minimal L_0-degrees of vectors in U_{d_1,d_2} is $2N - kd_1$ as we computed above. It is easy to see now that the conditions of Proposition 24 are satisfied for the space $W^{k\omega} \cap \pi_{k\omega-(k-r)a-sb}$. So, the vector bundle corresponding to this space is given by (26). Therefore, the bundle corresponding to the whole Demazure module $W^{k\omega}$ is the direct sum of such bundles for all possible r,s. We state this result as a proposition.

Proposition 25. *The vector bundle corresponding to the Demazure module of level* 1 $W^{k\omega}$ *for* $\mathfrak{sl}(3,\mathbb{C})$ *is isomorphic to*

$$\bigoplus_{\substack{r\geqslant s\geqslant 0\\ d_1\geqslant d_2\geqslant 0}} \mathcal{O}(2N-d_1k)^{\oplus\binom{r-d_2}{r-d_1}\binom{k-r+d_2}{d_1}\binom{s}{s-d_2}\binom{r-s}{d_2}} \tag{27}$$

In the remaining part of this section, we simplify this formula. Namely, we prove the following proposition.

Proposition 26. *The vector bundle corresponding to the Demazure module of level 1* $W^{k\omega}$ *for* $\mathfrak{sl}(3,\mathbb{C})$ *is isomorphic to*

$$\bigoplus_{\substack{r\geqslant s\geqslant 0\\ d_1\geqslant d_2\geqslant 0}} \mathcal{O}(2N-d_1k)^{\oplus\sum_{i\geqslant 0}\binom{k+1}{3d_1+2-i}\binom{3d_1+2-i}{i}}.$$

Proof. We need to prove that the following identity holds:

$$\sum_{r,s,d_2}\binom{r-d_2}{r-d_1}\binom{k-r+d_2}{d_1}\binom{s}{s-d_2}\binom{r-s}{d_2}=\sum_{i\geqslant 0}\binom{k+1}{3d_1+2-i}\binom{3d_1+2-i}{i}. \tag{28}$$

Let us work a bit with the left-hand side of this identity. First, we know that

$$\sum_{s}\binom{s}{s-d_2}\binom{r-s}{d_2}=\binom{r+1}{2d_2+1}$$

(see the end of Section 5). So, the left-hand side of (28) turns to

$$\sum_{r,d_2}\binom{r-d_2}{r-d_1}\binom{k-r+d_2}{d_1}\binom{r+1}{2d_2+1}.$$

Next, we have the following well-known identities:

$$\binom{r-d_2}{r-d_1}=\sum_{p=0}^{d_2}\binom{r-2d_2}{d_1-d_2-p}\binom{d_2}{p}$$

and

$$\binom{r+1}{2d_2+1}\binom{r-2d_2}{d_1-d_2-p}=\binom{r+1}{d_1+d_2-p+1}\binom{d_1+d_2-p+1}{2d_2+1}.$$

Applying them, we transform our sum to

$$\sum_{r,d_2,p}\binom{k-r+d_2}{d_1}\binom{r+1}{d_1+d_2-p+1}\binom{d_2}{p}\binom{d_1+d_2-p+1}{2d_2+1}=$$

$$=\sum_{d_2,p}\binom{k+d_2+2}{k-2d_1+p}\binom{d_2}{p}\binom{d_1+d_2-p+1}{2d_2+1}$$

(to sum over r, we applied a generalized version of the identity from the end of Section 5). Next, we have

$$\binom{k+d_2+2}{k-2d_1+p}=\sum_{q}\binom{k+1}{k-2d_1+p-q}\binom{d_2+1}{q},$$

so our sum turns to

$$\sum_{p,q,d_2}\binom{k+1}{k-2d_1+p-q}\binom{d_2+1}{q}\binom{d_2}{p}\binom{d_1+d_2-p+1}{2d_2+1}.$$

Comparing the coefficients of $\binom{k+1}{k-2d_1+p-q}$ in this expression and the right-hand side of the identity we are proving, we see that it suffices to prove the following (we denote $a=q-p$)

$$\sum_{p,d_2}\binom{d_2+d_1-p+1}{2d_2+1}\binom{d_2+1}{p+a}\binom{d_2}{p}=\binom{2d_1+a+1}{d-a+1}. \tag{29}$$

The left-hand side of (29) is equal to the coefficient of x^{2d_2+1} in the expression

$$\sum_{p,d_2}\binom{d_2}{d_2-p}\binom{d_2+1}{p+a}(1+x)^{d_2+d_1-p+1}=(1+x)^{d_1+1}\sum_{p,d_2}\left(\binom{d_2}{d_2-p}\binom{d_2+1}{p+a}(1+x)^{d_2-p}\right).$$

The last sum is the coefficient of y^{d_2+a} in $\sum_{d_2}(1+y(1+x))^{d_2}(1+y)^{d_2+1}$. At the same time, we have

$$(1+y(1+x))^{d_2}(1+y)^{d_2+1}=\sum_i((1+y)^{d_2-i}\binom{d_2}{d_2-i}x^iy^i)(1+y)^{d_2+1}=$$

$$=\sum_i(1+y)^{2d_2+1-i}x^iy^i\binom{d_2}{i},$$

so the left-hand side of (29) is the coefficient of $x^{2d_2+1}y^{d_2+a}$ in

$$\sum_{d_2,i}(1+x)^{d_1+1}(1+y)^{2d_2+1-i}x^iy^i\binom{d_2}{i}.$$

On the other hand, the coefficient of $x^{2d_2+1}y^{d_2+a}$ is (denoting $j=2d_2+1-i$)

$$\sum_{j,d_2}\binom{d_1+1}{j}\binom{d_2}{2d_2+1-j}\binom{j}{a-d_2+j-1}.$$

So, denoting $q=j-d_2-1$, we need to prove now that

$$\sum_{q,d_2}\binom{d_1+1}{q+d_2+1}\binom{q+d_2+1}{q+a}\binom{d_2}{d_2-q}=\binom{2d_1+a+1}{d_1-a+1}.$$

We have

$$\binom{d_1+1}{q+d_2+1}\binom{q+d_2+1}{q+a}=\binom{d_1-d_2+a}{q+a}\binom{d_1+1}{d_2-a+1},$$

and we need to prove that

$$\sum_{q,d_2}\binom{d_1-d_2+a}{q+a}\binom{d_2}{q}\binom{d_1+1}{d_2-a+1}=\binom{2d_1+a+1}{d_1-a+1}.$$

It follows from a well-known combinatorial identity that

$$\sum_q\binom{d_1-d_2+a}{q+a}\binom{d_2}{q}=\sum_q\binom{d_1-d_2+a}{d_1-d_2-q}\binom{d_2}{q}=\binom{d_1+a}{d_1-d_2}$$

and similarly

$$\sum_{d_2} \binom{d_1+1}{d_2-a+1}\binom{d_1+a}{d_1-d_2} = \binom{2d_1+a+1}{d_1-a+1}$$

which proves the proposition. □

9 Explicit computation of vector bundles corresponding to Demazure modules of level 1 for $\mathfrak{sl}(n,\mathbb{C})$

In this section, we compute the vector bundles associated with the Demazure modules $W^{k\omega}$ for $\mathfrak{sl}(n,\mathbb{C})$ in case of arbitrary n. In this case, however, instead of working with the polynomials from Proposition 21, we will consider another basis of $W^{k\omega}$. This basis does not have such a compact form as (14) and the formulas for its elements are rather bulky but it will be more convenient for computation of the corresponding vector bundles. So, we start with describing this basis.

Recall that the space $W^{k\omega}$ can be obtained from $|k\omega\rangle$ by applying all possible operators $V_{-\alpha^i}[m]$, $m \geqslant 0$. So, all elements of $W^{k\omega}$ are polynomials in α^i_{-s}, $s>0$, $i=1,2,\dots,n-1$. We are going to describe these polynomials explicitly.

As before, denote by P_t polynomials (7). Then, we have the following statement similar to Lemmas 3, 4 and 5.

Lemma 10. *For any $i=1,2,\dots,n-1$*

$$\widetilde{V}_{-\alpha^i}(z)(P_t(\alpha^i)) = (P_t(\alpha^i) - 2z^{-1}P_{t-1}(\alpha^i) + z^{-2}P_{t-2}(\alpha^i))\widetilde{V}_{-\alpha^i}(z),$$

for any $i,j=1,2,\dots,n-1$, $|i-j|=1$

$$\widetilde{V}_{-\alpha^i}(z)(P_t(\alpha^j)) = (P_t(\alpha^j) + z^{-1}P_{t-1}(\alpha^j) + z^{-2}P_{t-2}(\alpha^j) + \dots)\widetilde{V}_{-\alpha^i}(z),$$

and for any $i,j=1,2,\dots,n-1$, $|i-j|\geqslant 2$

$$\widetilde{V}_{\alpha^i}(z)(P_t(\alpha^j)) = (P_t(\alpha^j))\widetilde{V}_{-\alpha^i}(z),$$

Proof. The same as the proof of Lemma 3. Here, we need to use the fact that $(\alpha_i,\alpha_i)=2$, $(\alpha_i,\alpha_{i+1})=-1$ and $(\alpha_i,\alpha_j)=0$ if $|i-j|\geqslant 2$. □

Using this lemma, we can prove inductively in the same way as in Sections 4, 5 and 7 the following proposition.

Proposition 27. *$W^{k\omega}\cap\pi_{k\omega-r_1\alpha^1+\dots-r_{n-1}\alpha^{n-1}}$ is a linear combination of polynomials of the form*

$$R_{\lambda^1;\lambda^2;\dots;\lambda^{n-1}} = \sum P^1_{\lambda^1_1-A^1_1,\dots,\lambda^1_{r_1}-A^1_{r_1}}\cdot P^2_{\lambda^2_1+B^1_1-A^2_1,\dots,\lambda^2_{r_2}+B^1_{r_2}-A^2_{r_2}}\cdot$$
$$\cdot P^3_{\lambda^3_1+B^2_1-A^3_1,\dots,\lambda^3_{r_3}+B^2_{r_3}-A^3_{r_3}}\cdot\ldots\cdot P^{n-1}_{\lambda^{n-1}_1+B^{n-2}_1,\dots,\lambda^{n-1}_{r_{n-1}}+B^{n-2}_{r_{n-1}}},$$

$\lambda^j_i\leqslant r_{j-1}-1$ (assuming that $r_0=k$). Here, by λ^i we denote a tuple $(\lambda^i_1,\lambda^i_2,\dots,\lambda^i_{r_i})$ of integer numbers not exceeding $k-1$. The notation P^i_λ means that the polynomial P_λ depends on the variables α^i_{-s}, $s>0$. The sum is taken over all matrices of the form

$$C_i = \begin{pmatrix} c^i_{1,1} & c^i_{1,2} & \dots & c^i_{1,r_i} \\ c^i_{2,1} & c^i_{2,2} & \dots & c^i_{2,r_i} \\ \vdots & \vdots & \ddots & \vdots \\ c^i_{r_{i+1},1} & c^i_{r_{i+1},2} & \dots & c^i_{r_{i+1},r_i} \end{pmatrix},$$

$c^i_{t_1,t_2} \in \mathbb{Z}_{\geqslant 0}$, $i = 1, 2, \ldots, n-2$. *The numbers* $B^i_s = \sum_{t=1}^{r_i} c^i_{s,t}$ *and* $A^i_s = \sum_{t=1}^{r_{i+1}} c^i_{t,s}$ *are the sums of the elements of rows and columns of these matrices respectively.*

Proof. To avoid bulky notation, we sketch a proof for $n = 3$. Denote by α, β the two simple roots. Also, denote by P_μ, Q_λ the polynomials P_μ, P_λ depending on α_{-s}, β_{-s} respectively. Then, we need to show that $W^{k\omega} \cap \pi_{k\omega - r\alpha - s\beta}$ is linearly spanned by the polynomials of the form

$$R_{\mu_1,\ldots,\mu_r;\lambda_1,\ldots,\lambda_s} = \sum P_{\mu_1 - A_1,\ldots,\mu_r - A_r} Q_{\lambda_1 + B_1,\ldots,\lambda_s + B_s},$$

where A_i, B_j are the sums of the elements of the i-th row and the j-th column of the matrix

$$\begin{pmatrix} c_{1,1} & c_{1,2} & \ldots & c_{1,r} \\ c_{2,1} & c_{2,2} & \ldots & c_{2,r} \\ \vdots & \vdots & \ddots & \vdots \\ c_{s,1} & c_{s,2} & \ldots & c_{s,r} \end{pmatrix}$$

whose elements are non-negative integers. Suppose that the statement is true for (r, s). Then, applying $V_{-\beta}(z)$ to any such polynomial, we obtain a linear combination of the following expressions:

$$\sum P_{\mu_1 - A_1 - c_{s+1,1},\ldots,\mu_r - A_r - c_{s+1,r}} z^{-B_{s+1}} \cdot$$

$$\cdot \det \begin{pmatrix} Q_{\lambda_1+B_1-s+1} - Q_{\lambda_1+B_1-s} z^{-1} & \ldots & Q_{\lambda_1+B_1} - Q_{\lambda_1+B_1-1} z^{-1} \\ \vdots & \ddots & \vdots \\ Q_{\lambda_s+B_s-s+1} - Q_{\lambda_s+B_s-s} z^{-1} & \ldots & Q_{\lambda_s+B_s} - Q_{\lambda_s+B_s-1} z^{-1} \end{pmatrix} \cdot (Q_0 + Q_1 z + Q_2 z^2 + \ldots).$$

The coefficient of any power of z in this expression is a polynomial of the same form as stated in the proposition. So, any element of $W^{k\omega} \cap \pi_{k\omega - r\alpha - (s+1)\beta}$ is a linear combination of such polynomials. To show that all such polynomials themselves lie in $W^{k\omega}$ for any r, s, we consider the polynomial of maximal degree lying in $W^{k\omega} \cap \pi_{k\omega - r\alpha - s\beta}$, i.e. the polynomial $R_{k-1,k-2,\ldots,k-r;r-1,r-2,\ldots,r-s}$. Then, all other polynomials $R_{\mu,\lambda}$ can be obtained from this one by applying the «positive half» of the Heisenberg Lie algebra.

For arbitrary n, the proof is similar. The only difference is that we also need to use the third relation from Lemma 10. □

As for the polynomials from Proposition 19, the elements we just described are linearly dependent. A linearly independent set of such elements spanning $W^{k\omega} \cap \pi_{k\omega - r_1\alpha^1 - \ldots - r_{n-1}\alpha^{n-1}}$ is given in the following proposition.

Proposition 28. *A basis of* $W^{k\omega} \cap \pi_{k\omega - r_1\alpha^1 - \ldots - r_{n-1}\alpha^{n-1}}$ *is formed by all polynomials from Proposition 27 such that each tuple* λ^j *consists of non-negative integers.*

Proof. The fact that such polynomials are linearly independent can be proved in the same manner as in the proof of Proposition 21. Indeed, suppose that there is a non-trivial linear combination of such polynomials equal to zero. Consider all terms of this linear combination with the highest degree in α^1_{-s}. For these terms, $A^1_1 = A^1_2 = \ldots = A^1_{r_1} = 0$, $B^1_1 = \ldots = B^1_{r_2} = 0$ and the sum $\lambda^1_1 + \lambda^1_2 + \ldots + \lambda^1_{r_1}$ is maximal among the terms of this linear combination. The sum of these terms must also be equal to zero. But the sum of these terms with any fixed λ^1 is the product of $P^1_{\lambda^1}$ and a linear combination of polynomials $R_{\lambda^2;\ldots;\lambda^{n-1}}$. Using induction on n, we conclude that all terms of this linear

combination must be zero. So, the initial linear combination of $R_{\lambda^1;\lambda^2;\ldots;\lambda^{n-1}}$ must also be trivial.

To show that such $R_{\lambda^1;\lambda^2;\ldots;\lambda^{n-1}}$ form a basis of $W^{k\omega} \cap \pi_{k\omega - r_1\alpha^1 - \ldots - r_{n-1}\alpha^{n-1}}$, it suffices to show now that the number of these polynomials is equal to the dimension of this space. The number of these polynomials is the number of choices of $\lambda^1, \ldots, \lambda^{n-1}$. Taking into account the restrictions $0 \leqslant \lambda_i^j \leqslant r_{j-1} - 1$, we see that it is equal to

$$\binom{k}{r_1} \cdot \binom{r_1}{r_2} \cdot \ldots \cdot \binom{r_{n-2}}{r_{n-1}} = \frac{k!}{(k-r_1)!(r_1-r_2)!\ldots(r_{n-2}-r_{n-1})!r_{n-1}!}.$$

Finally, recalling that $\alpha^i = a^i - a^{i-1}$ for any i, we obtain

$$k\omega - r_1\alpha^1 - \ldots - r_{n-1}\alpha^{n-1} = k\omega - (r_1 - r_2)a^1 - (r_2 - r_3)a^2 - \ldots - (r_{n-2} - r_{n-1})a^{n-1},$$

we see that this multinomial coefficient coincides with the dimension of $W^{k\omega} \cap \pi_{k\omega - r_1\alpha^1 - \ldots - r_{n-1}\alpha^{n-1}}$ computed in the proof of Corollary 6. □

The next proposition describes linear relations that the polynomials $R_{\lambda^1;\ldots;\lambda^{n-1}}$ satisfy.

Proposition 29. *If $\lambda_{r_i}^i < 0$, then for any i we have*

$$\sum_{\substack{\delta_1,\ldots,\delta_r=0,1 \\ \Delta=\delta_1+\ldots+\delta_{r_i}}} (-1)^{\Delta} R_{\lambda^1;\ldots;\lambda_1^{i-1}-\delta_1,\ldots,\lambda_{r_{i-1}}-\delta_{r_{i-1}};\lambda_1^i,\ldots,\lambda_{r_i-1}^i,\lambda_{r_i}^i+\Delta;\ldots;\lambda^{r_{n-1}}} = 0. \tag{30}$$

Proof. First, we are going to prove this formula for $n = 3$. In this case, we need to show that

$$\sum_{\substack{\delta_1,\ldots,\delta_r=0,1 \\ \Delta=\delta_1+\ldots+\delta_r}} (-1)^{\Delta} R_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda_1,\ldots,\lambda_{s-1},\Delta-p} = 0$$

for any $p > 0$. First, we have

$$\sum_{\substack{\delta_1,\ldots,\delta_r=0,1 \\ \Delta=\delta_1+\ldots+\delta_r}} (-1)^{\Delta} R_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda_1,\ldots,\lambda_{s-1},\Delta-p} =$$

$$= \sum_{\substack{\delta_1,\ldots,\delta_r=0,1 \\ \Delta=\delta_1+\ldots+\delta_r}} (-1)^{\Delta} \sum P_{\mu_1-\delta_1-A_1,\ldots,\mu_r-\delta_r-A_r} Q_{\lambda_1+B_1,\ldots,\lambda_{s-1}+B_{s-1},\Delta-p+B_s},$$

where A_i and B_j are sums of the elements of columns and rows of all possible matrices

$$C = \begin{pmatrix} c_{1,1} & c_{1,2} & \ldots & c_{1,r} \\ c_{2,1} & c_{2,2} & \ldots & c_{2,r} \\ \vdots & \vdots & \ddots & \vdots \\ c_{s,1} & c_{s,2} & \ldots & c_{s,r} \end{pmatrix}$$

whose elements are non-negative integers. Denote by A_i' the sum of all elements of the i-th column of C except the last one, i.e. $A_i' = A_i - c_{s,i}$. Then, the last sum is equal to

$$\sum_{\substack{\delta_1,\ldots,\delta_r=0,1 \\ \Delta=\delta_1+\ldots+\delta_r}} (-1)^{\Delta} \sum P_{\mu_1-\delta_1-A_1'-c_{s,1},\ldots,\mu_r-\delta_r-A_r'-c_{s,r}} Q_{\lambda_1+B_1,\ldots,\lambda_{s-1}+B_{s-1},\Delta-p+B_s}.$$

Consider all terms of this sum corresponding to matrices C with fixed elements in all rows except the last one, i.e. except $c_{s,1}, c_{s,2}, \ldots, c_{s,r}$. It can be seen that all such terms mutually cancel except for the one corresponding to $c_{s,1} = c_{s,2} = \ldots = c_{s,r} = 0$ and $\delta_1 = \delta_2 = \ldots = \delta_r = 0$. So, the remaining sum is

$$\sum_{\substack{\delta_1,\ldots,\delta_r=0,1\\ \Delta=\delta_1+\ldots+\delta_r}} (-1)^{\Delta} \sum P_{\mu_1-A'_1,\ldots,\mu_r-A'_r} Q_{\lambda_1+B_1,\ldots,\lambda_{s-1}+B_{s-1},-p}.$$

This sum is zero since $Q_{\lambda_1+B_1,\ldots,\lambda_{s-1}+B_{s-1},-p}$ is zero for $-p < 0$.

Finally, the statement for $\mathfrak{sl}(n,\mathbb{C})$ follows from the case $n = 3$. Indeed, we can expand each term of (30) using the formula from Proposition 27. After that, consider all terms of the sum we obtained with fixed matrices C_j, $j \neq i-1$. Note that the sum of such terms is the product of $\prod_{t\neq i,i-1} P^t_{\lambda^t_1+B^{t-1}_1-A^t_1,\ldots,\lambda^t_{r_t}+B^{t-1}_{r_t}-A^t_{r_t}}$ and the sum

$$\sum_{\substack{\delta_1,\ldots,\delta_{r_{i-1}}=0,1\\ \Delta=\delta_1+\ldots+\delta_{r_{i-1}}}} (-1)^{\Delta-1} \sum P^{i-1}_{\lambda^{i-1}_1+B^{i-2}_1-A^{i-1}_1-\delta_1,\ldots,\lambda^{i-1}_{r_{i-1}}+B^{i-2}_{r_{i-1}}-A^{i-1}_{r_{i-1}}-\delta_{r_{i-1}}} P^{i}_{\lambda^{i}_1+B^{i-1}_1-A^{i}_1,\ldots,\lambda_{r_i}+B^{i-1}_{r_i}-A^{i}_{r_i}}$$

which is of the same type as we considered above (when working with the case $n = 3$). So, it follows from what we have already done that this sum is zero. □

It is easy to write an explicit formula for the action of L_1 on $R_{\lambda^1;\lambda^2;\ldots;\lambda^{n-1}}$.

Proposition 30. *For any $\lambda^1, \ldots, \lambda^{n-1}$ we have (assuming that $r_0 = k$)*

$$L_1(R_{\lambda^1;\ldots;\lambda^{n-1}}) = \sum_{j=1}^{n-1}\sum_{t=1}^{r_j} (\lambda^j_t - r_{j-1} + r_j) R_{\lambda_1;\ldots;\lambda^j_1,\ldots,\lambda^j_t-1,\ldots,\lambda^j_{r_j};\ldots;\lambda^{n-1}}.$$

Proof. It is easy to see from a direct computation that $L_1(R_{\lambda^1;\ldots;\lambda^{n-1}})$ is a linear combination of $R_{\lambda_1;\ldots;\lambda^j_1,\ldots,\lambda^j_t-1,\ldots,\lambda^j_{r_j};\ldots;\lambda^{n-1}}$ with coefficients equal to $\lambda^j_t - C_j$, where C_j is some constant. The value $C_j = r_{j-1} - r_j$ is determined uniquely from the fact that the polynomial $R_{\lambda^1;\ldots;\lambda^{n-1}}$ of the maximal degree is sent to 0 by L_1. □

If we look at the formulas we obtained in the case $n = 3$, we can see that they coincide with (20) and (21). For arbitrary n, they can be viewed as a generalization of (20) and (21). In Section 8, we replaced the basis $R_{\mu;\lambda}$ with $S_{\mu;\lambda}$ in such a way that the space $W^{k\omega} \cap \pi_{k\omega-r\alpha-s\beta}$ splits naturally into the direct sum of its subspaces U_{d_1,d_2} after this change. Recall that U_{d_1,d_2} are not L_1-invariant which was not a problem for $n = 3$. Now, however, we are going to show how to change $S_{\mu;\lambda}$ with yet another basis $T_{\mu;\lambda}$ and obtain an explicit decomposition into a direct sum of L_1-invariant subspaces. After that, we are going to extend this construction to arbitrary n and compute the corresponding vector bundles on $\mathbb{P}^1$.

First, we describe this change for the space linearly spanned by elements of the form $\widetilde{S}_{\mu;\lambda}$, where $\mu = (\mu_1, \ldots, \mu_r)$ is an r-tuple of integers, $\mu_i \leqslant k-1$, and $\lambda \leqslant r-1$ is just one number, with the following action of L_1

$$L_1(\widetilde{S}_{\mu;\lambda}) = \sum_{t=1}^{r} (\mu_t - C)\widetilde{S}_{\mu_1,\ldots,\mu_t-1,\ldots,\mu_r;\lambda} + (\lambda - a)S_{\mu;\lambda-1} \text{ if } \lambda \geqslant a,$$

$$L_1(\widetilde{S}_{\mu;\lambda}) = \sum_{t=1}^{r}(\mu_t - C - 1)\widetilde{S}_{\mu_1,\ldots,\mu_t-1,\ldots,\mu_r;\lambda} + (\lambda - a)\widetilde{S}_{\mu;\lambda-1} \text{ if } \lambda \leqslant a-1$$

and the following relations:

$$\widetilde{S}_{\mu;-1} = \sum_{\substack{\delta_1,\ldots,\delta_r=0,1\\ r\geqslant\Delta\geqslant a+1}} (-1)^{\Delta-1}\widetilde{S}_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\Delta-1}. \tag{31}$$

Here, a is any integer from 0 to $r-1$ and C is any constant. These formulas are a generalized version of the formulas from Lemma 8 (in Lemma 8, $a = r-1$ and $C = k-r$). Our aim is to replace the polynomials $\widetilde{S}_{\mu;\lambda}$ ($\mu = (\mu_1,\ldots,\mu_r)$, λ is just one number) with $\widetilde{T}_{\mu;\lambda}$ in such a way that $\widetilde{T}_{\mu;-1} = 0$. The main idea is to search for such a change in the following form:

$$\widetilde{S}_{\mu;\lambda} = \widetilde{T}_{\mu;\lambda} \text{ if } \lambda \geqslant a,$$

$$\widetilde{S}_{\mu;\lambda} = \widetilde{T}_{\mu;\lambda} + \sum_{a-\lambda\leqslant\Delta\leqslant r-1-\lambda} A_{\Delta,a-\lambda}\widetilde{T}_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda+\Delta} \text{ if } \lambda \leqslant a-1.$$

We want to choose the coefficients $A_{\Delta,a-\lambda}$ in such a way that the formulas for the L_1-action remain unchanged in the new basis. Formally speaking, we want to ensure that the formulas for the L_1-action in the new basis can be obtained from the formulas written above simply replacing $\widetilde{S}$ with $\widetilde{T}$. We need to know what conditions this imposes on $A_{\Delta,a-\lambda}$. We have

$$L_1(\widetilde{T}_{\mu;\lambda}) = L_1(\widetilde{S}_{\mu;\lambda} - \sum_{a-\lambda\leqslant\Delta\leqslant r-1-\lambda} T_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda+\Delta}) =$$

$$= \sum_{t=1}^{r}(\mu_t - C - 1)\Big(\widetilde{T}_{\mu_1,\ldots,\mu_t-1,\ldots,\mu_r;\lambda} + \sum_{a-\lambda\leqslant\Delta\leqslant r-1-\lambda} A_{\Delta,a-\lambda}\widetilde{T}_{\mu_1-\delta_1,\ldots,\mu_t-\delta_t-1,\ldots,\mu_r-\delta_r;\lambda+\Delta}\Big) -$$

$$-(a-\lambda)\Big(\widetilde{T}_{\mu;\lambda-1} + \sum_{a-\lambda+1\leqslant\Delta\leqslant r-\lambda} A_{\Delta,a-\lambda+1}\widetilde{T}_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda+\Delta-1}\Big) -$$

$$- \sum_{a-\lambda\leqslant\Delta\leqslant r-1-\lambda} A_{\Delta,a-\lambda}\Big(\sum_{t=1}^{r}(\mu_t-\delta_t-C)\widetilde{T}_{\mu_1-\delta_1,\ldots,\mu_t-\delta_t-1,\ldots,\mu_r-\delta_r;\lambda+\Delta}+(\lambda+\Delta-a)\widetilde{T}_{\mu_1-\delta_1,\ldots,\mu_r-\delta_r;\lambda+\Delta-1}\Big).$$

The coefficients of $\widetilde{T}_{\mu_1,\ldots,\mu_t-1,\ldots,\mu_r;\lambda}$ and $\widetilde{T}_{\mu;\lambda-1}$ in this expression are equal to $\mu_t - C - 1$ and $\lambda - a$ respectively as we need. The coefficient of $\widetilde{T}_{\mu_1-2,\mu_2-1,\ldots,\mu_m-1,\mu_{m+1},\ldots,\mu_r;\lambda+\Delta}$ is equal to

$$(\mu_1 + \ldots + \mu_m - Cm - m)A_{\Delta,a-\lambda} - A_{\Delta,a-\lambda}(\mu_1 + \ldots + \mu_m - m - Cm) = 0,$$

while the coefficient of $\widetilde{T}_{\mu_1-1,\ldots,\mu_m-1;\mu_{m+1},\ldots,\mu_r;\lambda+\Delta}$ for $m \geqslant 2$ is

$$A_{\Delta,a-\lambda}(\mu_1 + \ldots + \mu_m - Cm - m) - (a-\lambda)A_{\Delta+1,a-\lambda+1} -$$

$$-A_{\Delta,a-\lambda}(\mu_1 + \ldots + \mu_m - Cm) - A_{\Delta+1,a-\lambda}(\lambda + \Delta - a) =$$

$$= -mA_{\Delta,a-\lambda} - (a-\lambda)A_{\Delta+1,a-\lambda+1} - (\lambda + \Delta - a)A_{\Delta+1,a-\lambda}.$$

We want the last expression to be zero which leads to the following conditions:

$$A_{\Delta+1,a-\lambda+1} = -\frac{m}{a-\lambda}A_{\Delta,a-\lambda} - \frac{\lambda+\Delta-a}{a-\lambda}A_{\Delta+1,a-\lambda} \text{ if } \Delta \neq r-1-\lambda,$$

$$A_{r-\lambda,a-\lambda+1} = -\frac{m}{a-\lambda} A_{r-\lambda-1,a-\lambda} \text{ if } \Delta = r-1-\lambda.$$

For a more convenient notation, let $b^i_{a-\lambda} = A_{a-\lambda+i,a-\lambda}$ $(i = 0, 1, \dots, r-a)$. Then, denoting $t = a - \lambda$, we have

$$b^i_{t+1} = -\frac{m}{a-\lambda} b^i_t - \frac{\lambda+\Delta-a}{a-\lambda} b^{i+1}_t, \ \ \Delta = a-\lambda+i,$$

$$b^{r-a}_{t+1} = -\frac{m}{a-\lambda} b^{r-a}_t.$$

These formulas show that the transition matrix from b^i_t to b^i_{t+1}, $i = 0, 1, \dots, r-a$ is invertible upper-triangular. So, if we choose arbitrary values of b^i_t for one t, then all other numbers $b^i_{t'}$ are determined uniquely and the operator L_1 acts on $\widetilde{T}_{\mu;\lambda}$ by the same formulas as on $\widetilde{S}_{\mu;\lambda}$.

Also, we require $\widetilde{T}_{\mu;-1}$ to vanish. We have

$$\widetilde{S}_{\mu;-1} = \widetilde{T}_{\mu;-1} + \sum_{\Delta \geqslant a+1} A_{\Delta,a+1} \widetilde{T}_{\mu_1-\delta_1,\dots,\mu_r-\delta_r;\Delta-1} =$$

$$= \widetilde{T}_{\mu;-1} + \sum_{\Delta \geqslant a+1} A_{\Delta,a+1} \widetilde{S}_{\mu_1-\delta_1,\dots,\mu_r-\delta_r;\Delta-1}$$

since $\widetilde{S}_{\mu;\lambda} = \widetilde{T}_{\mu;\lambda}$ if $\lambda \geqslant a$. Using expression (31), we see that $A_{\Delta,a+1} = b^{\Delta-a-1}_{a+1}$ must be equal to $(-1)^{\Delta-1}$. So, we are done.

For arbitrary s, we can state the following proposition.

Proposition 31. *The basis $S_{\mu;\lambda}$ of $W^k\omega \cap \pi_{k\omega-r\alpha-s\beta}$ can be replaced with $T_{\mu;\lambda}$ related to $S_{\mu;\lambda}$ by the formula*

$$S_{\mu;\lambda} = \sum B_{\Delta,\lambda} T_{\mu_1-\Delta^1,\dots,\mu_r-\Delta^r;\lambda_1+\Delta_1,\dots,\lambda_s+\Delta_s},$$

where $B_{0,\lambda} = 0$ for any λ. The sum is taken over all matrices of the form

$$\Delta = \begin{pmatrix} \delta^1_1 & \delta^2_1 & \dots & \delta^r_1 \\ \delta^1_2 & \delta^2_2 & \dots & \delta^r_2 \\ \vdots & \vdots & \ddots & \vdots \\ \delta^1_s & \delta^2_s & \dots & \delta^r_s \end{pmatrix},$$

$\delta^j_i \in \{0,1\}$, $\Delta_i = \sum_{t=1}^r \delta^t_i$, $\Delta^j = \sum_{t=1}^s \delta^j_t$. The formulas of the action of L_1 on $T_{\mu;\lambda}$ can be obtained from (23) simply replacing S with T. Also, we have $T_{\mu;\lambda_1,\dots,\lambda_s} = 0$ if $\lambda_s = -1$.

Proof. The construction described above gives us a desired change of basis of $S_{\mu;\lambda}$ if λ is one number, i. e. if $s = 1$. For arbitrary s, we can change the basis $S_{\mu;\lambda}$ in several steps similarly to what we described in Section 8 after the proof of Lemma 9. Namely, we replace $S_{\mu;\lambda}$ consecutively with $S^s_{\mu;\lambda}$, $S^{s-1}_{\mu;\lambda}$, …, $S^1_{\mu;\lambda}$. At each step, when replacing $S^{i+1}_{\mu;\lambda}$ with $S^i_{\mu;\lambda}$, we use the change of basis described above with respect to $\mu = (\mu_1, \dots, \mu_r)$ and λ_i while λ_j, $j \neq i$, remain fixed (as in Section 8, we temporarily forget about anticommutativity in λ_i). Eventually, we obtain the change of basis given in the proposition. □

An important observation is that the following conditions hold:

1. The coefficients $B_{\Delta,\lambda}$ do not depend on μ.

2. There are no terms $T_{\mu';\lambda'}$ in the right-hand side of the given formula such that $\mu'_i > \mu_i$.

Moreover, we can make the same observation for the transition from the basis $R_{\mu;\lambda}$ to the basis $S_{\mu;\lambda}$ (see formula (25)). This gives us the following result.

Proposition 32. *The space $W^{k\omega} \cap \pi_{k\omega - r\alpha - s\beta}$ is spanned linearly by the polynomials $T_{\mu;\lambda}$, $\mu_i \leqslant k-1$, $\lambda_i \leqslant r-1$. These polynomials are related to $R_{\mu;\lambda}$ as follows:*

$$R_{\mu;\lambda} = \sum C_{\mu',\lambda',\lambda} T_{\mu-\mu';\lambda+\lambda'},$$

where μ', λ' are tuples of non-negative integers and $C_{\mu',\lambda',\lambda}$ are coefficients satisfying the following conditions:

1. *$C_{0,0,\lambda} = 1$ for any μ, λ.*

2. *The coefficients $C_{\mu',\lambda',\lambda}$ do not depend on μ.*

The construction we just described can be directly generalized to the case of arbitrary n. The idea is to replace the polynomials $R_{\lambda^1;\ldots;\lambda^{n-1}} = T^{n-1}_{\lambda^1;\ldots;\lambda^{n-1}}$ successively with $T^{n-2}_{\lambda^1;\ldots;\lambda^{n-1}}, \ldots, T^{1}_{\lambda^1;\ldots;\lambda^{n-1}}$. At each step, when replacing T^{j+1} with T^j, we apply the change described in Proposition 31 to T^{j+1} with respect to the pair $(\lambda^j, \lambda^{j+1})$ (all other indices remain fixed). The only thing that might be unclear is why relations (30) remain unchanged when we pass from T^{j+1} to T^j, $j > i$. To avoid bulky notation, we show this for $n = 4$, $i = 1$, $j = 2$ (for the general case, the proof is the same). In this particular case, we have

$$\sum_{\substack{\delta_1,\ldots,\delta_r=0,1\\ \Delta=\delta_1+\ldots+\delta_r}} (-1)^{\Delta} R_{\lambda_1-\delta_1,\ldots,\lambda_r-\delta_r;\mu_1,\ldots,\mu_s+\Delta;\nu} = 0$$

if $\mu_s < 0$. We need to show that this equality remains true if we replace R with T^2. After this replacement, we obtain the expression

$$\sum_{\substack{\delta_1=\ldots=\delta_r=0,1\\ \Delta=\delta_1+\ldots+\delta_r}} (-1)^{\Delta} T^2_{\lambda_1-\delta_1,\ldots,\lambda_r-\delta_r;\mu_1,\ldots,\mu_s+\Delta;\nu} =$$

$$= \sum_{\substack{\delta_1=\ldots=\delta_r=0,1\\ \Delta=\delta_1+\ldots+\delta_r}} (-1)^{\Delta} \sum C_{\mu',\nu',\nu} R_{\lambda_1-\delta_1,\ldots,\lambda_r-\delta_r;\mu_1-\mu'_1,\ldots,\mu_s-\mu'_s+\Delta;\nu+\nu'}.$$

Recall that $\mu'_s \geqslant 0$, so $\mu_s - \mu'_s < 0$. So, the sum of all terms in this expression with fixed μ', ν' is zero, since such sum is of the same form as (30). This implies that the whole sum also vanishes and we are done.

As a result, the space $W^{k\omega} \cap \pi_{k\omega - r_1\alpha^1 - \ldots - r_{n-1}\alpha^{n-1}}$ decomposes as the direct sum of its subspaces $U_{d_1,d_2,\ldots,d_{n-1}}$. For any $(n-1)$-tuple $(d_1, d_2, \ldots, d_{n-1})$ of non-negative integers, the space $U_{d_1,d_2,\ldots,d_{n-1}}$ is defined as the space with a basis formed by $R_{\lambda^1;\ldots;\lambda^{n-1}}$ such that each tuple λ^i has d^i «small» indices. Speaking precisely, assuming that the indices in each tuple λ^i are given in descending order, we have

$$r_{i-1} - 1 \geqslant \lambda^i_1 > \ldots > \lambda^i_{r_i-d_i} \geqslant r_{i-1} - r_i + d_{i+1} > \lambda_{r_i-d_i+1} > \ldots > \lambda_{r_i} \geqslant 0$$

for all $i = 1, 2, \ldots, n-1$ (for convenience, we assume that $d_n = 0$ and $r_0 = k$). Each such subspace is L_1-invariant and isomorphic to tensor product of exterior powers of single Jordan blocks. So, to compute the corresponding vector bundles we need to know only the maximal and the minimal L_0-degrees of vectors and the dimension of this subspace. For $U_{d_1,d_2,\ldots,d_{n-1}}$, these degrees are equal to

$$\sum_{t=1}^{n-1}\Big(\sum_{i=1}^{r_t-d_t}(r_{t-1}-i)+\sum_{j=1}^{d_t}(r_{t-1}-r_t+d_{t+1}-j)\Big)$$

and

$$\sum_{t=1}^{n-1}\Big(\sum_{i=1}^{r_t-d_t}(r_{t-1}-d_t+d_{t+1}-i)+\sum_{j=1}^{d_t}(d_t-j)\Big)$$

respectively. Their sum is equal to

$$\sum_{t=1}^{n-1}\Big(2r_{t-1}-r_t+d_{t+1}-1)(r_t-d_t)+(r_{t-1}-r_t+d_{t+1}-1)d_t\Big)=$$

$$=\sum_{t=1}^{n-1}\Big(r_{t-1}(r_t-d_t)+(r_{t-1}-r_t+d_{t+1}-1)r_t\Big)=\widetilde{C}(r_1,\ldots,r_{n-1})+\sum_{t=1}^{n-1}(r_td_{t+1}-r_{t-1}d_t)=$$

$$=\widetilde{C}(r_1,\ldots,r_{n-1})+r_{n-1}d_n-r_0d_1=C(r_1,\ldots,r_{n-1})-kd_1,$$

where $\widetilde{C}(r_1,\ldots,r_{n-1})$ and $C(r_1,\ldots,r_{n-1})$ are polynomials not depending on d_i.

The dimension of $U_{d_1,\ldots,d_{n-1}}$ is also easy to compute and is equal to

$$\prod_{t=1}^{n-1}\binom{r_t-d_{t+1}}{r_t-d_t}\binom{r_{t+1}-r_t+d_{t+1}}{d_t}.$$

So, the vector bundle corresponding to $W^{k\omega}\cap\pi_{k\omega-r_1\alpha^1-\ldots-r_{n-1}\alpha^{n-1}}$ is isomorphic to

$$\bigoplus_{d_1\geqslant\ldots\geqslant d_{n-1}\geqslant 0}\mathcal{O}(2N-kd_1)^{\oplus\prod_{i=1}^{n-1}\binom{r_i-d_{i+1}}{r_i-d_i}\binom{r_{i+1}-r_i+d_{i+1}}{d_i}},\quad N\in\mathbb{Z}.$$

To obtain the vector bundle corresponding to the whole Demazure module $W^{k\omega}$, we need to take the sum over all $r_1, r_2, \ldots, r_{n-1}$. Therefore, the following proposition is proved.

Proposition 33. *The vector bundle corresponding to the Demazure module of level* 1 $W^{k\omega}$ *for* $\mathfrak{sl}(n,\mathbb{C})$ *is isomorphic to (assuming* $d_n = 0$, $r_0 = k$*)*

$$\bigoplus_{\substack{d_1\geqslant\ldots\geqslant d_{n-1}\geqslant 0\\ r_1\geqslant\ldots\geqslant r_{n-1}\geqslant 0}}\mathcal{O}(2N-kd_1)^{\oplus\prod_{i=1}^{n-1}\binom{r_i-d_{i+1}}{r_i-d_i}\binom{r_{i+1}-r_i+d_{i+1}}{d_i}},\quad N\in\mathbb{Z}. \tag{32}$$

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